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(54) **DEVELOPMENT AND APPLICATION OF
T-CELL ENGAGER THERAPEUTIC AGENT**

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C07K 2317/31 (2013.01); **C07K 2317/92**

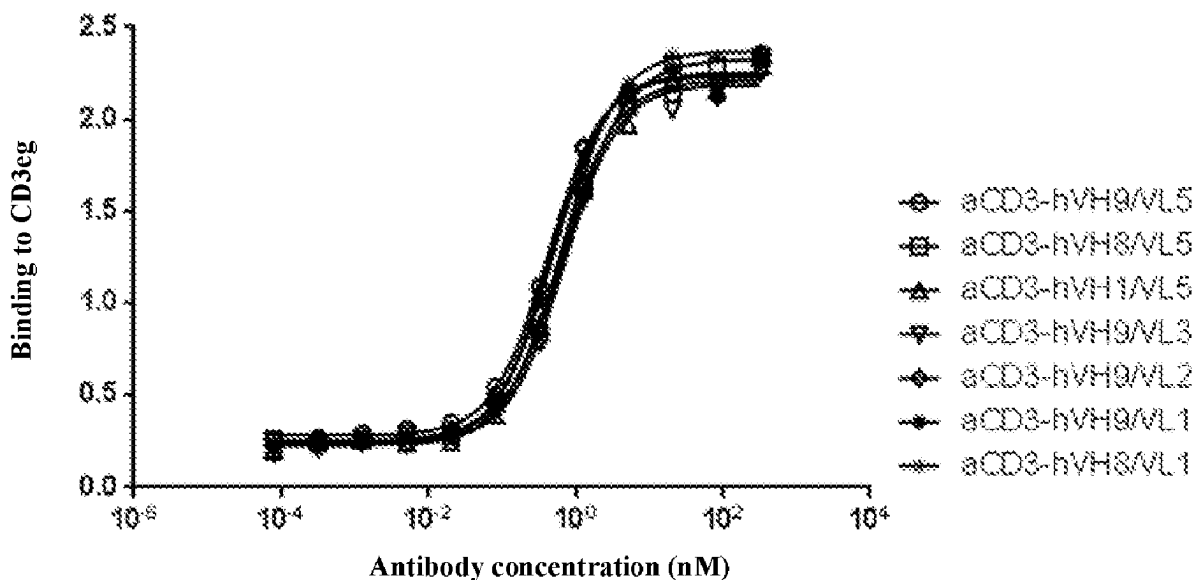
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(57)

ABSTRACT

The present disclosure relates to the development and application of a T-cell engager therapeutic agent. A T-cell engager is a T-cell bispecific antibody, and the bispecific antibody contains a first binding domain bound to BCMA on the surface of a target cell and a second binding domain bound to CD3 on the surface of a T cell. The bispecific antibody has good tolerance, can effectively block the binding of the BCMA to a receptor APRIL, and has no adverse reaction, such as weight loss.

Specification includes a Sequence Listing.



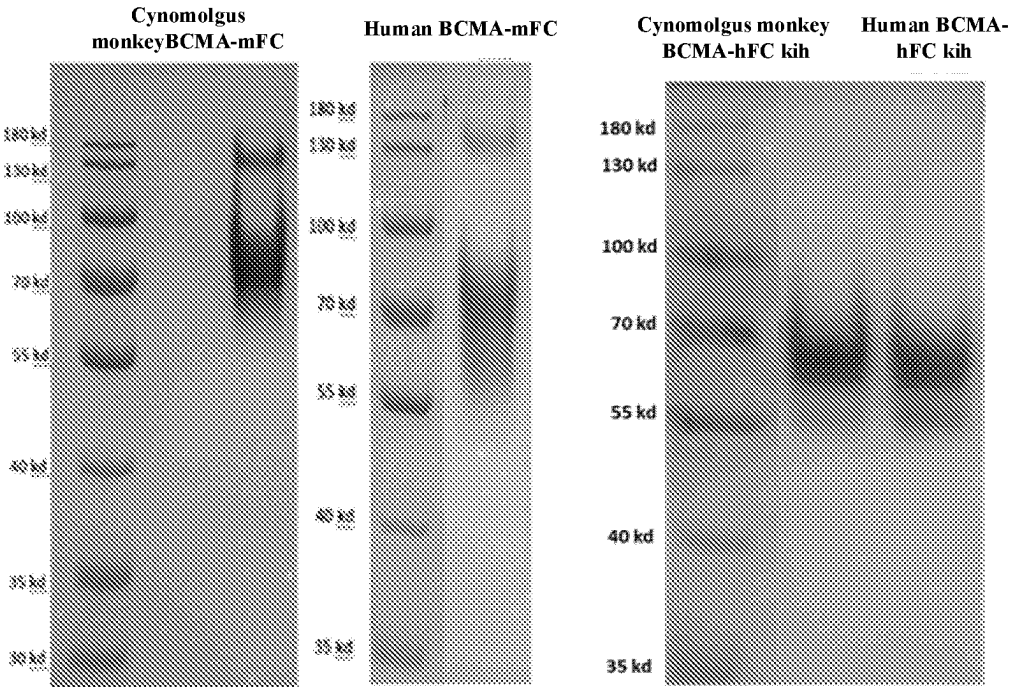


FIG 1

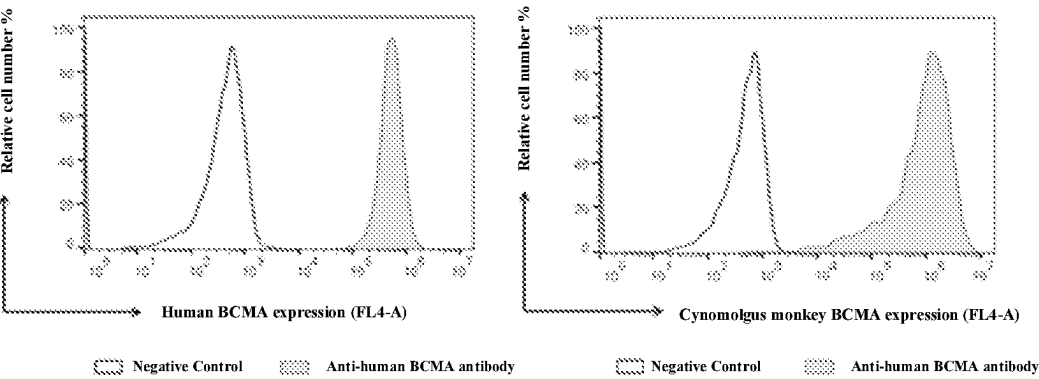


FIG 2

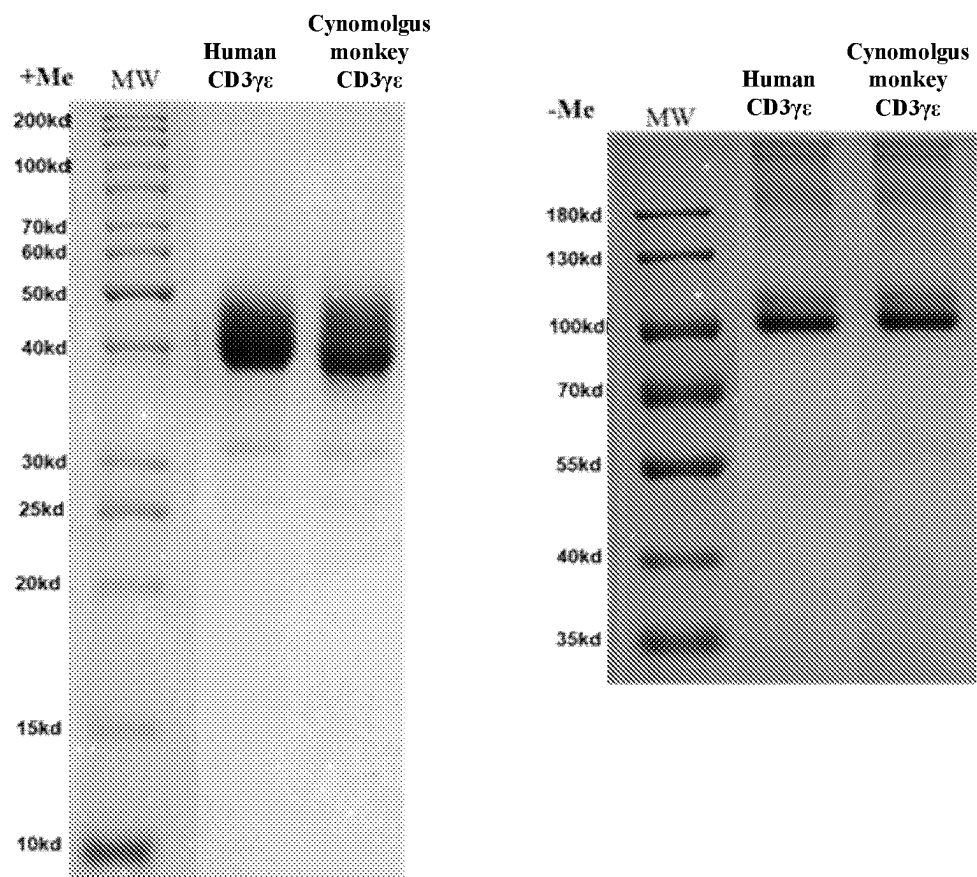


FIG 3

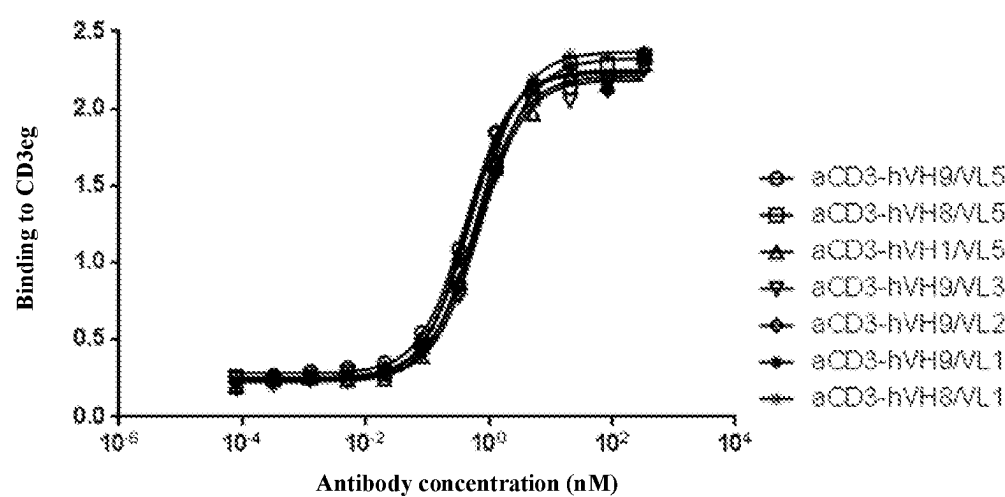


FIG 4

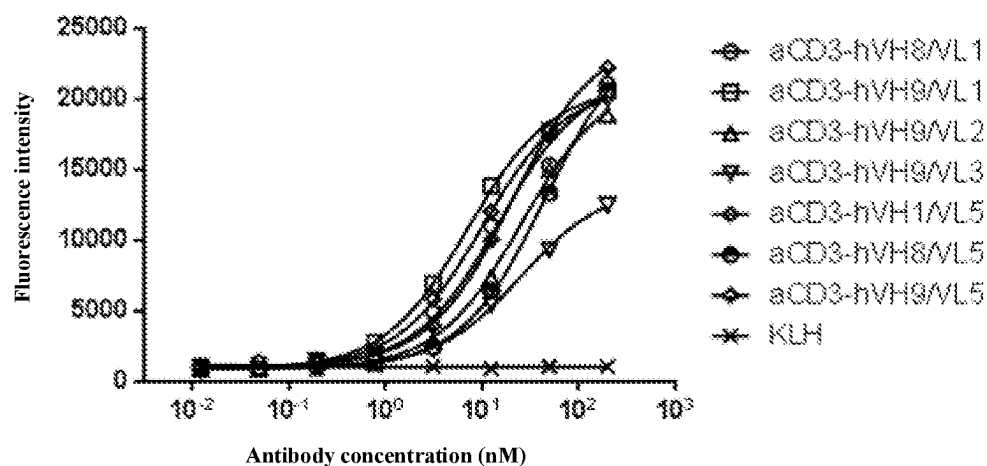
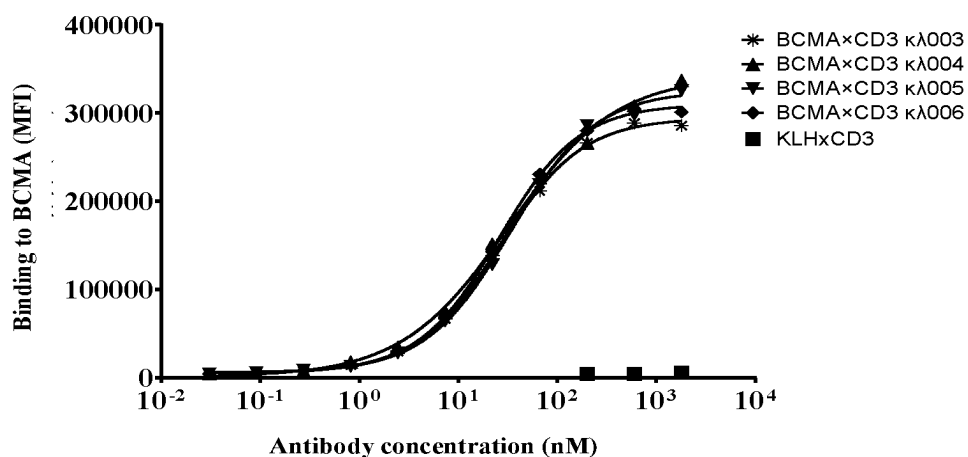


FIG 5

CHO-hBCMA



CHO-cynoBCMA

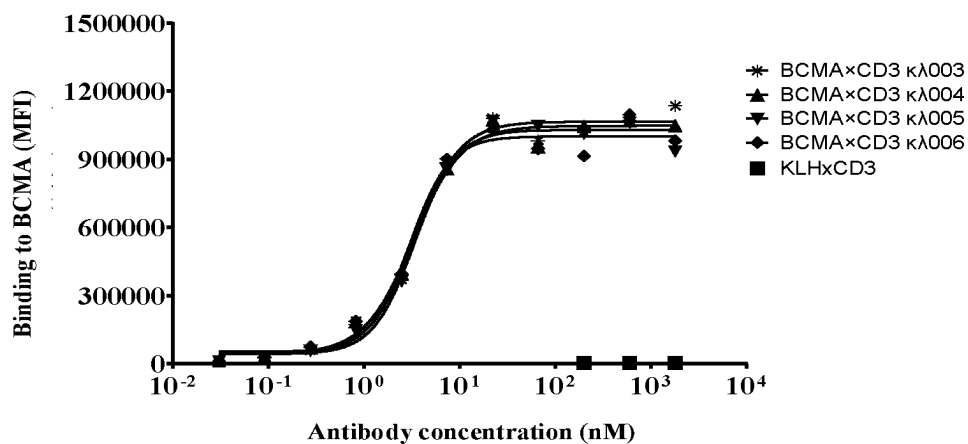


FIG 6

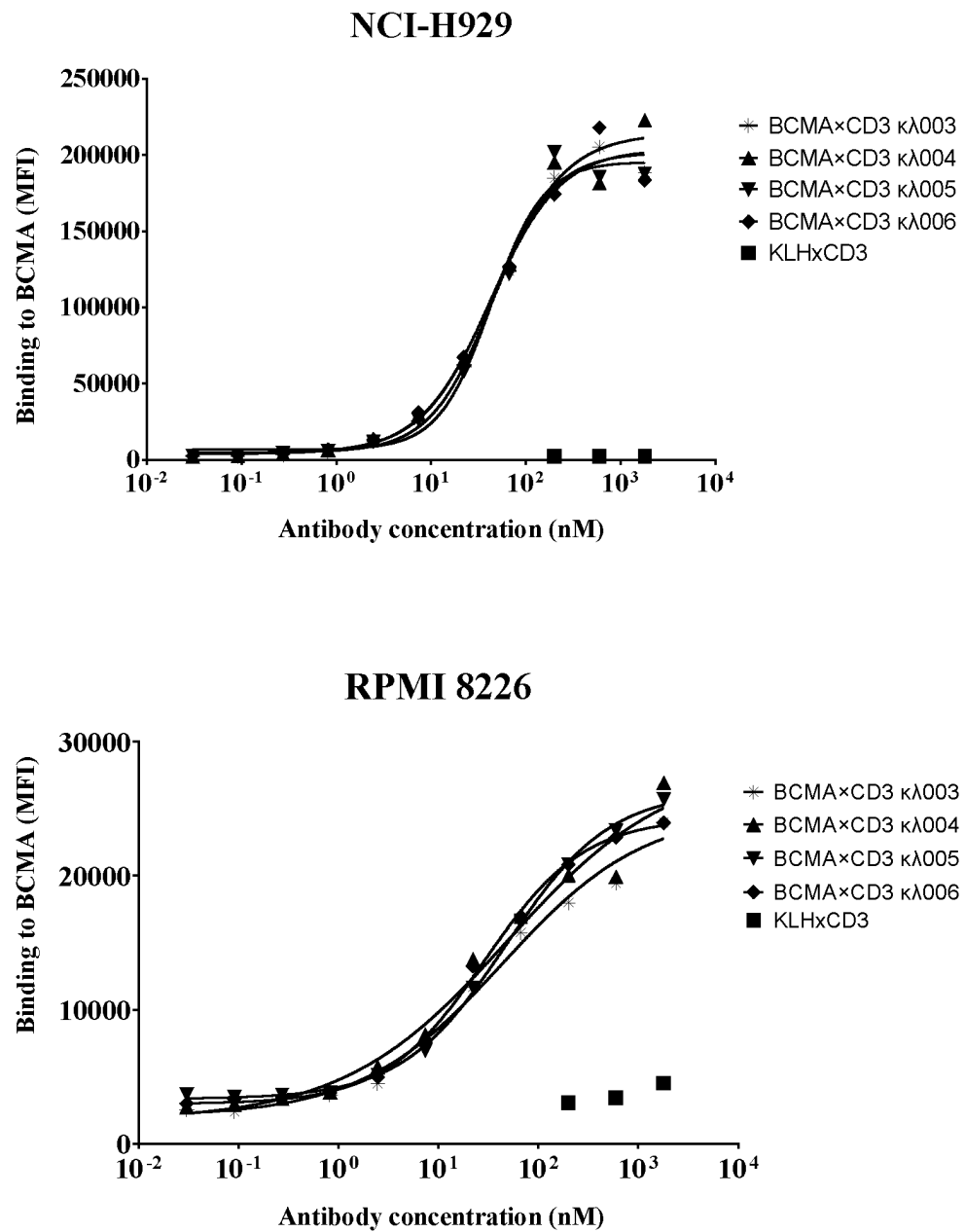


FIG 7

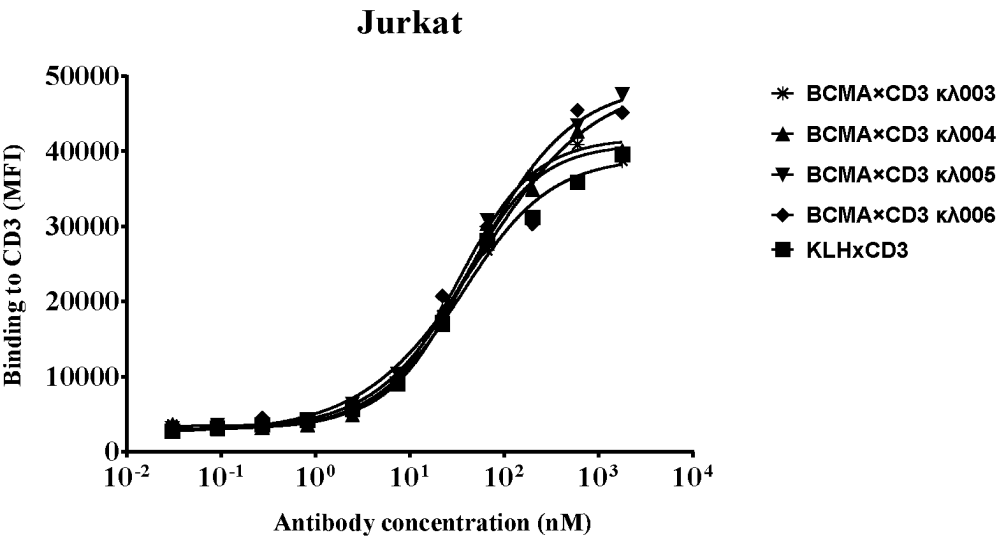


FIG 8

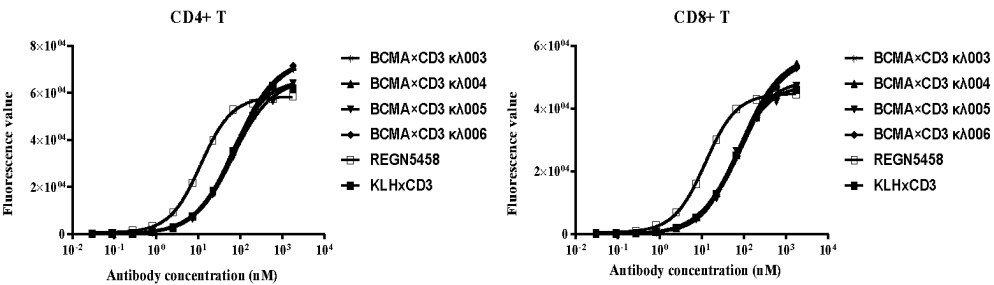


FIG 9

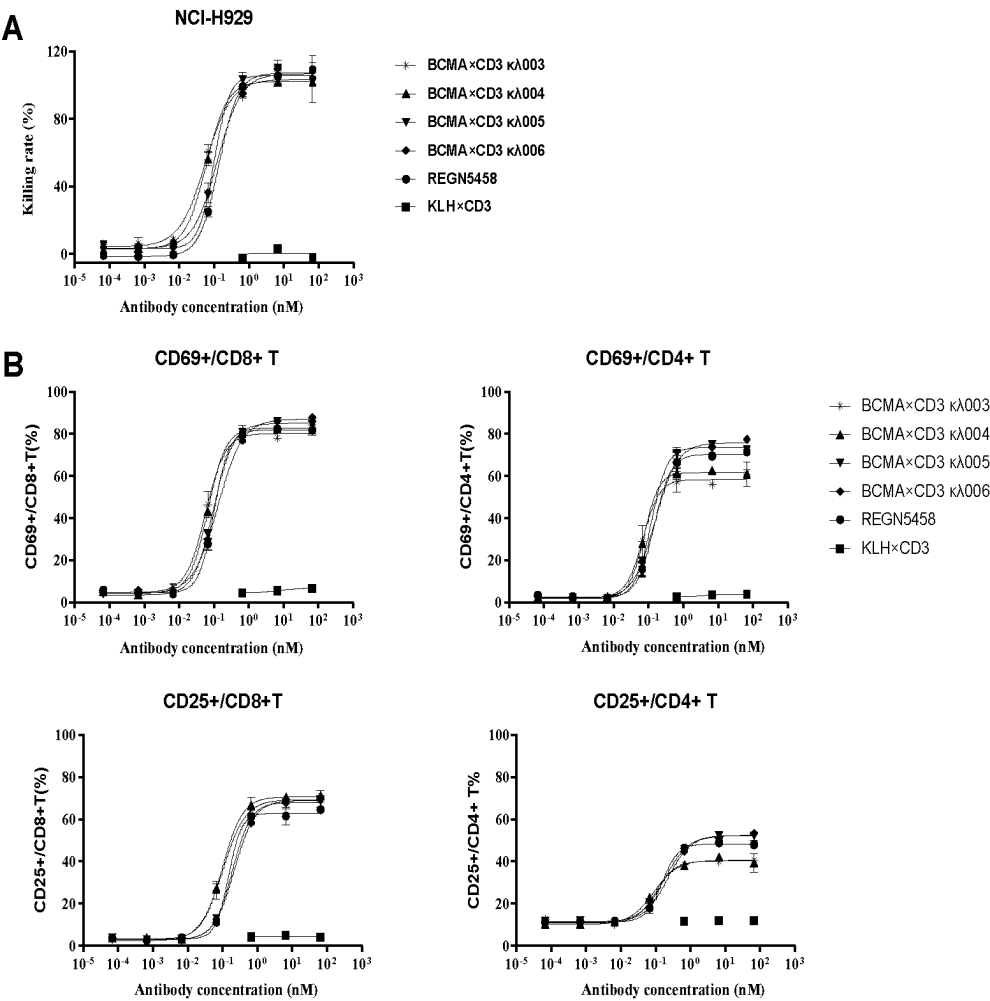


FIG. 10

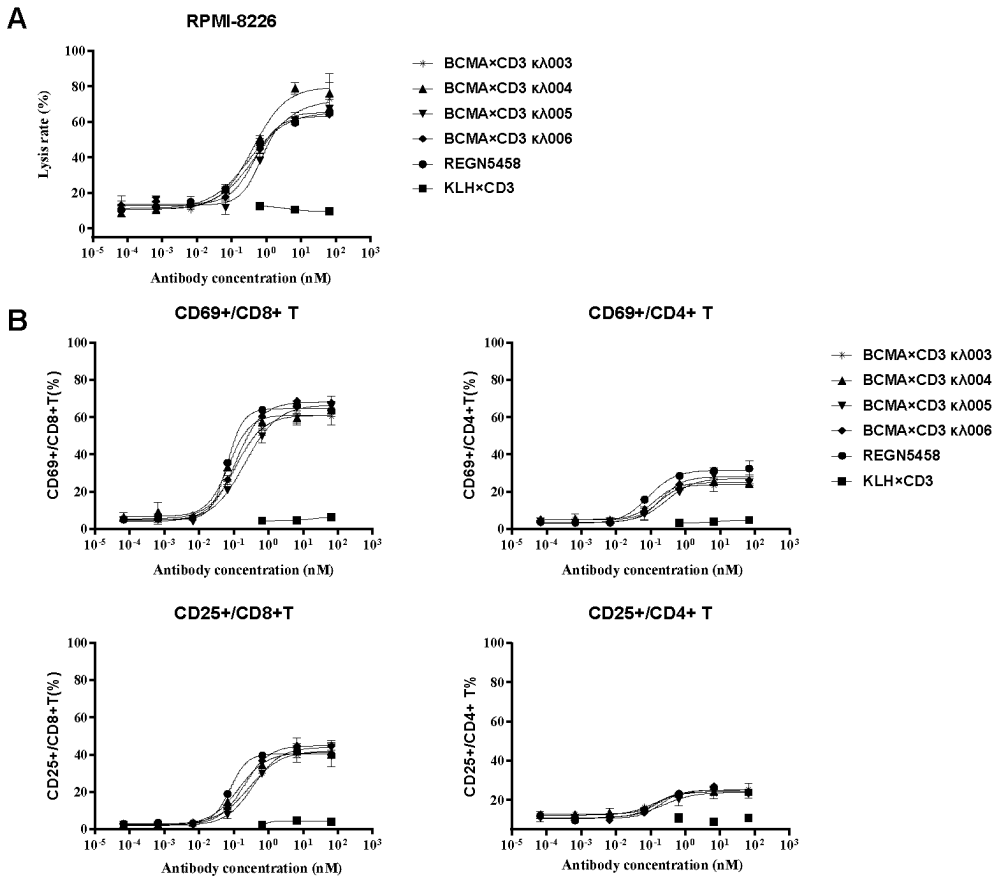


FIG. 11

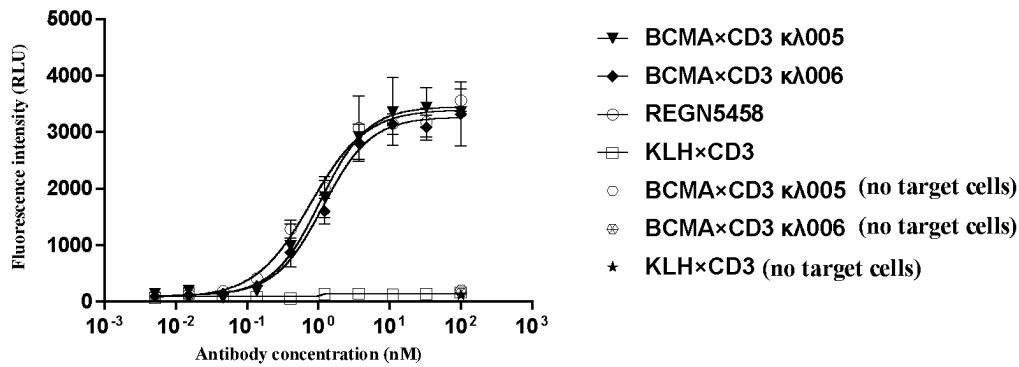


FIG. 12

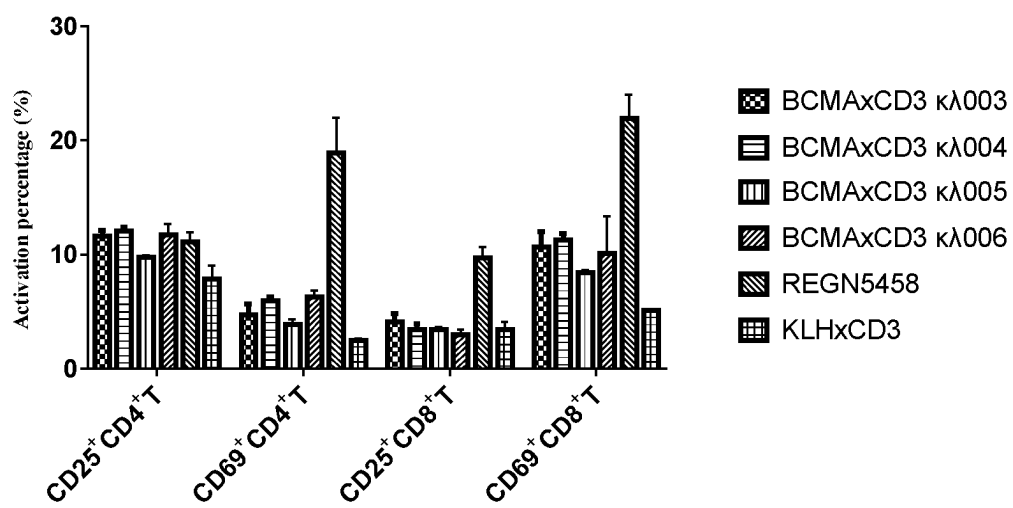


FIG. 13

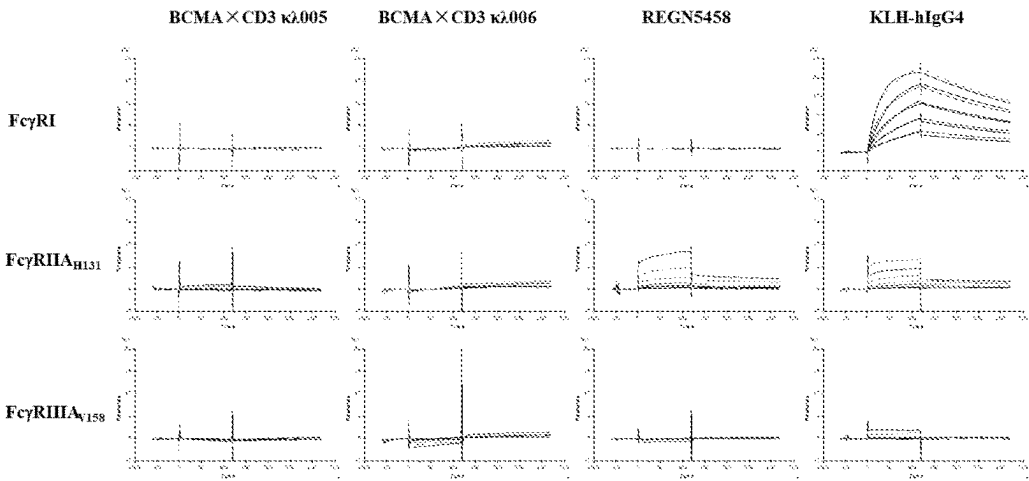


FIG. 14

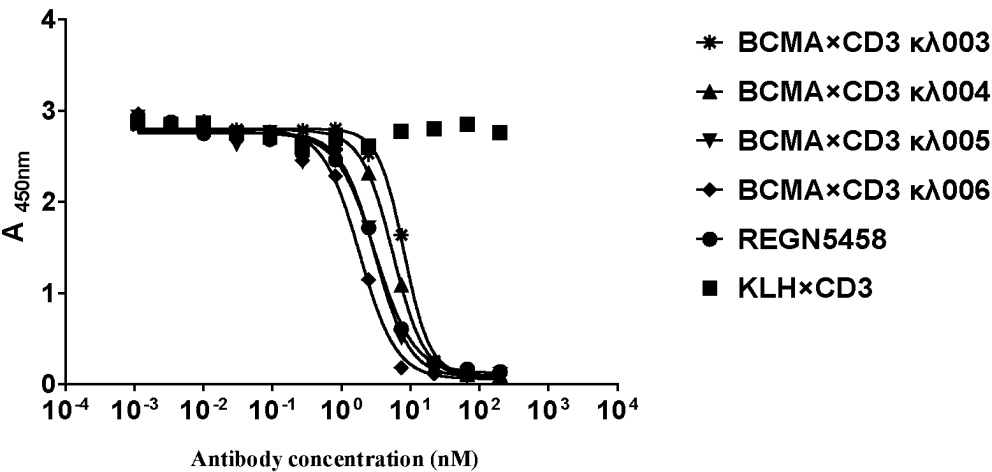


FIG. 15

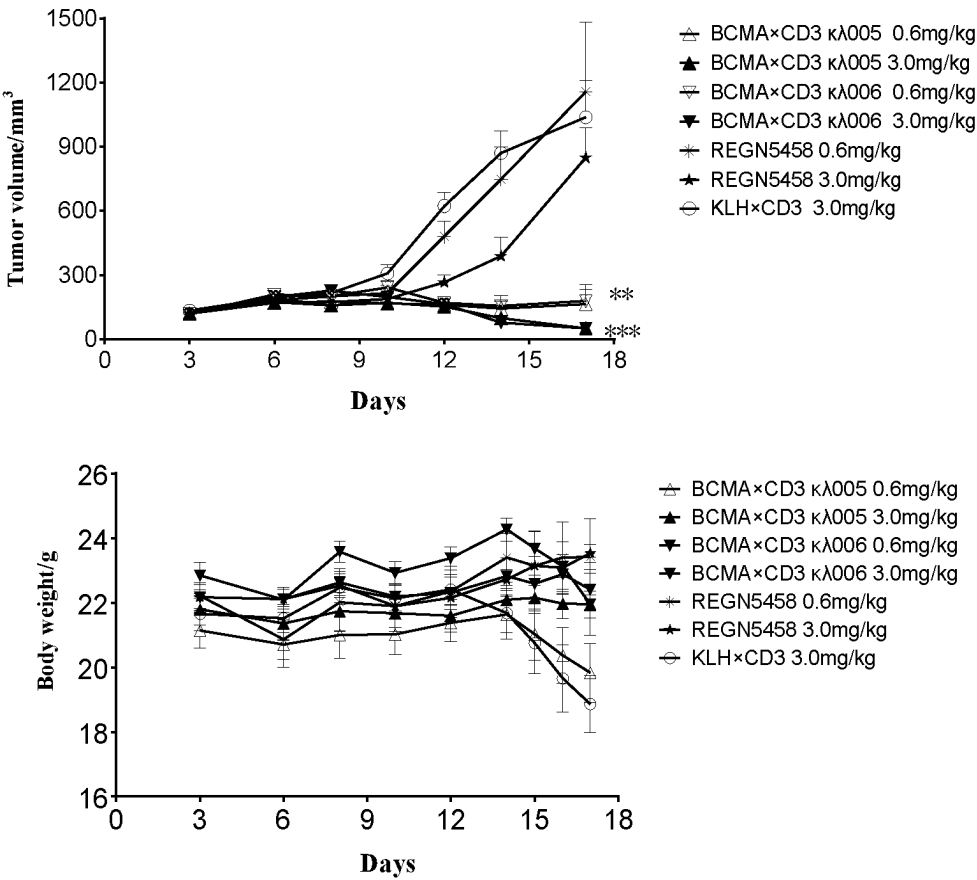


FIG. 16

DEVELOPMENT AND APPLICATION OF T-CELL ENGAGER THERAPEUTIC AGENT

CROSS REFERENCE TO THE RELATED APPLICATIONS

[0001] This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/CN2021/135121, filed on Dec. 2, 2021, which is based upon and claims priority to Chinese Patent Application No. 202011402180.8, filed on Dec. 2, 2020, the entire contents of which are incorporated herein by reference.

SEQUENCE LISTING

[0002] This application contains a Sequence Listing that has been submitted electronically as an ASCII text file named “51426-0007US1_UPDATEDSL_ST25.txt.” The ASCII text file, created on Aug. 21, 2024, is 149,177 bytes in size. The material in the ASCII text file is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0003] The present disclosure relates to the development and application of a T-cell engager therapeutic agent, particularly a bispecific antibody bound to BCMA and CD3, and applications thereof.

BACKGROUND ART

[0004] B-cell maturation antigen (BCMA), as a member of the tumor necrosis factor receptor superfamily, is widely expressed on the surface of myeloma cells, only expressed on plasma cells in normal tissues, but not on naive B cells, CD34+ hematopoietic stem cells, and other normal tissues. BCMA-deficient mice have normal B cell development and humoral immune response. In multiple myeloma patient samples, after the ligand APRIL (A Proliferation Inducing Ligand) binds to BCMA, it may stimulate myeloma cell growth and proliferation, and prevent tumor cell apoptosis simultaneously. Since the expression of BCMA in malignant plasma cells is significantly higher than that in normal plasma cells, BCMA has become an important target for immunotherapy.

[0005] In hematological malignancies, multiple myeloma is the second most common disease, frequently occurring in the elderly. Accumulation of malignantly proliferating plasma cells in the patient's bone marrow may lead to bone marrow failure and loss of immune function, and is often accompanied by multiple osteolytic lesions, hypercalcemia, anemia, and renal damage simultaneously. Although new drugs, including proteasome inhibitors, immunomodulators, and targeted antibodies, have significantly prolonged patient survival in recent years, nearly all patients relapse. New therapies that are being developed such as CAR-T cells (chimeric antigen receptor T-cells) have a complicated preparation process, high treatment cost, and high recurrence rate; antibody drug conjugates (ADC) are limited to low doses due to the safety of conjugated toxins; although BCMA bispecific antibodies previously developed are clinically shown to be quite effective, increased doses are accompanied with increased side effects, which may lead to a strong cytokine storm syndrome and endanger patient life. Thus, there remains a need in the art to develop new, safer, and more effective drugs.

[0006] A T-cell bispecific antibody (or referred to as T-cell engager) is a specially constructed antibody molecule that recognizes a target cell surface antigen (antigen arm) at one end and binds to a T-cell CD3 receptor (CD3 arm) at the other end. Using antibodies that target the ϵ chain of CD3, the CD3 on T cells is aggregated in a manner like TCR/peptide/HLA, thereby activating T cells and killing tumors. The BCMA $\times$ CD3 bispecific molecule is a T-cell TCB antibody that targets BCMA expressed on myeloma cells and the CD3 ϵ chain (CD3 ϵ) present on T cells. The action mechanism of the BCMA $\times$ CD3 bispecific molecule involves binding to both BCMA+ myeloma cells and CD3+ T cells, resulting in T cell activation and T cell-mediated cell killing.

[0007] During a normal immune response, TCR binds with low affinity (about 1-100 μ M) to an exogenous peptide-human leukocyte antigen complex (HLLA) on infected or mutated cells, transduces an activation signal into the nucleus via CD3, activates the expression of transcription factors and their downstream proteins (cytokines, granzymes, perforin, etc.), wherein the signal intensity produced by the TCR complex will determine the fate of T cells. The ϵ , γ , δ , and ζ subunits of the signaling complex associate with each other to form CD3 ϵ - γ heterodimers, CD3 ϵ - δ heterodimers, and CD3 ζ - ζ homodimers. The early developed CD3 bispecific antibodies were mostly based on a few murine antibodies such as OKT3, L2K, UCHT1, and TR66, and have high affinity. In clinical studies, it was found that excessively strong affinity of CD3 antibody would lead to excessive activation of T cells and release of a large number of cytokines, forming cytokine storm syndrome; at the same time, high affinity also leads to the enrichment of bispecific antibodies in secondary lymphoid organs, reducing the exposure to tumor tissue. The binding ability of the Fc portion of the antibody to the Fc γ receptor is another important factor affecting drug safety. Since the Fc γ receptor is expressed in various normal tissues, after bispecific antibody binds to the Fc γ receptor on cell membrane through Fc, it may result in cross-linking activation of CD3 receptor bound at the other end due to Fc γ receptor aggregation, thus resulting in severe off-target toxicity, for example early marketed catumaxomab triggered rapid cytokine release due to Fc fragment binding to Fc γ receptors expressed by liver Kupffer cells. By using a human IgG2 subtype or IgG4 subtype which has a weak Fc γ receptor binding ability, or further performing amino acid substitutions at the corresponding position in CH2, for example, Armour et al. substituted positions 233-236 (EU sequence numbering) of IgG1 and IgG4 with the corresponding sequence of IgG2, reduced binding to Fc γ receptors (Armour K L et al., Recombinant human IgG molecules lacking Fc gamma receptor I binding and monocyte triggering activities, Eur J Immunol. 1999 August; 29(8):2613-24), Newman et al. introduced Ser228Pro and Leu235Glu mutations in IgG4 to stabilize the IgG4 structure while reducing binding to Fc γ receptors (Newman R. et al., Modification of the Fc region of a primatized IgG antibody to human CD4 retains its ability to modulate CD4 receptors but does not deplete CD4(+) T cells in chimpanzees, Clin Immunol. 2001 February; 98(2):164-74), Idusogie et al. found that alanine substitution at positions Asp270, Lys322, Pro329, or Pro331 significantly reduced the binding ability of IgG1 to complement component C1q (Idusogie E E et al., Mapping of the

C1q Binding Site on Rituxan, a Chimeric Antibody with a Human IgG1 Fc, *J Immunol* Apr. 15, 2000, 164 (8) 4178-4184), etc.

SUMMARY OF THE INVENTION

[0008] In order to solve the problems in the prior art, the present disclosure provides a novel BCMA antibody having a high affinity to the BCMA protein. Also, the present disclosure provides novel BCMA-CD3 $\kappa\lambda$ bispecific antibodies formed by BCMA antibodies and CD3 antibodies. By adopting a full-length IgG configuration, introducing charge, and optimizing affinity, BCMA-CD3 $\kappa\lambda$ bispecific antibody allows the $\kappa\lambda$ bispecific antibody preferentially to localize in BCMA+ tumor tissues, can recruit activated T cells at a low concentration, and thus causes effective killing to target cells, but does not activate T cells in the absence of target cells. At the same time, $\kappa\lambda$ bispecific antibodies do not bind to receptors such as Fc γ RI, Fc γ RIIA, Fc γ RIIIA, etc. thereby reducing the risk of cytokine storm occurring. It is proved in a mouse model with reconstituted human immune system that, this novel BCMA-CD3 $\kappa\lambda$ bispecific antibody can effectively inhibit tumor growth, and the efficacy was superior to the same class of antibodies.

[0009] In one aspect, the present disclosure provides a BCMA-binding antibody or antigen-binding portion thereof.

[0010] In one aspect, the present disclosure provides a bispecific antibody or antigen-binding portion thereof.

[0011] In one aspect, the present disclosure provides a nucleic acid encoding bispecific antibodies or antigen-binding portions thereof as previously described.

[0012] In one aspect, the present disclosure provides a vector comprising a nucleic acid as previously described.

[0013] In one aspect, the present disclosure provides a cell comprising a nucleic acid or vector as previously described.

[0014] The antibody or antigen-binding portion thereof according to the foregoing aspects, wherein the antibody or antigen-binding portion thereof as previously described is humanized.

[0015] In one aspect, the present disclosure provides a pharmaceutical composition or kit comprising an antibody or antigen-binding portion thereof, or nucleic acid encoding the same as previously described, and a pharmaceutically acceptable carrier.

[0016] In one aspect, the present disclosure provides antibody-drug conjugates comprising the foregoing antibodies or antigen-binding portions thereof as previously described, bispecific or multispecific molecules covalently attached to a therapeutic moiety.

[0017] In one aspect, the present disclosure provides a method of treating a disorder associated with BCMA, comprising the steps of administering to a mammal a therapeutically effective amount of an antibody or antigen-binding fragment thereof, nucleic acid, vector, cell, and/or pharmaceutical composition as previously described.

[0018] In one aspect, the present disclosure provides a use of the foregoing antibodies or antigen-binding fragments thereof, nucleic acids, vectors, cells, and/or pharmaceutical compositions in the manufacture of a drug or kit for treating a BCMA-associated disorder in a mammal.

[0019] The antibodies of the present disclosure can be used in a variety of applications, including the detection of BCMA proteins, diagnosis, treatment, or prevention of diseases associated with BCMA.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 shows SDS-PAGE results of human, cynomolgus monkey BCMA-mFc bivalent recombinant protein and BCMA-hFc kih monovalent recombinant protein.

[0021] FIG. 2 shows that, both human and cynomolgus monkey BCMA stably transfected cells are detected by flow cytometry, both human BCMA stably transfected cells (CHO-hBCMA-2C2) and cynomolgus monkey BCMA stably transfected cells (CHO-cynoBCMA-1A2) express high levels of BCMA.

[0022] FIG. 3 shows SDS-PAGE results for the CD3 ϵ -Fc recombinant protein.

[0023] FIG. 4 shows binding of humanized CD3 antibody to CD3 ϵ -Fc recombinant protein.

[0024] FIG. 5 shows binding of humanized CD3 antibodies to Jurkat cells.

[0025] FIG. 6 shows binding of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies to BCMA stably transfected cells.

[0026] FIG. 7 shows high-affinity binding of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies to BCMA+ tumor cells.

[0027] FIG. 8 shows binding of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies to Jurkat cells.

[0028] FIG. 9 shows binding of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies to human peripheral blood T cells.

[0029] FIG. 10 shows killing of NCI-H929 cells and activation of T cells by the BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody, wherein FIG. 10A shows killing of NCI-H929 cells, and FIG. 10B shows activation of T cells.

[0030] FIG. 11 shows killing of RPMI-8226 cells and activation of T cells by the BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody, wherein FIG. 11A shows killing of RPMI-8226 cells, and FIG. 11B shows activation of T cells.

[0031] FIG. 12 shows activation of the T cell signaling pathway by the BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody.

[0032] FIG. 13 shows activation of peripheral blood T cells by the BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody.

[0033] FIG. 14 shows binding of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies to activated Fc γ R receptors.

[0034] FIG. 15 shows blocking of BCMA binding to the receptor APRIL by a BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody.

[0035] FIG. 16 shows inhibitory effect of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies on the subcutaneous NCI-H929 xenograft tumor model in immunodeficient mice.

DETAILED DESCRIPTION OF EMBODIMENTS

I. Definitions

[0036] In the present disclosure, the scientific and technical terms used herein have the meanings commonly understood by those skilled in the art, unless otherwise specified. Also, as used herein, protein and nucleic acid chemistry, molecular biology, cell and tissue culture, microbiology, immunology, and laboratory procedures used herein are terms and routine procedures widely used in the corresponding fields. Meanwhile, to better understand the present disclosure, definitions and explanations of related terms are provided below.

[0037] As used herein, the term “BCMA” may refer to the concept of BCMA itself and any variant, isotype, and paralog thereof, which are present together in animals and preferably in humans.

[0038] The term “human BCMA” refers to BCMA of human origin and may preferably have, but is not limited to, the amino acid sequence of Genbank accession number AB052772.1.

[0039] The terms “anti-BCMA antibody” and “BCMA-binding antibody” refer to an antibody that is capable of binding to BCMA with sufficient affinity such that the antibody is useful as a diagnostic and/or therapeutic agent in targeting BCMA. In one embodiment, the degree of binding of an anti-BCMA antibody to an unrelated non-BCMA protein is less than about 10% of the binding of the antibody to BCMA, as measured, for example, by a radioimmunoassay (RIA). In certain embodiments, a BCMA-binding antibody has a dissociation constant (K_d) of $\leq 1 \mu\text{M}$, $\leq 100 \text{ nM}$, $\leq 10 \text{ nM}$, $\leq 1 \text{ nM}$, $\leq 0.1 \text{ nM}$, $\leq 0.01 \text{ nM}$, or $\leq 0.001 \text{ nM}$ (e.g. 10^{-8} M or less, e.g. from 10^{-8} M to 10^{-13} M , e.g. from 10^{-9} M to 10^{-13} M). In certain embodiments, anti-BCMA antibodies bind BCMA epitopes conserved among BCMA from different species.

[0040] “CD3” refers to any native CD3 from any vertebrate source, including mammals, such as primates (e.g. humans), non-human primates (e.g. cynomolgus monkey), and rodents (e.g. mice and rats), unless otherwise specified. The term encompasses “full-length” unprocessed CD3 as well as any form of CD3 derived from processing in cells. The term also encompasses naturally occurring variants of CD3, such as splicing variants or allelic variants. In one embodiment, the CD3 is human CD3, in particular, the α subunit (CD3 ϵ) of human CD3. The amino acid sequence of human CD3 ϵ is shown under UniProt (www.uniprot.org) accession number P07766 (version 144), or NCBI (www.ncbi.nlm.nih.gov/) RefSeq NP_000724.1. The amino acid sequence of cynomolgus monkey CD3 ϵ is shown under NCBI GenBank no. BAB71849.1.

[0041] The term “cell surface” is used in accordance with its normal meaning in the art and thus includes cell exterior accessible by binding to proteins and other molecules.

[0042] As used herein, unless otherwise specified, the term “about” or “approximately” means within plus or minus 10% of a given value or range. Where an integer is required, the term refers to an integer rounding up or down to the nearest integer within plus or minus 10% of the given value or range.

[0043] The phrase “substantially identical” with respect to polypeptide chain sequences of an antibody is understood to mean an antibody that exhibits at least 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or more sequence identity to a reference polypeptide chain sequence. With respect to a nucleic acid sequence, the term is understood to mean a nucleotide sequence that exhibits at least greater than 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or more sequence identity to a reference nucleic acid sequence.

[0044] Sequence “similarity” or “identity” has well-known meanings in the field, and the percentage of sequence identity between two nucleic acid or polypeptide molecules or regions can be calculated using published techniques. Sequence identity can be measured along the full-length polynucleotide or polypeptide, or along a region of the molecule. Although there are many methods of measuring the identity between two polynucleotides or polypeptides, the term “identity” is well known to technicians (Carrillo, H. & Lipman, D. SIAM J Applied Math 48: 1073 (1988)).

[0045] A “substituted” variant is a variant in which at least one amino acid residue in the native sequence has been removed and a different amino acid is inserted at its same position. The substitution may be single, wherein only one amino acid in the molecule is substituted; or maybe multiple amino acids are substituted, wherein two or more amino acids are substituted in the same molecule. Multiple substitutions may be at consecutive sites. Likewise, an amino acid may be substituted with multiple residues, wherein such variants include both substitutions and insertions. An “insertion” variant is a variant in which one or more amino acids are inserted immediately adjacent to an amino acid at a particular position in a native sequence. The immediately adjacent amino acids is meant attached to the α -carboxyl or α -amine functional groups of the amino acid. A “deletion” variant is a variant in which one or more amino acids in the native amino acid sequence have been removed. Typically, deletion variants have one or two amino acids deleted in a particular region of the molecule.

[0046] With respect to the variable domains of antibodies, the term “variable” refers to certain portions of related molecules that differ widely in sequence among antibodies and are used for specific recognition and binding of a particular antibody to its specific target. However, the variability is not uniformly distributed throughout the variable domain of the antibody. The variability is concentrated in what is known as the complementarity determining regions (CDRs; i.e. CDR1, CDR2, and CDR3) or three segments of the hypervariable region, all of which are located within the variable domains of the light chains and heavy chains. The highly conserved portions within the variable domains are referred to as framework (FR) regions or framework sequences. Each variable domain of a native heavy chain and light chain comprises four FR regions, which predominantly adopt a β -folded configuration, they are linked by three CDRs, the CDRs form loops, the loops are linked to the β -folded configuration and in some cases the loops form part of the β -folded configuration. The CDRs of each chain are typically linked together in close proximity by FR regions and are helpful for the formation of target binding sites (epitopes or determinants) of antibodies by use of CDR from other chains. As used herein, the numbering of amino acid residue of immunoglobulin is performed in accordance with the amino acid residue numbering system of immunoglobulin edited by Kabat et al., unless otherwise indicated. One CDR may have the ability to specifically bind to an associated epitope.

[0047] As used herein, an “antibody fragment” or “antigen-binding fragment” of an antibody refers to any portion of a full-length antibody that is less than full-length, but comprises at least a portion of the variable region (e.g. one or more CDR and/or one or more antibody binding sites) of the antibody that binds antigen, and thus the “antibody fragment” or “antigen-binding fragment” retains binding specificity as well as at least a portion of the specific binding capacity of the full-length antibody. Therefore, an antigen-binding fragment refers to an antibody fragment that comprises an antigen-binding portion that binds to the same antigen as the antibody from which the antibody fragment is derived. Antibody fragments include antibody derivatives produced by enzymatic treatment of full-length antibodies, as well as synthetically produced derivatives, e.g. recombinantly produced derivatives. Antibodies include antibody fragments. Examples of antibody fragments include, but are

not limited to, Fab, Fab', F(ab')₂, single chain Fv (scFv), Fv, dsFv, diabodies, Fd and Fd' fragments, and other fragments, including modified fragments. The fragments may comprise multiple chains linked together, for example by disulfide bonds and/or by peptide linkers. Antibody fragments generally comprise at least or about 50 amino acids, and typically at least or about 200 amino acids. An antigen-binding fragment includes any antibody fragment that, when the antibody fragment is inserted into an antibody framework (e.g. by displacement of the corresponding region), it results in an antibody that immunospecifically binds (i.e. exhibits a K_a of at least or at least about 10⁷-10⁹M⁻¹) to an antigen. A "functional fragment" or "analog of an anti-BCMA antibody" is a fragment or analog that prevents or substantially reduces the binding ability of the receptor to a ligand or initiate signal transduction. As used herein, functional fragments generally have the same meaning as "antibody fragments", and in the case of antibodies, fragments that prevent or substantially reduce the binding ability of the receptor to a ligand or initiate signal transduction, e.g. Fv, Fab, F(ab')₂, and the like. A "Fv" fragment consists of a dimer (V_H-V_L dimer) formed by a variable domain of a heavy chain and a variable domain of a light chain in a non-covalent association. In this configuration, the three CDRs of each variable domain interact with each other to determine a target binding site on the surface of the V_H-V_L dimer, as is the case with intact antibodies. The six CDRs jointly confer target binding specificity to the intact antibody. However, even a single variable domain (or half of an Fv comprising only three target-specific CDRs) may still have the ability to recognize and bind targets.

[0048] As used herein, the term "bispecific antibody (BsAb)" refers to antibodies and/or antigen-binding molecules capable of specifically binding to two different antigene determinants. Generally, a bispecific antibody and/or antigen-binding molecule comprises two antigen-binding sites, each of which is specific for a different antigene determinant. In certain embodiments, the bispecific antibody and/or antigen-binding molecule is capable of binding two antigene determinants simultaneously, particularly two antigene determinants expressed on two different cells.

[0049] As used herein, "monoclonal antibody" refers to a population of identical antibodies, meaning that each individual antibody molecule in the population of monoclonal antibodies is identical to another antibody molecule. This property is in contrast to the property of a polyclonal population of antibodies, the polyclonal population of antibodies comprises antibodies having a plurality of different sequences. Monoclonal antibodies can be prepared by many well-known methods (Smith et al. (2004) J. Clin. Pathol. 57, 912-917; and Nelson et al., J Clin Pathol (2000), 53, 111-117). For example, monoclonal antibodies can be made by immortalizing B cells, for example, by fusion with myeloma cells to produce hybridoma cell lines, or by infecting B cells with a virus such as EBV. Recombinant techniques can also be used to prepare antibodies from clonal populations of host cells in vitro by transforming host cells with plasmids carrying artificial sequences encoding the nucleotides of the antibodies.

[0050] As used herein, the term "hybridoma" or "hybridoma cell" refers to a cell or cell line (typically a myeloma or lymphoma cell) resulting from the fusion of antibody-producing lymphocytes and non-antibody-producing cancer cells. As is known to those skilled in the art,

hybridomas can be propagated and continuously supplied to produce a particular monoclonal antibody. Methods for producing hybridomas are known in the art. When referring to the term "hybridoma" or "hybridoma cell", it also includes subclones and progeny cells of hybridomas.

[0051] As used herein, a full-length antibody is an antibody having two full-length heavy chains (e.g. VH-CH1-CH2-CH3 or VH-CH1-CH2-CH3-CH4), two full-length light chains (VL-CL) and a hinge region, e.g. an antibody naturally produced by an antibody-secreting B cell as well as a synthetically produced antibody having the same domain.

[0052] The term "chimeric antibody" refers to an antibody in which the variable region sequences are derived from one species and the constant region sequences are derived from another species, such as an antibody in which the variable region sequences are derived from a mouse antibody and the constant region sequences are derived from a human antibody.

[0053] A "humanized" antibody refers to a form of non-human (e.g. mouse) antibody that is a chimeric immunoglobulin, immunoglobulin chain or fragment thereof (e.g. Fv, Fab, Fab', F(ab')₂ or other antigen-binding subsequences of an antibody) that contains a minimal sequence derived from non-human immunoglobulin. Preferably, the humanized antibody is a human immunoglobulin (recipient antibody) in which residues of the complementarity-determining region (CDR) of the recipient antibody are replaced by CDR residues from a non-human species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity.

[0054] Furthermore, in humanization it is also possible to mutate amino acid residues within the CDR1, CDR2, and/or CDR3 regions of VH and/or VL, thereby improving one or more binding properties (e.g. affinity) of the antibody. Mutations can be introduced, e.g. by PCR-mediated mutagenesis, and their effect on antibody binding or other functional properties can be assessed using the in vitro or in vivo assays described herein. Typically, conservative mutations are introduced. Such mutations may be amino acid substitutions, additions, or deletions. In addition, mutations within CDR typically do not exceed one or two. Thus, the humanized antibodies of the present disclosure also encompass antibodies comprising one or two amino acid mutations within CDR.

[0055] As used herein, the term "CDR" refers to a complementarity-determining region, as is known that, each heavy chain and light chain of an antibody molecule has three CDRs. CDR is also known as a hypervariable region and is present in the variable region of each of the heavy chain and light chains of an antibody, and has sites with very high variability in the primary structure of CDR. In the present specification, the heavy chain CDR of is represented by CDR1, CDR2, and CDR3 from the amino terminus of the amino-terminal sequence of the heavy chain, and the light chain CDR is represented by CDR1, CDR2, CDR3 from the amino terminus of the amino-terminal sequence of the light chain. These sites are adjacent to each other in the tertiary structure and determine the specificity of the antigen to which the antibody binds.

[0056] As used herein, the term "epitope" refers to any antigene determinant on an antigen bound to the paratope of an antibody. Epitope determinants usually comprise chemically active surface types of molecules, such as amino acids

or sugar side chains, and usually have specific three-dimensional structural characteristics as well as specific charge characteristics.

[0057] As used herein, “specific binding” or “immunospecifically binding” with respect to an antibody or antigen-binding fragment thereof is used interchangeably herein and refers to the ability of an antibody or antigen-binding fragment to form one or more non-covalent bonds with the same antigen through non-covalent interactions between the antibody and the antibody binding site of the antigen. The antigen may be an isolated antigen or present in a tumor cell. Generally, an antibody that immunospecifically binds (or specifically binds) to an antigen binds the antigen with an affinity constant K_a of about $1 \times 10^7 M^{-1}$ or $1 \times 10^8 M^{-1}$ or more (or a dissociation constant (K_d) of $1 \times 10^{-7} M$ or $1 \times 10^{-8} M$ or less). Affinity constants can be determined by standard kinetic methods for antibody reactions, e.g. immunoassays, surface plasmon resonance (SPR) (Rich and Myszka (2000) *Curr. Opin. Biotechnol.* 11:54; Englebienne (1998) *Analyst.* 123:1599), isothermal titration calorimetry (ITC), or other kinetic interaction assays known in the art; see also U.S. Pat. No. 7,229,619 which describes an exemplary SPR and ITC method for calculating the binding affinity of an antibody). Instruments and methods for detecting and monitoring the rate of binding in real-time are known and commercially available (see, *BiaCore 2000*, *Biacore AB*, *Uppsala, Sweden* and *GE Healthcare Life Sciences*; *Malmqvist (2000) Biochem. Soc. Trans.* 27:335).

[0058] As used herein, the terms “polynucleotide” and “nucleic acid molecule” refer to an oligomer or polymer comprising at least two linked nucleotides or nucleotide derivatives, including deoxyribonucleic acid (DNA) and ribonucleic acid (RNA), typically linked together by phosphodiester bonds. As used herein, the term “nucleic acid molecule” is intended to include DNA molecules and RNA molecules. The nucleic acid molecule may be single-stranded or double-stranded and may be cDNA.

[0059] As used herein, an isolated nucleic acid molecule is a nucleic acid molecule isolated from other nucleic acid molecules present in the natural source of the nucleic acid molecule. An “isolated” nucleic acid molecule, such as a cDNA molecule, can be substantially free of other cellular material or culture medium when prepared by recombinant techniques, or substantially free of chemical precursors or other chemical ingredients when chemically synthesized. Exemplary isolated nucleic acid molecules provided herein include isolated nucleic acid molecules encoding a provided antibody or antigen-binding fragment.

[0060] As used herein, “operably linked” with respect to a nucleic acid sequence, region, element or domain means that the nucleic acid regions are functionally related to one another. For example, a promoter may be operably linked to a nucleic acid encoding a polypeptide such that the promoter regulates or mediates transcription of the nucleic acid.

[0061] Also provided are “conservative sequence modifications” with respect to the sequences set forth in the sequence listings described herein, i.e. nucleotide sequence and amino acid sequence modifications that do not eliminate binding of an antibody encoded by a nucleic acid or containing an amino acid sequence to an antigen. These conservative sequence modifications include conservative nucleotide substitutions and amino acid substitutions, and nucleotide additions and amino acid additions, and nucleotide deletions and amino acid deletions. For example, modi-

fications can be introduced into the sequence listings described herein by standard techniques known in the art, such as site-directed mutagenesis and PCR-mediated mutagenesis. Conservative sequence modifications include conservative amino acid substitutions, wherein an amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g. lysine, arginine, histidine), amino acids with acidic side chains (e.g. aspartic acid, glutamic acid), amino acids with uncharged polar side chains (e.g. glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine, tryptophan), amino acids with nonpolar side chains (e.g. alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine), amino acids with β -branched side chains (e.g. threonine, valine, isoleucine) and amino acids with aromatic side chains (e.g. tyrosine, phenylalanine, tryptophan, histidine). Thus, a predicted nonessential amino acid residue in an anti-BCMA antibody is preferably replaced with another amino acid residue from the same side chain family. Methods for identifying conservative substitutions of nucleotides and amino acids that do not eliminate antigen binding are well known in the art (For example, see Brummell et al., *Biochem.* 32: 1180-1187 (1993); Kobayashi et al., *Protein Eng.* 12(10): 879-884 (1999); Burks et al., *Proc. Natl. Acad. Sci. USA* 94: 412-417 (1997)).

[0062] Alternatively, in another embodiment, mutations can be randomly introduced along all or a portion of the anti-BCMA antibody coding sequence, e.g. by saturation mutagenesis, and the resulting modified anti-BCMA antibodies can be screened for improved binding activity.

[0063] As used herein, “expression” refers to a process of producing a polypeptide by transcription and translation of a polynucleotide. The expression level of a polypeptide can be assessed using any method known in the art, including, for example, methods of determining the amount of polypeptide produced from a host cell. Such methods may include, but are not limited to, quantitation of polypeptides in cell lysates by ELISA, gel electrophoresis followed by Coomassie blue staining, Lowry protein assay, and Bradford protein assay.

[0064] As used herein, a “host cell” is a cell for receiving, maintaining, replicating, and amplifying a vector. Host cells may also be used to express a polypeptide encoded by the vector. As the host cell divides, the nucleic acid contained in the vector replicates, thereby amplifying the nucleic acids. The host cell may be a eukaryotic cell or a prokaryotic cell. Suitable host cells include, but are not limited to, CHO cells, various COS cells, HeLa cells, and HEK cells such as HEK 293 cells.

[0065] As used herein, a “vector” is a replicable nucleic acid from which one or more heterologous proteins can be expressed when the vector is transformed into an appropriate host cell. With respect to vectors, they include vectors into which a nucleic acid encoding a polypeptide or fragment thereof may be introduced, typically by restriction enzyme digestion and ligation, with respect to vectors, they also include vectors comprising a nucleic acid encoding a polypeptide. Vectors are used to introduce a nucleic acid encoding a polypeptide into a host cell, to amplify the nucleic acid, or to express/display the polypeptide encoded by the nucleic acid. The vectors usually remain episomal, but can be designed such that the gene or part thereof is integrated into

the chromosome of the genome. Also contemplated are vectors for artificial chromosomes, such as yeast artificial chromosome vectors and mammalian artificial chromosomes. The selection and use of such vehicles are well-known to those skilled in the art.

[0066] As used herein, a vector also includes a “virus vector” or a “viral vector”. A viral vector is an engineered virus operably linked to an exogenous gene to transfer the exogenous gene (as a vehicle or shuttle) into a cell.

[0067] As used herein, an “expression vector” includes a vector capable of expressing a DNA, the DNA is operably linked to regulatory sequences, such as a promoter region, capable of affecting the expression of such a DNA fragment. Such additional fragments may include promoter and terminator sequences and optionally may include one or more origins of replication, one or more selectable markers, enhancers, polyadenylation signals, and the like. Expression vectors are typically derived from plasmid or viral DNA, or may contain elements of both. Thus, an expression vector refers to a recombinant DNA or RNA construct, such as a plasmid, phage, recombinant virus, or other vector. When introduced into an appropriate host cell, it results in the expression of a cloned DNA. Suitable expression vectors are well known to those skilled in the art and include expression vectors that are replicable in eukaryotic and/or prokaryotic cells and expression vectors that remain episomal or that are integrated into the host cell genome.

[0068] As used herein, the “treatment” of an individual with a disease or disease condition indicates that the symptoms of the individual are partially or completely relieved, or remain unchanged after treatments. Therefore, treatment includes prevention, treatment, and/or cure. Prevention refers to preventing the underlying disease and/or preventing the worsening of symptoms or the progression of the disease. Treatment also includes any pharmaceutical use of any antibody or antigen-binding fragment thereof provided and the compositions provided herein.

[0069] As used herein, “therapeutic effect” refers to an effect resulting from the treatment of a subject that alters, typically ameliorates, or ameliorates a symptom of a disease or condition, or cures a disease or condition.

[0070] As used herein, a “therapeutically effective amount” or “therapeutically effective dose” refers to an amount of a substance, compound, material, or composition comprising a compound that is at least sufficient to produce a therapeutic effect after administration to a subject. Thus, it is an amount necessary to prevent, cure, ameliorate, block, or partially block the symptoms of a disease or disorder.

[0071] As used herein, a “prophylactically effective amount” or “prophylactically effective dose” refers to an amount of a substance, compound, material, or composition comprising a compound that, when administered to a subject, has the desired prophylactic effect, e.g. prevents or delays the occurrence or recurrence of a disease or condition, reduces the likelihood of the occurrence or recurrence of a disease or condition. A fully prophylactically effective dose does not need to occur by administration of one dose, and may occur only after administration of a series of doses. Thus, a prophylactically effective amount can be administered in one or more administrations.

[0072] As used herein, the term “patient” refers to a mammal, such as a human.

II. Detailed Description of Specific Embodiments

[0073] In one aspect, the present disclosure provides a bispecific antibody or antigen-binding portion thereof, comprising:

[0074] (a) a first antigen-binding portion or an antigen-binding fragment thereof bound to a BCMA antigen on the surface of a target cell, the first antigen-binding portion comprises a first heavy chain and a first light chain, the first antigen-binding portion comprises a first binding domain that binds to the first antigen, wherein the first binding domain comprises a heavy chain CDR selected from the amino acid sequences of SEQ ID NOs: 12, 13, 14, 26, 27, 28, 36, 37, 43, 44, 50, 55, 56, 57, 65, 66, 71, 72, 73, 147, and 186, or any variant thereof, and/or a light chain CDR selected from the amino acid sequences of SEQ ID NOs: 17, 18, 19, 22, 23, 31, 32, 33, 40, 47, 60, 61, 62, 76, 77, 78, 83, 84, 87, 88, 89, 92 and 144 or any variant; and

[0075] (b) a second antigen-binding portion or antigen-binding fragment thereof bound to a CD3 antigen on the surface of a T cell, the second antigen-binding portion comprises a second heavy chain and a second light chain, the second antigen-binding portion comprises a second binding domain that binds to a second antigen.

[0076] The antibody or antigen-binding portion thereof according to the preceding aspect, the first binding domain comprises a heavy chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 12, 26, 55, 65, 71 or any variant thereof, a heavy chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 13, 27, 36, 43, 50, 56, 66, 72, 147, 186 or any variant thereof, a heavy chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 14, 28, 37, 44, 57, 73 or any variant thereof; and/or the first binding domain comprises a light chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 17, 22, 31, 47, 60, 76, 83, 87, 92, 144 or any variant thereof, a light chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 18, 23, 32, 40, 61, 77, 84, 88 or any variant thereof, a light chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 19, 33, 62, 78, 89 or any variant thereof.

[0077] The antibody or antigen-binding portion thereof according to any of the preceding aspects, the heavy chain CDR1, CDR2, and CDR3 sequences of the first binding domain are selected from: the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 186 and 14, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 26, 27 and 28, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 36 and 37, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 43 and 44, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 50 and 14, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 55, 56 and 57, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 65, 66 and 14, the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 72 and 73; the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of

SEQ ID NOs: 71, 147 and 73; and the light chain CDR1, CDR2, and CDR3 sequences of the first binding domain are selected from: the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 17, 18 and 19, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 22, 23 and 19, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 31, 32 and 33, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 17, 40 and 19, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 47, 18 and 19, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 60, 61 and 62, the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 76, 77 and 78, and the sequences of the first light chain CDR1, CDR2 and CDR3 of the amino acid sequences of SEQ ID NOs: 83, 84 and 19, the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 87, 88 and 89, the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 92, 23 and 19, the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 144, 77 and 78.

[0078] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 18 and 19.

[0079] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 22, 23 and 19.

[0080] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 26, 27 and 28 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 31, 32 and 33.

[0081] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 36 and 37 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19.

[0082] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 43 and 44 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 47, 18 and 19.

[0083] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 50 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19.

[0084] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3

sequences of the amino acid sequences of SEQ ID NOs: 55, 56 and 57 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 60, 61 and 62.

[0085] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19.

[0086] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 65, 66 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19.

[0087] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 72 and 73 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 76, 77 and 78.

[0088] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 83, 84 and 19.

[0089] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13, and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 87, 88 and 89.

[0090] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 92, 23 and 19.

[0091] In some embodiments, the first binding domain comprises the first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 147 and 73 and the first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 144, 77 and 78.

[0092] In some embodiments, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 186 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19.

[0093] In some embodiments, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 186 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 22, 23 and 19.

[0094] The antibody or antigen-binding portion thereof according to any of the preceding aspects, the first binding domain comprises a first heavy chain variable region selected from the amino acid sequences of SEQ ID NOs: 10, 24, 34, 41, 48, 53, 63, 69, 79, 145, 150, 154, 158, 162, 166, 170 and 187, or any variant thereof, and/or comprises a first

NOs: 96, 103, 111 or any variant thereof, a second light chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 91, 108 or any variant thereof.

[0118] In some embodiments, the heavy chain CDR1, CDR2, and CDR3 sequences of the second binding domain are selected from: second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 116, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 122, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 125, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 128, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 131, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 134, 135 and 116; and the light chain CDR1, CDR2, and CDR3 sequences of the second binding domain are selected from: second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96 and 97, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 102, 103 and 97, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96 and 108, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 111 and 108.

[0119] In some embodiments, the second binding domain comprises the second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, and the second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96 and 97.

[0120] In some embodiments, the second binding domain comprises the second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, and the second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 102, 103 and 97.

[0121] In some embodiments, the second binding domain comprises the second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, and the second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96 and 108.

[0122] In some embodiments, the second binding domain comprises the second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, and the second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 111 and 108.

[0123] In some embodiments, the second binding domain comprises a second heavy chain variable region selected from the amino acid sequences of SEQ ID NOs: 112, 117, 120, 123, 126, 129, 132, 136, 138, 140, or any variant thereof, and/or comprises a second light chain variable region selected from the amino acid sequences of SEQ ID NOs: 93, 98, 100, 104, 106, 109 or any variant thereof.

[0124] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 136 or any variant thereof,

and a second light chain variable region of the amino acid sequence of SEQ ID NO: 93 or any variant thereof.

[0125] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 93 or any variant thereof.

[0126] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 98 or any variant thereof.

[0127] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 100 or any variant thereof.

[0128] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 112 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0129] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 136 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0130] In some embodiments, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0131] The antibody or antigen-binding portion thereof according to any of the preceding aspects, the first light chain of the first antigen-binding portion is a κ light chain and the second light chain of the second antigen-binding portion is a λ light chain.

[0132] In some embodiments, the second light chain variable region of the second antigen-binding portion has a Gln₄₀Glu mutation (V λ_{CD3} :Gln₄₀Glu); the second heavy chain variable region of the second antigen-binding portion has a Gln₃Lys mutation (VH $_{CD3}$:Gln₃Lys).

[0133] In some embodiments, the first light chain variable region of the first antigen-binding portion has a Gln₄₂Lys mutation (V κ_{BCMA} :Gln₄₂Lys). In some embodiments, the first heavy chain variable region of the first antigen-binding portion has a Gln₃₉Glu mutation (VH $_{BCMA}$:Gln₃₉Glu).

[0134] In some embodiments, the Fc portion of the first antigen-binding portion and the second antigen-binding portion of the bispecific antibody adopts a knob-into-hole structure. In some preferred embodiments, the Fc portion adopts a knob-into-hole structure of human IgG4.

[0135] In some embodiments, the first heavy chain comprises a heavy chain selected from SEQ ID NOs: 174, 178, or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NOs: 172, 176, or any variant thereof.

[0136] In some preferred embodiments, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 174 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 172 or any variant thereof.

[0137] In some preferred embodiments, the first heavy chain comprises a heavy chain selected from SEQ ID NO:

174 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 176 or any variant thereof.

[0138] In some preferred embodiments, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 172 or any variant thereof.

[0139] In some preferred embodiments, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 176 or any variant thereof.

[0140] In some embodiments, the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof.

[0141] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 145 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 142 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0142] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 150 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 148 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0143] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 154 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 152 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0144] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 158 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 156 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0145] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 162 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 160 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino

acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0146] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 166 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 164 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

[0147] In some preferred embodiments, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 170 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 168 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof; In a specific embodiment, the first heavy chain comprises SEQ ID NO: 174, the first light chain of the bispecific antibody comprises SEQ ID NO: 172; the second heavy chain comprises SEQ ID NO: 182, the second light chain comprises SEQ ID NO: 180.

[0148] In a specific embodiment, the first heavy chain comprises SEQ ID NO: 174, the first light chain of the bispecific antibody comprises SEQ ID NO: 176; the second heavy chain comprises SEQ ID NO: 182, the second light chain comprises SEQ ID NO: 180.

[0149] In a specific embodiment, the first heavy chain comprises SEQ ID NO: 178, the first light chain of the bispecific antibody comprises SEQ ID NO: 172; the second heavy chain comprises SEQ ID NO: 182, the second light chain comprises SEQ ID NO: 180.

[0150] In a specific embodiment, the first heavy chain comprises SEQ ID NO: 178, the first light chain of the bispecific antibody comprises SEQ ID NO: 176; the second heavy chain comprises SEQ ID NO: 182, the second light chain comprises SEQ ID NO: 180.

[0151] In one aspect, the present disclosure provides a BCMA-binding antibody or antigen-binding portion thereof, the antibody comprises a first antigen-binding portion as previously described.

[0152] In some embodiments, the BCMA-binding antibody or antigen-binding portion thereof has at least greater than 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or more sequence identity to the antibody or antigen-binding portion thereof of any of the preceding aspects.

[0153] In one aspect, the present disclosure provides a nucleic acid encoding the antibody or antigen-binding portion thereof according to any of the preceding aspects, or a nucleic acid molecule having at least greater than 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or more sequence identity thereto.

[0154] In some embodiments, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion is selected from the nucleotide sequences of SEQ ID NOs: 11, 25, 35, 42, 49, 54, 64, 70, 80, 146, 151, 155, 159, 163, 167, 171, and 189 or any variant thereof;

[0193] In some preferred embodiments, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 159; and the nucleic acid encoding the first light chain variable region is the nucleotide sequence of SEQ ID NO: 157; the nucleic acid encoding the second heavy chain variable region of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107.

[0194] In some preferred embodiments, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 163; and the nucleic acid encoding the first light chain variable region is the nucleotide sequence of SEQ ID NO: 161; the nucleic acid encoding the second heavy chain variable region of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107.

[0195] In some preferred embodiments, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 167; and the nucleic acid encoding the first light chain variable region is the nucleotide sequence of SEQ ID NO: 165; the nucleic acid encoding the second heavy chain variable region of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107.

[0196] In some preferred embodiments, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 171; and the nucleic acid encoding the first light chain variable region is the nucleotide sequence of SEQ ID NO: 169; the nucleic acid encoding the second heavy chain variable region of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107.

[0197] In some preferred embodiments, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

[0198] In some preferred embodiments, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 175, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 173, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

[0199] In some preferred embodiments, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence

of SEQ ID NO: 175, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 177, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

[0200] In some preferred embodiments, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 179, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 173, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

[0201] In some preferred embodiments, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 179, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 177, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

[0202] In one aspect, the present disclosure provides a BCMA-binding antibody or antigen-binding portion thereof, the antibody comprises a first antigen-binding portion as previously described.

[0203] In one aspect, the present disclosure provides a vector comprising the foregoing nucleic acids.

[0204] In one aspect, the present disclosure provides a cell comprising the foregoing nucleic acids or vector.

[0205] In one aspect, the present disclosure provides compositions comprising the foregoing bispecific antibodies or antigen-binding portions thereof, nucleic acids, vectors, and/or cells.

[0206] In one aspect, the present disclosure provides an antibody-drug conjugate comprising the foregoing bispecific antibody or antigen-binding portion thereof covalently attached to a therapeutic moiety.

[0207] Preferably, the therapeutic moiety is selected from a cytotoxic moiety, a chemotherapeutic agent, a cytokine, an immunosuppressant, an immunostimulant, a lytic peptide, or a radioisotope.

[0208] The antibodies of the present disclosure are useful as therapeutic or diagnostic tools in a variety of diseases in which BCMA is adversely expressed or found.

[0209] In one embodiment of a disease associated with BCMA, the expression of BCMA in cells of the diseased tissue or organ is increased, as compared to the state in a healthy tissue or organ. An increase means an increase of at least 10%, in particular at least 20%, at least 50%, at least 100%, at least 200%, at least 500%, at least 1000%, at least 10000%, or even more. In one embodiment, expression is found only in diseased tissues, whereas expression in corresponding healthy tissues is suppressed. According to the present disclosure, diseases associated with BCMA include

B-cell diseases, e.g. B-cell disorders such as plasma cell disorders and/or autoimmune diseases.

[0210] In some embodiments, the disease condition is cancer. In some embodiments, the cancer is a B-cell related cancer selected from the group consisting of multiple myeloma, malignant plasmacytoma, Hodgkin lymphoma, nodular lymphocyte predominant Hodgkin lymphoma, kahl-er's disease and myeloid leukemia, plasma cell leukemia, plasmacytoma, B-cell prolymphocytic leukemia, hairy cell leukemia, B-cell non-Hodgkin lymphoma (NHL), acute myelogenous leukemia (AML), chronic lymphocytic leukemia (CLL), acute lymphocytic leukemia (ALL), chronic myelogenous leukemia (CML), follicular lymphoma, burkitt lymphoma, marginal zone lymphoma, mantle cell lymphoma, large cell lymphoma, precursor B-lymphocyte lymphoma, myeloid leukemia, waldenstrom's macroglobulinaemia, diffuse large B cell lymphoma, follicular lymphoma, marginal zone lymphoma, mucosa-associated lymphoid tissue lymphoma, small cell lymphocytic lymphoma, mantle cell lymphoma, burkitt lymphoma, primary mediastinal (thymic) large B cell lymphoma, lymphoplasmacytic lymphoma, waldenstrom's macroglobulinaemia, nodal marginal zone B cell lymphoma, splenic marginal zone lymphoma, intravascular large B-cell lymphoma, primary effusion lymphoma, lymphomatoid granulomatosis, large B-cell lymphoma rich in T cells/histiocytes, primary central nervous system lymphoma, primary cutaneous diffuse large B-cell lymphoma (leg type), EBV-positive diffuse large B-cell lymphoma of the elderly, diffuse large B-cell lymphoma associated with inflammation, intravascular large B-cell lymphoma, ALK-positive large B-cell lymphoma, plasmablastic lymphoma, large B-cell lymphoma arising in HHV8-associated multicenter Castleman's disease, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and Burkitt's lymphoma, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and classic Hodgkin lymphoma, and other B-cell-related lymphomas.

[0211] In plasma cell disorders, one plasma cell clone reproduces uncontrollably. As a result, this clone produces a large number of single (monoclonal) antibodies called M-proteins. In some cases, for example, a subject with monoclonal gammopathy, the antibody produced is incomplete, consisting only of light chains or heavy chains. These abnormal plasma cells and the produced antibodies thereof are generally limited to one type. Preferably, the plasma cell disorder is selected from multiple myeloma, plasmacytoma, plasma cell leukemia, macroglobulinaemia, amyloidosis, waldenstrom's macroglobulinaemia, solitary plasmacytoma of the bone, extramedullary plasmacytoma, osteosclerotic myeloma, heavy chain disease, monoclonal gammopathy of undetermined significance, and multiple myeloma of stasis type.

[0212] In some embodiments, the disease condition is an autoimmune disorder, such as systemic lupus erythematosus or rheumatoid arthritis. In some embodiments, the therapeutic agent comprises an antibody that specifically binds to an activated T-cell antigen.

[0213] In one embodiment, the therapeutic agent comprises an antibody that specifically binds to CD3, particularly CD3e.

[0214] Methods of treating diseases and conditions using bispecific antibodies of the present disclosure include the steps of administering to a mammal a therapeutically effec-

tive amount of an antibody or antigen-binding fragment thereof or a nucleic acid molecule or vector or cell or pharmaceutical composition of any of the preceding aspects.

[0215] In some embodiments, the present disclosure provides a method of treating or preventing a cancer disease comprising administering to a patient an antibody capable of binding to BCMA, wherein the antibody is administered to provide a serum level of at least 40 µg/ml. In various embodiments, the antibody is administered to provide a serum level of at least 50 µg/ml, at least 150 µg/ml, at least 300 µg/ml, at least 400 µg/ml, or at least 500 µg/ml. In various embodiments, the antibody is administered to provide a serum level of no more than 800 µg/ml, 700 µg/ml, 600 µg/ml, 550 µg/ml, or 500 µg/ml. In one embodiment, the serum level provided is from 40 µg/ml to 700 µg/ml, preferably from 40 µg/ml to 600 µg/ml, preferably from 50 µg/ml to 500 µg/ml, such as from 150 µg/ml to 500 µg/ml or from 300 µg/ml to 500 µg/ml. The term "serum level" as used in the present description means the concentration of the discussed substance in serum. In one embodiment, serum levels are provided for at least 7 days or at least 14 days. In one embodiment, the method comprises administering a dose of the antibody of at least 300 mg/m², such as at least 600 mg/m², and preferably at most 1500 mg/m², at most 1200 mg/m², or at most 1000 mg/m².

[0216] In some embodiments, the disclosure provides a method of treating or preventing a cancer disease, comprising administering to a patient an antibody capable of binding to BCMA, wherein the antibody is administered at a dose of at least 300 mg/m², such as at least 600 mg/m², and preferably at most 1500 mg/m², at most 1200 mg/m², or at most 1000 mg/m².

[0217] In some embodiments, the disclosure provides a method of treating or preventing a cancer disease comprising administering to a patient an antibody capable of binding to BCMA, wherein at least 50%, preferably 60%, 70%, 80%, or 90% of the cancer cells of the patient are positive for BCMA and/or at least 40%, preferably 50% or 60% of the cancer cells of the patient are positive for surface expression of BCMA. In this aspect, the present disclosure also provides a method of treating or preventing a cancer disease, the method comprises: a. identifying patients showing at least 50%, preferably 60%, 70%, 80%, or 90% of BCMA-positive cancer cells and/or at least 40%, preferably 50% or 60% of cancer cells that are positive for surface expression of BCMA; and b. administering to the patient an antibody capable of binding to BCMA. In one embodiment, at least 95% or at least 98% of the cancer cells of the patient are BCMA-positive. In one embodiment, at least 70%, at least 80%, or at least 90% of the cancer cells of the patient are positive for surface expression of BCMA.

[0218] In one embodiment of the methods of any of the aspects described herein, the therapeutic outcome of the cancer disease is the achievement of stable disease conditions. In one embodiment, stable disease conditions are achieved for at least 2 months, at least 3 months, or at least 6 months.

[0219] In some embodiments, the present disclosure provides methods of achieving stable disease in a cancer patient comprising administering to a patient an antibody capable of binding to BCMA. In one embodiment, stable disease is achieved for at least 2 months, at least 3 months, or at least 6 months.

[0220] In one embodiment of the method of any of the aspects described herein, the antibody is administered in a single dose or multiple doses.

[0221] In some embodiments, the present disclosure provides a method of treating or preventing a cancer disease comprising administering to a patient an antibody capable of binding to BCMA, wherein the antibody is administered in multiple doses.

[0222] If the antibody is administered in multiple doses according to the present disclosure, the antibody is preferably administered in at least 3 doses, at least 4 doses, at least 5 doses, at least 6 doses, at least 7 doses, at least 8 doses, at least 9 doses, or at least 10 doses, and preferably at most 30 doses, 25 doses, 20 doses, 15 doses, or 10 doses. Preferably, the dose of antibody is administered at intervals of at least 7 days, at least 10 days, at least 14 days, or at least 20 days. The dose of antibody is preferably administered at intervals of from 7 to 30 days, from 10 to 20 days, and preferably about 14 days.

[0223] In one embodiment, the antibody is administered so as to provide a serum level of at least 40 µg/ml.

[0224] In various embodiments, the antibody is administered so as to provide a serum level of at least 50 µg/ml, at least 150 µg/ml, at least 300 µg/ml, at least 400 µg/ml, or at least 500 µg/ml. In various embodiments, the antibody is administered so as to provide a serum level of no more than 800 µg/ml, 700 µg/ml, 600 µg/ml, 550 µg/ml, or 500 µg/ml. In one embodiment, the serum level provided is from 40 µg/ml to 700 µg/ml, preferably from 40 µg/ml to 600 µg/ml, preferably from 50 µg/ml to 500 µg/ml, such as from 150 µg/ml to 500 µg/ml or from 300 µg/ml to 500 µg/ml. In one embodiment, serum levels are provided for at least 7 days or at least 14 days. In one embodiment, the method comprises administering a dose of at least 300 mg/m², such as at least 600 mg/m² and preferably at most 1500 mg/m², at most 1200 mg/m², or at most 1000 mg/m² of the antibody.

[0225] Use of an antibody or antigen-binding fragment thereof or a nucleic acid molecule or a vector or a cell or a pharmaceutical composition of any of the preceding aspects in the manufacture of a drug for treating a BCMA-related disorder in a mammal.

[0226] According to any of the preceding aspects, optionally, the antibody is conjugated to other drugs, such as a labeled or cytotoxic conjugate.

[0227] In one aspect, the disclosure also includes kits, e.g. kits comprising an antibody, fragment thereof, homologue, derivative, etc. of the disclosure, e.g. a labeled or cytotoxic conjugate, as well as instructions for use of the antibody, a conjugate that kills a particular type of cell, etc. The instructions can include directions for using the antibody, conjugate, etc. in vitro, in vivo, or ex vivo. The antibodies may be in liquid form or solid form, usually lyophilized. The kit may contain other suitable reagents, such as buffers, reconstitution solutions, and other necessary ingredients for the intended use. Combinations of reagents packaged in predetermined amounts with instructions for their use, e.g. for therapeutic use or for conducting diagnostic assays, are contemplated. When the antibody is labeled, e.g. with an enzyme, then the kit can include a substrate and cofactors required for the enzyme (e.g. a substrate precursor that provides a detectable chromophore or fluorophore).

[0228] In addition, other additives such as stabilizers, buffers (e.g. blocking buffers or lysis buffers), and the like may also be included. The relative amounts of the various

reagents can be varied to provide a concentrate of reagent solution, thereby providing user flexibility, space savings, reagent savings, etc. These reagents may also be provided in dry powder form, usually in lyophilized form, including excipients which provide a reagent solution having the appropriate concentration when dissolved.

[0229] Use of an antibody or functional fragment thereof or a nucleic acid molecule or a vector or a cell or a pharmaceutical composition or kit according to any of the preceding aspects for the preparation of a reagent for inhibiting BCMA binding.

[0230] In addition, the antibodies of the present disclosure can be used in immunoassays, purification methods, and other methods using immunoglobulins or fragments thereof. Such uses are well-known in the art.

[0231] Accordingly, the present disclosure also provides compositions comprising the anti-BCMA antibodies of the present disclosure or fragments thereof, the antibody is conveniently combined with a pharmaceutically acceptable carrier, diluent, or excipient, which belongs to conventional practices in the art.

[0232] As used in the present disclosure, the term “pharmaceutical composition” refers to formulations of various prepared materials. Formulations containing a therapeutically effective amount of multivalent antibodies are in a sterile liquid solution, liquid suspension, or lyophilized form, optionally containing stabilizers or excipients.

[0233] The antibodies of the present disclosure can be used as a composition administered alone or can be used in combination with other active agents.

[0234] In some embodiments, the humanized antibodies of the disclosure are conjugated to a therapeutic moiety (i.e. a drug). The therapeutic moiety can be, for example, a cytotoxin, a chemotherapeutic agent, a cytokine, an immunosuppressant, an immunostimulant, a lytic peptide, or a radioisotope. Such conjugates are referred to herein as “antibody-drug conjugates” or “ADC”.

[0235] In some embodiments, the antibody is conjugated to a cytotoxic moiety. The cytotoxic moiety may for example be selected from the following: paclitaxel; cytochalasin B; gramicidin D; ethidium bromide; emetine; mitomycin; etoposide; teniposide; vincristine; vinblastine; colchicine; doxorubicin; daunorubicin; dihydroxy anthracenedione; tubulin inhibitors such as maytansine or analogs or derivatives thereof; antimitotic agents such as monomethyl auristatin E or F or analogues or derivatives thereof; dolastatin 10 or 15 or an analogue thereof; irinotecan or an analog thereof; mitoxantrone; mithramycin; actinomycin D; 1-dehydrotestosterone; glucocorticoids; procaine; tetracaine; lidocaine; propranolol; puromycin; calicheamicin or an analogue or derivative thereof; antimetabolites such as methotrexate, 6 mercaptopurine, 6 thioguanine, cytarabine, fludarabine, 5 fluorouracil, decadiazine, hydroxyurea, asparaginase, gemcitabine or cladribine; alkylating agents such as mechlorethamine, thiopurine, chlorambucil, melphalan, carmustine (BSNU), lomustine (CCNU), cyclophosphamide, busulfan, dibromomannitol, streptozotocin, dacarbazine (DTIC), procarbazine, mitomycin C; platinum derivatives such as cisplatin or carboplatin; duocarmycin A, duocarmycin SA, rachelymycin (CC-1065) or analogs or derivatives thereof; antibiotics such as actinomycin, bleomycin, daunorubicin, doxorubicin, idarubicin, mithramycin, mitomycin, mitoxantrone, plicomycin, anthramycin (AMC); pyrrolo [2,1-c][1,4]-benzodiazepine

(PDB); diphtheria toxin and related molecules such as diphtheria A chain and active fragments thereof and hybrid molecules, ricin toxin such as ricin A or deglycosylated ricin A chain toxin, cholera toxin, Shiga-like toxin such as SLT I, SLT II, SLT IIV, LT toxin, C3 toxin, Shiga toxin, pertussis toxin, tetanus toxin, soybean Bowman-Birk protease inhibitor, *pseudomonas* exotoxin, azlocillin, saporin, modeccin, gelsolin, abrin A chain, modeccin A chain, α -sarcin, *Aleu-rites fordii* proteins, dianthin proteins, *Phytolacca ameri-cana* proteins such as PAPI, PAPII, and PAP-S, *Momordica charantia* inhibitor, curcin, crotin, *Sapaonaria officinalis* inhibitor, gelonin, mitomycin, restrictocin, phenomycin, and enomycin toxins; ribonuclease (RNase); DNase I, staphylo-coccal endotoxin A; pokeweed antiviral protein; diphtheria toxin and *pseudomonas* endotoxin.

[0236] In some embodiments, the antibody is conjugated to auristatin or a peptide analog, derivative, or prodrug thereof. It has been shown that auristatin interferes with microtubule dynamics, GTP hydrolysis, and nuclear and cell division and has anti-cancer and anti-fungal activity. For example, auristatin E can be reacted with p-acetylbenzoic acid or benzoylpentanoic acid to produce AEB and AEVB, respectively. Other typical auristatin derivatives include AFP, MMAF (monomethyl auristatin F), and MMAE (monomethyl auristatin E). Suitable auristatins and analogs, derivatives, and prodrugs of auristatins, as well as suitable linkers for conjugating auristatins to Ab, are described, for example, in U.S. Pat. Nos. 5,635,483, 5,780,588, and 6,214,345, and International Patent Application Publication Nos. WO02088172, WO2004010957, WO2005081711, WO2005084390, WO2006132670, WO03026577, WO200700860, WO207011968 and WO205082023.

[0237] In some embodiments, the antibody is conjugated to a pyrrolo [2,1-c][1,4]-benzodiazepine (PDB) or a peptide analog, derivative, or prodrug thereof. Suitable PDB and PDB derivatives and related art are described, for example, in Hartley J. A. et al., Cancer Res 2010; 70(17):6849-6858; Antonow D. et al., Cancer J 2008; 14(3):154-169; Howard P. W. et al., Bioorg Med Chem Lett 2009; 19:6463-6466 and Sagnou et al., Bioorg Med Chem Lett 2000; 10(18):2083-2086.

[0238] In some embodiments, the antibody is conjugated to a cytotoxic moiety selected from anthracyclines, may-tansines, calicheamicins, duocarmycins, rachelmycin (CC-1065), dolastatin 10, dolastatin 15, irinotecan, monomethyl auristatin E, monomethyl auristatin F, PDB, or any analog, derivative or prodrug thereof.

[0239] In some embodiments, the antibody is conjugated to anthracycline or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to may-tansine or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to a calicheamicin or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to a duocarmycin or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to rachelmycin (CC-1065) or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to dolastatin 10 or an analog, derivative, or prodrug thereof.

[0240] In some embodiments, the antibody is conjugated to dolastatin 15 or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to monomethyl auristatin E or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to

monomethyl auristatin F or an analog, derivative, or prodrug thereof. In some embodiments, the antibody is conjugated to a pyrrolo [2,1-c][1,4]-benzodiazepine or an analog, deriva-tive, or prodrug thereof. In some embodiments, the antibody is conjugated to irinotecan or an analog, derivative, or prodrug thereof.

[0241] In some embodiments, the antibody is conjugated to a cytokine (e.g. IL-2, IL-4, IL-6, IL-7, IL-10, IL-12, IL-13, IL-15, IL-18, IL-23, IL-24, IL-27, IL-28a, IL-28b, IL-29, KGF, IFN α , IFN β , IFN γ , GM-CSF, CD40L, Flt3 ligand, stem cell factor, ancestim, and TNF α).

[0242] In some embodiments, the antibody is conjugated to a radioisotope or a chelate containing a radioisotope. For example, the antibody can be conjugated to a chelator linker (e.g. DOTA, DTPA, or thirace) that allows the antibody to complex with the radioisotope. The antibody may also or alternatively comprise or be conjugated to one or more radiolabeled amino acids or other radiolabeled molecules. Non-limiting examples of radioisotopes include ^3H , ^{14}C , ^{15}N , ^{35}S , ^{90}Y , ^{99}Tc , ^{125}I , ^{131}I , ^{186}Re , ^{213}Bi , ^{225}Ac and ^{227}Th . For therapeutic purposes, radioisotopes emitting β or α particle radiation may be used, such as ^{131}I , ^{90}Y , ^{211}At , ^{212}Bi , ^{67}Cu , ^{186}Re , ^{188}Re , and ^{212}Pb .

[0243] Techniques for coupling molecules to antibodies are well-known in the art. Typically, the nucleic acid molecule is covalently linked to a lysine or cysteine on the antibody via an N-hydroxysuccinimide ester or maleimide functional group, respectively. Conjugation methods using engineered cysteines or incorporating unnatural amino acids have been reported to improve the homogeneity of conju-gates. In particular, those skilled in the art may also predict that reactive endogenous glutamine-engineered Fc-contain-ing polypeptides are generated by use of acyl donor gluta-mine-containing tags (e.g. Gin peptide-containing tags or Q-tags) or by polypeptide engineering (e.g. by amino acid deletions, insertions, substitutions, or mutations in the poly-peptide). Transglutaminase can then be covalently cross-linked with amine donors (e.g., small molecules containing or linked to reactive amines) to form stable and homoge-neous groups of engineered polypeptides containing Fc, wherein amine donors are specifically coupled to polypep-tides containing Fc through glutamine tags containing acyl donor or accessible/exposure/reactive endogenous gluta-mine sites (WO2012059882).

[0244] It should be understood that the therapeutic agents according to the above embodiments will be administered together with suitable pharmaceutically acceptable carriers, excipients, and other reagents incorporated into the formu-lation to provide improved transfer, delivery, tolerability, etc. A large number of suitable formulations can be found in all pharmacopoeias known to pharmaceutical chemists: Remington's Pharmaceutical Sciences (15th edition, Mack Publishing Company, Easton, Pa. (1975)), in particular Chapter 87 of Blaug, Seymour therein. These formulations include, for example, powders, pastes, ointments, gels, waxes, oils, lipids, lipid-containing (cationic or anionic) carriers (e.g. LipofectinTM), DNA conjugates, anhydrous syrups, oil-in-water and water-in-oil emulsions, emulsion polyethylene glycols (polyethylene glycols of various molecular weights), semi-solid gels, and semi-solid mix-tures containing polyethylene glycol.

[0245] Any of the foregoing mixtures may be suitable for treatments or therapies of the present disclosure, provided that the active ingredients in the formulation are not inac-

tivated by the formulation and that the formulation is physiologically compatible and tolerates the routes of administration.

[0246] In one embodiment, the antibody can be used as a therapeutic agent. Such reagents will generally be used to treat, alleviate and/or prevent a disease or pathology associated with aberrant BCMA expression, activity, and/or signaling in a subject. Treatment regimens can be performed using standard methods by identifying a subject, e.g. a human patient, having (or at risk of or developing) a disease or disorder associated with aberrant BCMA expression, activity, and/or signaling, e.g. a BCMA-related disorder. An antibody preparation, preferably one preparation with high specificity and affinity for its target antigen, is administered to a subject and will generally have an effect due to its binding to the target. The administered antibody can eliminate or inhibit or interfere with the expression, activity, and/or signaling function of the target (e.g. BCMA). The administered antibody can eliminate or inhibit or interfere with the binding of the target (e.g. BCMA) to the endogenous ligand to which it naturally binds. For example, an antibody binds to a target, and modulates, blocks, inhibits, reduces, antagonizes, neutralizes, and/or otherwise interferes with BCMA expression, activity, and/or signaling. In some embodiments, in order to treat a disease or disorder associated with aberrant BCMA expression, an antibody having a heavy chain and light chain CDR can be administered to a subject.

[0247] In another embodiment, antibodies directed against BCMA can be used in methods known in the art and related to localization and/or quantitation of BCMA (e.g. for determining BCMA and/or levels of BCMA in an appropriate physiological sample, for diagnostic methods, for protein imaging, etc.).

[0248] In a given embodiment, an antibody comprising an antibody-derived antigen-binding domain specific for BCMA, or a derivative, fragment, analog, or homolog thereof, is used as a pharmaceutically active compound (hereinafter “therapeutic agent”).

[0249] In another embodiment, BCMA polypeptides can be isolated using antibodies specific for BCMA by standard techniques such as immunoaffinity, chromatography, or immunoprecipitation.

[0250] Antibodies (or fragments thereof) directed against BCMA proteins can be used to detect proteins in biological samples. In some embodiments, BCMA can be detected in a biological sample as part of a clinical testing procedure, for example, to determine the efficacy of a given treatment regimen.

[0251] Detection may be facilitated by coupling (i.e. physically linking) the antibody to a detectable substance. Examples of detectable substances include various enzymes, prosthetic groups, fluorescent materials, luminescent materials, bioluminescent materials, and radioactive materials.

[0252] Examples of suitable enzymes include horseradish peroxidase, alkaline phosphatase, β -galactosidase or acetylcholinesterase; examples of suitable prosthetic group complexes include streptavidin/biotin and avidin/biotin; examples of suitable fluorescent materials include umbelliferone, fluorescein, fluorescein isothiocyanate, rhodamine, dichlorotriazine aminofluorescein, dansyl chloride, or phycoerythrin; one example of a luminescent material includes luminol; examples of bioluminescent materials include

luciferase, luciferin, and aequorin, and examples of suitable radioactive materials include ^{125}I , ^{131}I , ^{35}S or ^3H .

[0253] In another embodiment, antibodies according to the present disclosure can be used as reagents for detecting the presence of BCMA or protein fragments thereof in a sample. In some embodiments, the antibody comprises a detectable label. The antibody is a polyclonal antibody, or more preferably a monoclonal antibody. Whole antibodies or fragments thereof (e.g. Fab, scFv, or F(ab')_2) are used.

[0254] The term “labeling” with respect to an antibody is intended to encompass direct labeling of the antibody by coupling (i.e. physically linking) a detectable substance to the antibody, as well as indirect labeling of the antibody by reaction with another reagent that is directly labeled. Examples of indirect labeling include the detection of the first antibody using a fluorescently labeled second antibody, and end-labeling of the antibody with biotin so as to enable detection with fluorescently labeled streptavidin. The term “biological sample” is intended to include tissues, cells, and biological fluids isolated from a subject, as well as tissues, cells, and fluids present in a subject.

[0255] Thus, the term “biological sample” as used includes blood and fractions or components of blood, including serum, plasma, or lymph. In other words, the detection method of the above embodiments can be used to detect the analyte mRNA, protein, or genomic DNA in a biological sample in vitro as well as in vivo. For example, analyte mRNA in vitro detection techniques include Northern hybridization and in situ hybridization. In vitro techniques for the detection of analyte proteins include enzyme-linked immunosorbent assays (ELISA), western blots, immunoprecipitations, and immunofluorescence. Analyte genomic DNA in vitro detection techniques including Southern hybridization. Procedures for performing immunoassays are described, for example, in “ELISA: Theory and Practice: Methods in Molecular Biology”, vol. 42, J. R. Crowther (ed.) Human Press, Totowa, N. J., 1995; “Immunoassay”, E. Diamandis and T. Christopoulos, Academic Press, Inc., San Diego, Calif., 1996; and “Practice and Theory of Enzyme Immunoassays”, P. Tijssen, Elsevier Science Publishers, Amsterdam, 1985. In addition, in vivo techniques for the detection of an analyte protein include introducing into a subject a labeled anti-analyte protein antibody. For example, an antibody can be labeled with a radiolabel, and the presence and location of the radiolabel in the subject can then be detected by standard imaging techniques.

[0256] The antibodies and derivatives, fragments, analogs, and homologs thereof described herein can be incorporated into pharmaceutical compositions suitable for administration. The principles and considerations involved in preparing such compositions, as well as guidance in selecting components, are well known in the art, for example, see Remington’s Pharmaceutical Sciences: The Science And Practice Of Pharmacy 19th edition (Alfonso R. Gennaro et al. eds.) Mack Pub. Co., Easton, Pa.: 1995; Drug Absorption Enhancement: Concepts, Possibilities, Limitations, And Trends, Harwood Academic Publishers, Langhorne, Pa., 1994; and Peptide And Protein Drug Delivery (Advances In Parenteral Sciences, vol. 4), 1991, M. Dekker, New York.

[0257] Such compositions typically comprise an antibody and a pharmaceutically acceptable carrier. When antibody fragments are used, minimal inhibitory fragments that specifically bind to the target protein binding domain may be

preferred. For example, based on the variable region sequence of an antibody, peptide molecules can be designed so as to retain the ability to bind to a target protein sequence. Such peptides can be chemically synthesized and/or produced by recombinant DNA techniques (see, e.g. Marasco et al. Proc. Natl. Acad. Sci. USA, 90: 7889-7893 (1993)).

[0258] As used herein, the term “pharmaceutically acceptable carrier” is intended to include any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration. Suitable pharmaceutically acceptable carriers are described in the latest edition of Remington’s Pharmaceutical Sciences, a standard reference text in the art, which is incorporated herein by reference. Preferred examples of such carriers or diluents include, but are not limited to, water, saline, ringer’s solution, dextrose solution, and 5% human serum albumin. Liposomes and non-aqueous vehicles such as fixed oils can also be used. It is well-known in the art that such media and reagents are used as pharmaceutically active substances. Except insofar as any conventional media or reagent is incompatible with the antibody, its use in the compositions is contemplated.

[0259] The pharmaceutical compositions of the above embodiments are formulated to be compatible with their intended routes of administration. Examples of routes of administration include parenteral, e.g. intravenous, intradermal, subcutaneous, oral (e.g. inhalation), transdermal (i.e. topical), transmucosal, and rectal administration. Solutions or suspensions for parenteral, intradermal, or subcutaneous administration may include the following components: sterile diluents for injection such as water, saline solutions, fixed oils, polyethylene glycols, glycerol, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl 4-hydroxybenzoate; antioxidants such as ascorbic acid or sodium bisulfite; chelators such as ethylene diamine tetraacetic acid (EDTA); buffers, such as acetates, citrates or phosphates, and reagents to adjust the osmotic pressure, such as sodium chloride or dextrose. The pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide. The parenteral preparation can be enclosed in ampoules, disposable syringes, or multiple dose vials made of glass or plastic.

[0260] Pharmaceutical compositions suitable for injection use include sterile aqueous solutions (where water is soluble) or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. For intravenous administration, suitable pharmaceutically acceptable carriers include physiological saline, bacteriostatic water, Cremophor EL™ (BASF, Parsippany, N.J.), or phosphate-buffered saline (PBS). In all cases, the composition must be sterile and should have a fluidity to the extent that is easy for injection. It must be stable under the conditions of manufacture and storage, and must be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, liquid polyethylene glycol, and the like), and suitable mixtures thereof. The proper fluidity can be maintained, for example, by use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion, and by use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents,

for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, polyols (such as mannitol, and sorbitol), and sodium chloride in the composition. Prolonged absorption of injectable compositions can be brought about by including in the composition a reagent that delays absorption, for example, aluminum monostearate and gelatin.

[0261] If needed, sterile injectable solutions can be prepared by incorporating the antibody in a required amount in an appropriate solvent with one or a combination of ingredients enumerated above (as required) followed by filtered sterilization. Generally, dispersions are prepared by incorporating the antibody into a sterile vehicle that contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, preparation methods are vacuum drying and freeze-drying that obtain a powder, the powder includes the active ingredient and any additional desired ingredient from a previously sterile-filtered solution of such ingredients.

[0262] For inhaled administration, the compound is delivered in the form of an aerosol spray from a pressurized container or distributor or sprayer containing suitable propellants such as carbon dioxide. Systemic administration may also be performed by transmucosal or transdermal ways. For transmucosal or transdermal administration, a penetrant suitable for a permeation barrier should be used in the preparation. Such penetrants are generally known in the art, and include, for example, transmucosal administration, detergents, bile salts, and fusidic acid derivatives. Transmucosal administration can be accomplished by use of nasal sprays or suppositories. For transdermal administration, one or more of the antibodies can be formulated into ointments, salves, gels, or creams as generally known in the art.

[0263] The compounds may also be prepared in a form of suppositories (e.g. with conventional suppository bases such as cocoa butter or other glycerides) or retention enemas for rectal delivery.

[0264] In one embodiment, the antibodies can be prepared with carriers that prevent their rapid elimination from the body, such as sustained/controlled release formulations, including implants and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene-vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and polylactic acid. Methods for preparing such formulations will be apparent to those skilled in the art.

[0265] It is especially advantageous to formulate parenteral compositions in dosage unit form for ease of administration and uniformity of dosage. The dosage unit form, as used herein, refers to physically discrete units suited as unitary dosages for the subject to be treated; each unit contains a predetermined quantity of one or more antibodies calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier. The specification for the dosage unit forms of the above embodiments is dictated by and directly dependent on the unique characteristics of antibodies and the specific therapeutic effects to be achieved, and the limitations inherent in the art of formulating such antibodies for treating individuals.

[0266] The pharmaceutical compositions can be presented in a container, pack, or dispenser together with instructions for administration.

[0267] The formulations described herein may also contain more than one antibody depending on the particular situation to be treated, preferably those which have complementary activities but do not adversely affect each other. Alternatively, or additionally, the composition may, for example, comprise a reagent that enhances its function, such as a cytotoxic reagent, a cytokine, a chemotherapeutic agent, or a growth inhibitor. Such molecules are suitably present in combination in amounts effective for the intended purpose. For example, they may also be present in combination in a kit or in combination for use.

[0268] In one embodiment, one or more antibodies can be administered in a combination therapy, i.e. in combination with other reagents such as therapeutic agents that can be used to treat pathological conditions or disorders, such as various forms of cancer, autoimmune disorders, and inflammatory diseases. The term “in combination” means herein that the reagents are administered substantially synchronously, simultaneously, or sequentially. If administered sequentially, the first of the two compounds is still preferably detected at an effective concentration at the treatment site when the second compound is initially administered. In one instance, “in combination” can also include both an antibody of the present disclosure and another therapeutic agent in a kit.

[0269] For example, a combination therapy can comprise coformulation and/or coadministration of one or more antibodies described herein with one or more additional therapeutic agents (e.g. one or more cytokine and growth factor inhibitors, immunosuppressants, anti-inflammatory agents, metabolic inhibitors, enzyme inhibitors, and/or cytotoxic or cytostatic agents, as described in more detail below). Such combination therapy may advantageously utilize lower dosages of the administered therapeutic agents, thus avoiding possible toxicities or complications associated with each monotherapy.

[0270] In one embodiment, the administration of the anti-BCMA antibody and the administration of the therapeutic agent can reduce the number of B cells in the subject.

[0271] In one embodiment, the treatment regimen is effective to reduce cytokine release associated with administration of the T cell activating therapeutic agent in the subject, as compared to a corresponding treatment regimen without administration of an anti-BCMA antibody.

[0272] For purposes of clarity and conciseness, features are described herein as part of the same or separate embodiments, however, it will be understood that the scope of the disclosure may include some embodiments having a combination of all or some of the features described.

EXAMPLES

Example 1. Antigen Expression and Construction of Stably Transfected Cells

[0273] Using human BCMA (Sino biological, HG10620-M) and rhesus monkey BCMA (Sino biological, CG90103-G, identical with the extracellular region sequence of cynomolgus monkey XP_005591343.1) as templates, respectively, human BCMA (Met1-Asn53) or rhesus monkey BCMA (Met1-Asn52) were amplified, fused with mouse IgG2a Fc at C-terminal, and transfected into HEK293E cells to express the bivalent recombinant protein of human or cynomolgus monkey BCMA-mFc (the amino acid sequence shown in SEQ ID NOs. 1-2); or, fused with human IgG Fc

(knob), and co-transfected into HEK293E cells with human IgG Fc (hole) by using knob-into-hole technology, the human or cynomolgus monkey BCMA-hFc kih monovalent recombinant protein (amino acid sequence shown in SEQ ID NOs. 3-5) was obtained. FIG. 1 shows SDS-PAGE results for human, cynomolgus monkey BCMA-mFc bivalent recombinant protein and BCMA-hFc kih monovalent recombinant protein.

[0274] The *Lenti* vector and packaging plasmid (Gene-coepia) containing the full-length sequences of human and cynomolgus monkey BCMA were co-transfected into HEK293T cells. 48 h after transfection, the culture supernatant containing pseudoviruses was collected. 10 μ L was used to infect 1×10^6 CHO cells, and different concentrations of puromycin were added to screen for resistance. Finally, stably transfected cells CHO-hBCMA (clone 2C2) and CHO-cynoBCMA (clone 1A2) highly expressing human or cynomolgus monkey BCMA were isolated. FIG. 2 shows that both human and monkey BCMA stably transfected cells detected by flow cytometry, both human BCMA stably transfected cells (CHO-hBCMA-2C2) and cynomolgus monkey BCMA stably transfected cells (CHO-cynoBCMA-1A2) express high levels of BCMA.

[0275] Construction of human or cynomolgus monkey CD3 ϵ heterodimers: the nucleotide sequences of the extracellular regions of human CD3 γ (UniProt P09693, Gln23-Asn116) and CD3 ϵ (UniProt P07766, Gln23-Asp126) are synthesized, fused with human IgG Fc hole or Fc knob respectively at the C-terminus, and to expressed to form a human CD3 ϵ -Fc heterodimer (the amino acid sequence of human CD3 γ IgG Fc (hole) is shown in SEQ ID NO. 6, and the amino acid sequence of human CD3 ϵ IgG Fc (knob) is shown in SEQ ID NO. 7); cynomolgus monkey CD3 γ (UniProt Q95LI7, Gln23-Asn110) and CD3 ϵ (UniProt Q95LI5, Gln22-Asp117) were also synthesized, C-terminus fused to cynomolgus monkey IgG Fc hole or Fc knob and expressed to form a cynomolgus monkey CD3 ϵ -Fc heterodimer (the amino acid sequence of cynomolgus monkey CD3 γ IgG Fc (hole) is shown in SEQ ID NO. 8 and the amino acid sequence of cynomolgus monkey CD3 γ IgG Fc (knob) is shown in SEQ ID NO. 9). Recombinant plasmids expressing CD3 γ -Fc and CD3 ϵ -Fc were mixed with 3 mg/mL PEI (Polysciences, #24765-2) and co-transfected into HEK293E cells (culture medium OPM-293 CD03 DPM). After 7 days at 37° C. under 120 rpm 5% CO₂, the culture supernatant was collected and purified by Protein A affinity chromatography to obtain human or cynomolgus monkey CD3 ϵ -Fc recombinant protein with an SDS-PAGE purity higher than 95% (FIG. 3).

Example 2. Animal Immunization and Preparation of BCMA Antibodies

[0276] Six-week-old female Balb/c mice (Charles River Beijing) were immunized with BCMA-mFc recombinant protein or CHO-BCMA stably transfected cells as immunogens, mixed with an equal volume of immune adjuvant (Titermax, Sigma-Aldrich), respectively, by footpad or subcutaneous immunization. After the primary immunization, the same dose of booster immunization is given every two weeks, and antigen immunization of human and monkey is alternated. After four immunizations, BCMA antibody titers in mouse serum greater than 1:10000 were detected. Three days after tail vein rush immunization, the spleen and part of the peripheral lymph nodes of mice were harvested, and total

RNA was extracted. Antibody cDNA was obtained by reverse transcription with IgG-specific primers. The light and heavy chain variable region fragments were amplified respectively, and the phage Fab library was constructed. The number of transformed colonies (pool capacity) was about 1.2×10^{10} .

1. Screening of Phage Antibody Libraries

[0277] Using classical panning, after mixing 1 mL BCMA immune library phage with 3.5 mL of 2% BSA/PBS solution, Maxisorp Immunotube (Nunc) coated with human or cynomolgus monkey BCMA recombinant protein was added. After binding for 1 h, multiple washing cycles were performed with PBS-Tween (0.5% v/v) and PBS to remove non-specific or low affinity bound phage. High affinity bound phage was eluted with 100 mM triethylamine and neutralized to infect TG1 cells. After two rounds of screening, a single clone was picked and inoculated into a 96-well U-shaped plate, 1 mM IPTG was added to induce expression, the induced supernatant containing Fab was used for ELISA detection screening, and 265 positive clones were obtained through primary screening.

[0278] After re-induction of expression, 10 µg/ml of biotinylated, labeled human BCMA recombinant protein was immobilized on an SA probe for 200 s of binding time and 300 s of dissociation time and tested for affinity to Fab (Octet, Fortebio). Fifty-nine clones with higher affinity were selected to detect the binding to cell membrane surface receptors using human and cynomolgus monkey BCMA-CHO stably transfected cells, wherein, fifty-two clones could cross-recognize human and cynomolgus monkey BCMA-CHO stably transfected cells.

2. Antibody Affinity Assay with Biacore

[0279] Partial high-affinity antibody clones were selected, plasmids were extracted and transformed into TG1 cells,

expression was induced overnight at 0.1 mM IPTG 30° C., recombinantly expressed Fab was extracted from a bacterial periplasmic cavity, and Fab antibody was purified by Ni-NTA affinity chromatography. The Biacore affinity assay utilized a Protein A chip (GE healthcare), which was set to capture human BCMA-hFc recombinant protein (0.5 µg/mL) at 200KRU and after the baseline was stable, a gradient dilution of antibody Fab (from 10 µg/mL, 2-fold dilutions for 7 gradients) was run through the chip at a flow rate of 30 µL/min for a binding time of 350 s and a dissociation time of 600 s. Kinetic constants were fitted using Biacore T200 evaluation software with a Langmuir 1:1 kinetics model. The results of the BCMA antibody affinity assay are shown in Table 1.

TABLE 1

The affinity of BCMA Antibodies			
Fab	ka (1/Ms)	kd (1/s)	KD (nM)
3G11	3.33E+04	4.32E-04	13.0
6G5	2.94E+04	6.13E-05	2.1
15G9	1.03E+06	1.26E-02	12.2
15H10	3.78E+04	6.90E-04	18.3
16B9	1.17E+05	4.76E-04	4.1
23F8	5.77E+04	3.22E-04	5.6
23G9	4.07E+03	3.13E-04	76.9
29A11	7.64E+04	2.43E-04	3.2
34G2	1.42E+05	7.05E-04	5.0
38E2	9.88E+04	7.98E-04	8.1
38D9	5.91E+04	3.29E-04	5.6
38F9	5.81E+03	2.00E-03	344
40C6	4.21E+04	2.71E-04	6.4

[0280] The BCMA antibody variable region sequences are as follows:

TABLE 2

Variable region sequences of BCMA murine antibody										
BCMA murine antibody	HCVR		HCDR1	HCDR2	HCDR3	LCVR		LCDR1	LCDR2	LCDR3
	Amino acid sequence	Nucleotide sequence	Amino acid Sequence	Amino acid Sequence	Amino acid Sequence	Amino acid sequence	Nucleotide sequence	Amino acid Sequence	Amino acid Sequence	Amino acid Sequence
3G11	10	11	12	13	14	15	16	17	18	19
6G5	10	11	12	13	14	20	21	22	23	19
15G9	24	25	26	27	28	29	30	31	32	33
15H10	34	35	12	36	37	38	39	17	40	19
16B9	41	42	12	43	44	45	46	47	18	19
23F8	48	49	12	50	14	51	52	17	40	19
23G9	53	54	55	56	57	58	59	60	61	62
29A11	10	11	12	13	14	38	39	17	40	19
34G2	63	64	65	66	14	67	68	17	40	19
38E2	69	70	71	72	73	74	75	76	77	78
38D9	79	80	12	13	14	81	82	83	84	19
38F9	10	11	12	13	14	85	86	87	88	89
40C6	10	11	12	13	14	90	91	92	23	19

3G11

[0281] The amino acid sequence of the 3011 heavy chain VH is shown in SEQ ID NO. 10, of which the encoding nucleic acid is shown in SEQ ID NO. 11, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 12, 13 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EIQLVQSGPELKPKGETVKISKASGYIFTNFGMNWVRQA

CDR2 <-----FR3----->
PGKALKWMGWINTYTGEOIYADDFKGRFAFSLETS

CDR3 <----FR4---->
STAYLQINNLIKNETATYFCARGEIIYGYDVGFFVYWGQGLTVTVSA

Nucleic acid sequence
GAGATCCAGTTGGTGAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATCTTCACAACTTTGGAA
TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
ATAAACACCTACACTGGAGAACAATATATGCTGATGACTTCAAGGGACG
GTTTGCCTTCTCTTTGGAACCTCTGCCAGCACTGCCTATTTGCAGATCA
ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
ATCTACTATGGTTACGAGCTGGGCTTTGTTTACTGGGCCAAGGGACTCT
GGTCACTGTCTCTGCA

[0282] The amino acid sequence of the 3G11 light chain VK is shown in SEQ ID NO. 15, of which the encoding nucleic acid is shown in SEQ ID NO. 16, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 17, 18 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLTSLGQRATISCRASKSVTSGYSYHMYQQ

CDR2 <-----FR3----->
KPGQPPKLLIYLASDLEAGVSPARFSGSGSDTFT

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGKLEIK

Nucleic acid sequence
GATATTGTGCTAACTAGTCTCCTGCTTCTTAACTGTATCTCTGGGCA
GAGGGCCACCATCTCATGCAAGGCCAGCAAAAGTGTAGTACATCTGGCT
ATAGTTATATGCACTGGTACCAACAGAAACAGGACAGCCACCCAACTC
CTCATCTATCTTGCATCCGACCTAGAACTCTGGGCTCTCTGCCAGGTTGAG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG
AGGAGGATGCTGCAACCTATTACTGTCAACACAGTAGGGAACCTCCGTGG
ACGTTTCGGTGGAGGCACCAAGCTGGAAATAAAA
6G5

[0283] The amino acid sequence of the 635 heavy chain VH is shown in SEQ ID NO. 10, of which the encoding nucleic acid is shown in SEQ ID NO. 11, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 12, 13 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EIQLVQSGPELKPKGETVKISKASGYIFTNFGMNWVRQ

-continued

CDR2 <-----FR3----->
APGKALKWMGWINTYTGEOIYADDFKGRFAFSLETS

CDR3 <----FR4---->
STAYLQINNLIKNETATYFCARGEIIYGYDVGFFVYWGQGLTVTVSA

Nucleic acid sequence
GAGATCCAGTTGGTGAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATCTTCACAACTTTGGAA
TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
ATAAACACCTACACTGGAGAACAATATATGCTGATGACTTCAAGGGACG
GTTTGCCTTCTCTTTGGAACCTCTGCCAGCACTGCCTATTTGCAGATCA
ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
ATCTACTATGGTTACGAGCTGGGCTTTGTTTACTGGGCCAAGGGACTCT
GGTCACTGTCTCTGCA

[0284] The amino acid sequence of the 6G5 light chain VK is shown in SEQ ID NO. 20, of which the encoding nucleic acid is shown in SEQ ID NO. 21, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 22, 23 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLAISLGQRATISCRASKSVTSGYSYHMY

CDR2 <-----FR3----->
QQKPGQPPKLLIYLASDLEAGVSPARFSGSGSDTFT

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGKLEIK

Nucleic acid sequence
GACATTGTGCTAACACAGTCTCCTGCTTCTTACGTATATCTCTGGGCA
GAGGGCCACCATCTCATGCAAGGCCAGCAAAAGTGTCACTACATCTGGCT
ATAGTTATATACACTGGTACCAACAGAAACAGGACAGCCACCCAACTC
CTCATCTATCTTGCATCCGACCTAGAAGCTGGGGTCCCTGCCAGGTTGAG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG
AAGAGGATGCTGCAACCTATTACTGTGACACAGTAGGAGCTTCCGTGG
ACGTTTCGGTGGAGGCACCAAGCTGGAGCTGAAA
15G9

[0285] The amino acid sequence of the 15G9 heavy chain VH is shown in SEQ ID NO. 24, of which the encoding nucleic acid is shown in SEQ ID NO. 25, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 26, 27 and 28, respectively.

<-----FR1-----> CDR1 <-----FR2----->
QVQLQQSGPELVKPGASVNLSCASGYFTTNYVIHWVKQ

CDR2 <-----FR3----->
KPGQGLEWIGYTIPTDVTKYNEKFKAKATLTSDKSS

CDR3 <----FR4---->
STAFMELSSLTSEDSAVYYCARYDFDGYFDYWGQGLTVSS

-continued

Nucleic acid sequence
 CAGGTCCAGCTGCAACAGTCTGGACCTGAGCTGGTGAAGCCTGGGGCTTC
 AGTGAATTTGTCTCTGCAAGGCTTCTGGATACACATTCACATAATTATGTTA
 TACACTGGGTAAGCAGAAAGCCTGGGCAGGGCCTTGAGTGGATTGGATAT
 ACTATCCCTTACACTGATGTTACTAAGTACAATGAGAAATTCAAAGCCAA
 GGCCACACTGACTTCAGACAAATCCTCCAGCACAGCCTTCATGGAGCTCA
 GCAGCTGACCTCTGAGGACTCTGCGGTCTATTACTGTGCAAGATATGAT
 TTCGACGGCTACTTTGACTACTGGGGCCAAGGCACCACTCTCACAGTCTC
 CTCA

[0286] The amino acid sequence of the 15G9 light chain VK is shown in SEQ ID NO. 29, of which the encoding nucleic acid is shown in SEQ ID NO. 30, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 31, 32 and 33, respectively.

<-----FR1-----> CDR1 <-----FR2----->
 DIVMTQAAPSNPVTLTGTSASISCRSSKSLLSNGITLYWY

CDR2 <-----FR3----->
 LQKPGQSPQLLIYQMSNLASGVPRFSSSGSGTDF

CDR3 <----FR4---->
 TLRISRVEAEDVGVYCAQNLLELPWTFGGGKLELK

Nucleic acid sequence
 GATATTGTGATGACTCAGGCTGCATTCTCCAATCCAGTCACTCTTGGAAAC
 ATCAGCTTCCATCTCCTGCAGGTCTAGTAAGAGTCTCTACATAGTAATG
 GCATCACTTATTTGTATTGGTATCTGCAGAACCCAGGCCAGTCTCCTCAG
 CTCCTGATTTATCAGATGTCCAACCTTGCTCAGGAGTCCCAGACAGGTT
 CAGTAGCAGTGGGTGAGGAAGTATTTCACTGAGAATCAGCAGAGTGG
 AGGCTGAGGATGTGGGTGTTTATTACTGTGCTCAAATCTAGAACTTCCG
 TGGACGTTTCGGTGGAGGCACCAAGCTGGAGCTGAAA
 15H10

[0287] The amino acid sequence of the 15H10 heavy chain VH is shown in SEQ ID NO. 34, of which the encoding nucleic acid is shown in SEQ ID NO. 35, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 12, 36 and 37, respectively.

<-----FR1-----> CDR1 <-----FR2----->
 EVQLQQSGPELKKPGETVKISKASGYTFNFGMNWVRQTPG

CDR2 <-----FR3----->
 KALKWMGWINTYTGDLIYGDDDFKGRFAFSLETS

CDR3 <----FR4---->
 STAYLQINNLIKNETATYFCARGEIYYGYDVGFAFWGQGLVTVSA

Nucleic acid sequence
 GAGGTCCAGCTGCAGCAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
 AGTCAAGATCTCTGCAAGGCTTCTGGGTATACCTTCACAACTTTGGAA
 TGAAGTGGGTGAGGCAGACTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
 ATAAACACCTACACTGGAGACTTAATATATGGTGATGACTTTAAGGGACG
 GTTTCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTACAGATCA

-continued

ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
 ATCTACTATGGTTACGACGTGGGGTTTGCTTACTGGGGCCAAGGGACTCT
 GGTCACGTCTCTGCA

[0288] The amino acid sequence of the 15H10 light chain VK is shown in SEQ ID NO. 38, of which the encoding nucleic acid is shown in SEQ ID NO. 39, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 17, 40 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
 DIVLTQSPASLAVSLGQRATISCRASKSVSTSGYSYMHWY

CDR2 <-----FR3----->
 QQKPGQPPKLLIYLASNLESGVPRFSGSGSGTDF

CDR3 <----FR4---->
 LNIHPVEEEDAATYYCQHSRELPWTFGGGKLEIK

Nucleic acid sequence
 GACATTGTGCTGACACAGTCTCCTGCTTCTTAGCTGTATCTCTGGGGCA
 GAGGGCCACCATCTCATGCAGGGCCAGCAAAAGTGTCACTACATCTGGCT
 ATAGTTATATGCACTGGTACCAACAGAAACAGGACAGCCACCCAACTC
 CTCATCTATCTTGCACTCAACCTAGAACTCTGGGTCTCCTGCCAGGTTTCA
 TGGTAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG
 AGGAGGATGCTGCAACCTATTACTGTGACACAGTAGGAGACTTCCGTGG
 ACGTTTCGGTGGAGGCACCAAGCTGGAAATCAAA
 16B9

[0289] The amino acid sequence of the 16B9 heavy chain VH is shown in SEQ ID NO. 41, of which the encoding nucleic acid is shown in SEQ ID NO. 42, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 12, 43 and 44, respectively.

<-----FR1-----> CDR1 <-----FR2----->
 EIQLVQSGPELKKPGETVKISKASGYIFTNFGMNWVR

CDR2 <-----FR3----->
 QTPGKALKWMGWINTYTGDIYADDFKGLFAFSLEVPA

CDR3 <----FR4---->
 STAYLQINNLIKNETATYFCARGEIYYGYDVGFAFWGQGLVTVSA

Nucleic acid sequence
 GAGATCCAGTTGGTGCAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
 AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATTTTCAAACTTTGGAA
 TGAAGTGGGTGAGGCAGACTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
 ATAAACACCTACACTGGAGAAACATATATGCTGATGACTTCAAGGGACT
 ATTTGCCTTCTCTTTGGAAGTGCCTGCCAGCACTGCCTATTTCAGATCA
 ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
 ATCTACTATGGTTTCGACGTGGGATTGCTTACTGGGGCCAAGGGACTCT
 GGTCACGTCTCTGCA

[0290] The amino acid sequence of the 16B9 light chain VK is shown in SEQ ID NO. 45, of which the encoding

nucleic acid is shown in SEQ ID NO. 46, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 47, 18 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLTSLGQRATISCRASKSVSTSGYSYIHWYQQ

CDR2 <-----FR3----->
KPGQPPKLLIYLASDLESQVPAFSGSGSDTFT

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGTKLELK

Nucleic acid sequence
GACATTGTGCTAACACAGTCTCCTGCTTCCTTAAGTGTATCTCTGGGGCA

GAGGGCCACCATCTCATGCAGGGCCAGCAAAAGTGTAGTACATCTGGCT
ATAGTTATATACACTGGTACCAACAGAAACCAGGACAGCCACCCAAACTC
CTCATCTATCTTGCATCCGACCTAGAATCTGGGGTCCCTGCCAGGTTTCTG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCTGTGGAGG
AGGAGGATGCTGCAACCTATTACTGTCAACACAGTAGGGAATCTCCGTGG

ACGTTCCGTGGAGGCACCAAGCTGGAGCTGAAA
23F8

[0291] The amino acid sequence of the 23F8 heavy chain VH is shown in SEQ ID NO. 48, of which the encoding nucleic acid is shown in SEQ ID NO. 49, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 12, 50 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EIQLVQSGPELKKPGETVKISCTASGYTFNFGMNWVRQ

CDR2 <-----FR3----->
APGKALRWMGWINTYTDQIYADDFKGRFAFSLTSA

CDR3 <----FR4---->
STAYLQINNKLKEDTATYFCARGEIYYGYDVGFIWQGTLLVTLSA

Nucleic acid sequence
GAGATCCAGTTGGTGCAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC

AGTCAAGATCTCCTGCACGGCTTCTGGGTATACCTTCACAAACTTTGGAA
TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAGGTGGATGGGCTGG
ATAAACACCTACACTGGAGACCAATATATGCTGATGACTTCAAGGGACG
GTTTGCCCTCTCTTTGGAACCTCTGCCAGCACTGCCTATTGTCAGATCA
ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
ATCTACTATGTTTACGACGTGGGCTTTGTTTATTGGGCCAAGGGACTCT
GGTCACTGTCTCTGCA

[0292] The amino acid sequence of the 23F8 light chain VK is shown in SEQ ID NO. 51, of which the encoding nucleic acid is shown in SEQ ID NO. 52, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 17, 40 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLAVSLGQRATISCRASKSVSTSGYSYIHWYQ

CDR2 <-----FR3----->
QKPGQPPKLLIYLASNLESQVPAFSGSGSDTFT

-continued

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGTKLELK

Nucleic acid sequence
GACATTGTGCTAACACAGTCTCCTGCTTCCTTAGTGTATCTCTGGGGCA

GAGGGCCACCATCTCATGCAGGGCCAGCAAAAGTGTAGTACATCTGGCT
ATAGTTATATGCACTGGTACCAACAGAAACCAGGACAGCCACCCAAACTC
CTCATCTATCTTGCATCCACCTAGAATCTGGGGTCCCTGCCAGGTTTCTG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCTGTGGAGG
AGGAGGATGCTGCAACCTATTACTGTGACAGTAGGAGCTTCCGTGG

ACGTTCCGTGGAGGCACCAAGCTGGAAATAAAA
23G9

[0293] The amino acid sequence of the 23G9 heavy chain VH is shown in SEQ ID NO. 53, of which the encoding nucleic acid is shown in SEQ ID NO. 54, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NO. 55, 56 and 57, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EVQLQQSGTSLVLRPGASVKLSCKASGYSFSSYWMHWLQK

CDR2 <-----FR3----->
RPGQGLEWIGAIYFGNSRTTYNQKFKGEAKLTADTSA

CDR3 <----FR4---->
STAYMDLSSLTNADSAVYFCTREETYWGQGTLLVTLSA

Nucleic acid sequence
GAGGTTCACTCCAGCAGTCTGGGACTGTGCTGACAAGGCCTGGGGCTTC
CGTGAAGTTGTCTGCAAGGCTTCTGGCTACAGCTTTTCCAGCTACTGGA

TGCAGTGGCTAAAACAGAGGCTTGGACAGGCTCTAGAATGGATTGGTGTCT
ATTTATCTGGAAATAGTCGTACTACCTACCAAGAGTTCAAGGGCGCA
GGCCAACTGACTGCGGACACTTCCGCCAGCACTGCCTACATGGACCTCA
GCAGCCTGACAAATGCCGACTCTGCGGTCTATTTCTGTACAAGAGAGGAG
ACTTACTGGGGCCCAAGGACTCTGGTCACTGTCTCTGCA

[0294] The amino acid sequence of the 23G9 light chain VK is shown in SEQ ID NO. 58, of which the encoding nucleic acid is shown in SEQ ID NO. 59, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOS. 60, 61 and 62, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DVVMTQTPSLSVLTIGQPASISCKSSQGLLDYGVVYTLNWL

CDR2 <-----FR3----->
FQRPQPPKRLIYLVSLEDSQVPAFSGSGSDTFT

CDR3 <----FR4---->
TLRISRVEADLGIYYCWQGSHPPLTFGAGTKLELK

Nucleic acid sequence
GATGTTGTGATGACCCAGACTCCACTCTCTTGTGCGGTACCATGGACA

ACCGACCTCCATCTCTTGAAGTCAAGTCAAGGCGCTCTTAGATAGTTATG
GAGTGACATATTTGAATTGGTTGTTTCAGAGGCCAGGCCAGCCTCCAAAG
CGCCTAATCTATCTAGTGTCTGAAGTGGACTCTGGAGTCCCTGACAGGTT

-continued

CACTGGCAGTGGATCAGGACAGATTTACACTGAGAATCAGCAGAGTGG
AGGCTGAGGATTGGGAATTTATTATTGCTGGCAAGGTTACATTTTCCT

CTCACGTTGCGTGCTGGGACCAAGCTGGAGCTGAAA
29A11

[0295] The amino acid sequence of the 29A11 heavy chain VH is shown in SEQ ID NO. 10, of which the encoding nucleic acid is shown in SEQ ID NO. 11, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EIQLVQSGPELKKPGETVKISKASGYIFTNFGMNWVRQAPGKAL

CDR2 <-----FR3----->
KWMGWINTYTTGEQIYADDFKGRFAFSLETS

CDR3 <----FR4---->
STAYLQINNLIKNETATYFCARGGEIYYGYDVGFFVYWGQGLTVTVSA

Nucleic acid sequence
GAGATCCAGCTGGTGACGCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC

AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATCTTCACAACTTTGGAA
TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
ATAAACACCTACACTGGAGAACAAATATATGCTGATGACTTCAAGGGACG
GTTTGCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTGTCAGATCA
ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
ATCTACTATGGTTACGACGTGGGCTTTGTTTACTGGGGCCAAGGGACTCT
GGTCACTGTCTCTGCA

[0296] The amino acid sequence of the 29A11 light chain VK is shown in SEQ ID NO. 38, of which the encoding nucleic acid is shown in SEQ ID NO. 39, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLAVSLGQRATISCRRASKSVSTSGYSYMHWY

CDR2 <-----FR3----->
QQKPGQPPKLLIYLASNLESGVPARFSGSGSDTDF

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGKLEIK

Nucleic acid sequence
GATATTGTGCTAACTCAGTCTCCTGCTTCCTTAGCCGTATCTCTGGGGCA

GAGGGCCACCATCTCATGCAGGGCCAGCAAAAGTGTCAGTACATCTGGCT
ATAGTTATATGCACTGGTACCAACAGAAACAGGACAGCCACCCAACTC
CTCATCTATCTTGATCCAACTAGAACTCTGGGGTCCCTGCCAGGTTGAG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG
AGGAGGATGCTGCAACCTATTACTGTCAGCACAGTAGGGAGCTTCCGTGG

ACGTTCCGGTGGAGGCCAACAGCTGGAAATCAAA
34G2

[0297] The amino acid sequence of the 34G2 heavy chain VH is shown in SEQ ID NO. 63, of which the encoding

nucleic acid is shown in SEQ ID NO. 64, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 65, 66 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EVQLQQSGPELKKPGETVKISKASGYTFTNYGMNWVKQTPGKGL

CDR2 <-----FR3----->
KWMGWINTYTTGERIYADDFKGRFAFSLETS

CDR3 <----FR3---->
STAYLQINNLIKNETATYFCARGGEIYYGYDVGFFVYWGQGLTVTVSA

Nucleic acid sequence
GAGGTCCAGTTGCAGCAGTCTGGACCTGAGCTGAGGAAGCCTGGAGAGAC

AGTCAAGATCTCCTGCAAGGCTTCTGGGTATACCTTCACAACTATGGAA
TGAAGTGGGTGAAGCAGACTCCAGGAAAGGTTTAAAGTGGATGGGCTGG
ATAAACACCTACACTGGAGAACGAATATATGCTGATGACTTCAAGGGACG
GTTTGCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTGTCAGATCA
ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
ATCTACTATGGTTACGACGTGGGCTTTGTTTACTGGGGCCAAGGGACTCT
GGTCACTGTCTCTGCA

[0298] The amino acid sequence of the 34G2 light chain VK is shown in SEQ ID NO. 67, of which the encoding nucleic acid is shown in SEQ ID NO. 68, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLVVSLGQRATISCRRASKSVSTSGYSYMHWYQ

CDR2 <-----FR3----->
QKPGQPPKLLIYLASNLESGVPARFSGSGSDTDF

CDR3 <----FR4---->
LNIHPVEEEDAATYYCQHSRELPWTFGGGTRLEIK

Nucleic acid sequence
GACATTGTGCTGACACAGTCTCCTGCTTCCTTAGTTGTATCTCTGGGGCA

GAGGGCCACCATCTCATGCAGGGCCAGCAAAAGTGTCAGTACATCTGGCT
ATAGTTATATGCACTGGTACCAACAGAAACAGGACAGCCACCCAACTC
CTCATCTATCTTGATCCAACTAGAACTCTGGGGTCCCTGCCAGGTTGAG
TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG
AGGAGGATGCTGCAACCTATTACTGTCAGCACAGTAGGGAGCTTCCGTGG
ACGTTCCGGTGGAGGCACAGACTGGAAATAAAA
38E2

[0299] The amino acid sequence of the 38E2 heavy chain VH is shown in SEQ ID NO. 69, of which the encoding nucleic acid is shown in SEQ ID NO. 70, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 71, 72 and 73, respectively.

<-----FR1-----> CDR1 <-----FR2----->
QVQLQQSGAELVRPGTSVKMSCKAAGYTFTNYIWIGWVKQ

CDR2 <-----FR3----->
RPGHGLEWIGDIYPGSGDTNNGKFKGKATLAADISS

-continued

CDR3 <----FR4---->
 NAAYMQLSSLTSEDSAIYYCARSYRNDAASFAYWGQGLVTVSA

Nucleic acid sequence
 CAGGTCCAGCTGCAGCAGTCTGGAGCTGAGCTGGTAAGGCCTGGGACTTC
 AGTGAAGATGTCTGCAAGGCTGCTGGATACACCTTCACTAACTACTGGA
 TAGGTTGGGTAAAGCAGAGGCCTGGACATGGCCTTGAGTGGATTGGAGAT
 ATTTACCCTGGAAGTGGTGATACTAATTACAATGGGAAGTTCAAGGGCAA
 GGCCACACTGGCTGCAGACATTTCTCCAACGCAGCCTACATGCAGCTCA
 GCAGCCTGACATCTGAGGACTCTGCCATCTATTACTGTGCAAGAAGCTAC
 TATAGGAACGACGCGGCCTCATTGCTTACTGGGCCAAGGACTCTGGT
 CACTGTCTCTGCA

[0300] The amino acid sequence of the 38E2 light chain VK is shown in SEQ ID NO. 74, of which the encoding nucleic acid is shown in SEQ ID NO. 75, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 76, 77 and 78, respectively.

<-----FR1-----> CDR1<-----FR2----->
 DIVMTQSHKFMSTSVGDRVNITCKASQDVDTAVAWYQQKP

CDR2 <-----FR3----->
 SQSPKLLIYWTSTRHTGVPDRFTGSGSGTDFTLTIS

CDR3 <---FR4--->
 NVQSEDLADYFCQFSSYPLTFGAGTKLELK

Nucleic acid sequence
 GACATTGTGATGACCCAGTCTCACAATTCATGTCCACATCAGTAGGAGA
 CAGGGTCAACATCACCTGCAAGGCCAGTCAGGATGTGGATACTGCTGTAG
 CCTGGTATCAACAGAAACCAAGTCAATCTCCTAACTACTGATTTACTGG
 ACATCCACCCGGCACACTGGAGTCCCTGATCGCTTCACAGGCAGTGGATC
 TGGGACAGATTTCACTCTCACCATTAGCAATGTGCAGTCTGAAGACTTGG
 CAGATTATTTCTGTCAACAATTTAGTAGCTATCCTCTCAGTTCGGTGCT
 GGGACCAAGCTGGAGCTGAAA
 38D9

[0301] The amino acid sequence of the 38D9 heavy chain VH is shown in SEQ ID NO. 79, of which the encoding nucleic acid is shown in SEQ ID NO. 80, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

<-----FR1-----> CDR1<-----FR2----->
 EVQLQQSGPELKKPGETVKISCKASGYIFTNFGMNWVRQ

CDR2<-----FR3----->
 APGKALKWMGWINTYTGEQIYADDFKGRFAFSLETS

CDR3 <----FR4---->
 STAYLQINNLIKNETATYFCARGEIYYGYDVGFFVWGQGLVTVSA

Nucleic acid sequence
 GAGGTCCAGTTGCAGCAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
 AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATCTTCACGAACCTTTGGAA
 TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG

-continued

ATAAACACCTACACTGGAGAACAAATATATGCTGATGACTTCAAGGGACG
 GTTTGCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTTCAGATCA
 ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
 ATCTACTATGGTTACGACGTGGGCTTTGTTTACTGGGCCAAGGGACTCT
 GGTCACCTGTCTCTGCA

[0302] The amino acid sequence of the 38D9 light chain VK is shown in SEQ ID NO. 81, of which the encoding nucleic acid is shown in SEQ ID NO. 82, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 83, 84 and 19, respectively.

<-----FR1-----> CDR1<-----FR2----->
 DIVLTQSPASLTVSLGQRATISCGASKSVSTSGYSYMHWYQ

CDR2 <-----FR3----->
 QKPGQPPKLLIYLGSDLESGVPARFSGSGSGTDF

CDR3 <---FR4--->
 LNIHPVEEEDAATYYCQHSRELPWTFGGGTKLEIK

Nucleic acid sequence
 GACATTGTGCTGACACAGTCTCCTGCTTCCTTAAGTATCTCTGGGCA
 GAGGGCCACCATCTCATGCGGGGCCAGCAAAAGTGTGAGTACATCTGGCT
 ATAGTTATATGCACTGGTACCAACAGAAACCAGGACAGCCACCCAACTC
 CTCATCTATCTTGGATCCGACCTAGAATCTGGGGTCCCTGCCAGGTTTCA
 TGGCAGTGGGTCTGGGACAGACTTCACCCCTCAACATCCATCCTGTGGAGG
 AGGAGGATGTGCAACCTATTACTGTCAACACAGTAGGGAACCTCCGTGG
 ACGTTCGGTGGAGGCACCAAGCTGGAAATCAA
 38F9

[0303] The amino acid sequence of the 38F9 heavy chain VH is shown in SEQ ID NO. 10, of which the encoding nucleic acid is shown in SEQ ID NO. 11, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

<-----FR1-----> CDR1<-----FR2----->
 EIQLVQSGPELKKPGETVKISCKASGYIFTNFGMNWVRQ

CDR2 <-----FR3----->
 APGKALKWMGWINTYTGEQIYADDFKGRFAFSLETS

CDR3 <----FR4---->
 STAYLQINNLIKNETATYFCARGEIYYGYDVGFFVWGQGLVTVSA

Nucleic acid sequence
 GAGATCCAGTTGGTGCAGTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC
 AGTCAAGATCTCCTGCAAGGCTTCTGGGTATATCTTCACAACTTTGGAA
 TGAAGTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG
 ATAAACACCTACACTGGAGAACAAATATATGCTGATGACTTCAAGGGACG
 GTTTGCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTTCAGATCA
 ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG
 ATCTACTATGGTTACGACGTGGGCTTTGTTTACTGGGCCAAGGGACTCT
 GGTCACCTGTCTCTGCA

[0304] The amino acid sequence of the 38F9 light chain VK is shown in SEQ ID NO. 85, of which the encoding nucleic acid is shown in SEQ ID NO. 86, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 87, 88 and 89, respectively.

<-----FR1-----> CDR1 <-----FR2----->
NIVLTQSPASLAVSLGQRATISCRASESVDSGNSFMHWYQQ

CDR2 <-----FR3----->
KPGQPPKLLIYLESNLESVGPAPRFGSGSGRTDFT

CDR3 <----FR4-->
LTIEPVEADDAATYYCQQSNEDPWTFGGGTKLEIK

Nucleic acid sequence

AACATTGTGCTGACCAATCTCCAGCTTCTTTGGCTGTGTCTCTAGGGCA

GAGGGCCACCATATCTGCAGAGCCAGTGAAAGTGTGATAGTTCTGGCA

ATAGTTTTATGCACTGGTACCAGCAGAAACCAGGACGCCACCCAACTC

CTCATCTATCTTGAATCAACCTAGAACTCTGGGGTCCCTGCCAGGTTTCTAG

TGGCAGTGGGTCTAGGACAGACTTCACCTCACCATTGAACCTGTGGAGG

CTGATGATGCTGCAACCTATTACTGTGAGCAAGTAATGAGGATCCGTGG

ACGTTCCGTGGAGGCACCAAGCTGGAAATCAAA
40C6

[0305] The amino acid sequence of the 40C6 heavy chain VH is shown in SEQ ID NO. 10, of which the encoding nucleic acid is shown in SEQ ID NO. 11, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

<-----FR1-----> CDR1 <-----FR2----->
EIQLVQSGPELKPGETVKISKASGYIFTNFGMNVVRQ

CDR2 <-----FR3----->
APGKALKWGMWINTTYTGEQIYADDFKGRFAFSLETS

CDR3 <----FR4-->
STAYLQINNLIKNEEDTATYFCARGEIYYGYDVGFFVWGQGLTVTVSA

Nucleic acid sequence

GAGATCCAGTTGGTGACCTCTGGACCTGAGCTGAAGAAGCCTGGAGAGAC

AGTCAAGATCTCTGCAAGGCTTCTGGGTATATCTTCACAACTTTGGAA

TGAACTGGGTGAGGCAGGCTCCAGGAAAGGCTTTAAAGTGGATGGGCTGG

ATAAACACCTACACTGGAGAACAATATATGCTGATGACTTCAAGGGACG

GTTTGCCTTCTCTTTGGAAACCTCTGCCAGCACTGCCTATTTGCAGATCA

ACAACCTCAAAAATGAGGACACGGCTACATATTTCTGTGCAAGAGGGGAG

ATCTACTATGTTTACGACGTGGGCTTTGTTTACTGGGCAAGGGACTCT

GGTCACTGTCTCTGCA

[0306] The amino acid sequence of 40C6 light chain VK is shown in SEQ ID NO. 90, of which the encoding nucleic acid is shown in SEQ ID NO. 91, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 92, 23 and 19, respectively.

<-----FR1-----> CDR1 <-----FR2----->
DIVLTQSPASLAISLGQRATISCRASRSVTTSGYSYIHWYQQ

CDR2 <-----FR3----->
KPGQPPKLLIYLASDLEAGVPARFSGSGSGTDFT

CDR3 <----FR4-->
LNHPVEEEDAATYYCQHSRELPTWFGGGTKLEIK

Nucleic acid sequence

GACATTGTGCTGACACAGTCTCCTGCTTCTTAGCTATATCTCTGGGGCA

GAGGGCCACCATCTCATGCAGGGCCAGCAGAAGTGTCACTACATCTGGCT

ATAGTTATATACACTGGTACCAACAGAAACCAGGACAGCCACCCAACTC

CTCATCTATCTTGCATCCGACCTAGAACTGGGGTCCCTGCCAGGTTTCTAG

TGGCAGTGGGTCTGGGACAGACTTCACCTCAACATCCATCCTGTGGAGG

AAGAGGATGCTGCAACCTATTACTGTGACACAGTAGGGAGCTTCCGTGG

ACGTTCCGTGGAGGCACCAAGCTGGAAATAAAA

Example 3. Construction of BCMA×CD3 Bispecific Antibody

1. CD3 Humanized Antibody

[0307] Murine hybridoma CD3 antibody (EMBO J. 1985. 4(2): 337-344; J. Immunol. 1986,137(4): 1097-100; J. Exp. Med. 1991, 174: 319-326; J. Immunol. 1991, 147(9): 3047-52) recognizes the human and cynomolgus monkey CD3 receptor, the sequence of which is as follows:

the anti-CD3 murine monoclonal antibody light chain amino acid sequence is shown in SEQ ID NO. 184:

QAVVTQESALTTSPGETVTLTCRSSTGAVTTSNYANWVQ

QKPDHLFTGLIGGTNKRAPGVPARFSGSLIGDKAALTIT

GAQTEDEAIYFCALWYSNLWVFGGGTKLTVL

the amino acid sequence of the anti-CD3 murine monoclonal antibody heavy chain is shown in SEQ ID NO. 185:

EVQLVESGGGLVQPKGSLKLSCAASGFTFTNTYAMNVVRQ

APGKGLEWVARIRSKYNNYATYYADSVKDRFTISRDDSQ

SILYLQMNLIKTEDTAMYYCVRHGNFGNSYVSWFAYWGQ

GTLTVTVSS

the anti-CD3 murine monoclonal antibody was humanized, the human germline gene IMGT_hVL7-43 with the highest homology was selected for light chain CDR transplantation and human IGLJ3*02 was selected for FM4; human IMGT_hVH3-73 was selected for heavy chain CDR transplantation and human IGHJ4*01 was selected for FM4. The different heavy chain and light chain variants were designed.

[0308] The sequence of the light chain variable region of the humanized CD3 antibody is as follows:

TABLE 3

Light chain variable region sequences of CD3 humanized antibodies					
SEQ ID NOs: 93-111					
CD3 humanized antibody	LCVR		LCDR1	LCDR2	LCDR3
	Amino acid sequence	Nucleotide sequence	Amino acid sequence	Amino acid sequence	Amino acid sequence
hVL1	93	94	95	96	97
hVL2	98	99	95	96	97
hVL3	100	101	102	103	97
hVL4	104	105	102	103	97
hVL5	106	107	95	96	108
hVL6	109	110	95	111	108

[0309] The amino acid sequence of hVL1 is shown in SEQ ID NO. 93, of which the encoding nucleic acid is shown in SEQ ID NO. 94, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 95, 96 and 97, respectively.

QAVVTQEPSTLVSPGGTTLTCSR**SSSTGAVTTSNYAN**WVQEKPGQAPRGLI

QKPGQAPRGLI**GGTNKRAP**WTPARFSGSLLGGKAALTLS

GAQPEDEAEYY**CALWYSNLWV**FGGGTKLTVL

[0310] The amino acid sequence of hVL2 is shown in SEQ ID NO. 98, of which the encoding nucleic acid is shown in SEQ ID NO. 99, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 95, 96 and 97, respectively.

QAVVTQEPSTLVSPGGTTLTCSR**SSSTGAVTTSNYAN**WVQEKPGQAPRGLI

GGTNKRAPWTPARFSGSLLGGKAALTLSGAQPEDEAEYY**CALWYSNLWV**F

GGGKLTTLV

[0311] The amino acid sequence of hVL3 is shown in SEQ ID NO. 100, of which the encoding nucleic acid is shown in SEQ ID NO. 101, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 102, 103 and 97, respectively.

EAVVTQEPSTLVSPGGTTLTCE**SSDGA**VTT**SNYAN**WVQEKPGQAPRGLI

GGTNKEAPWTPARFSGSLLGGKAALTLSGAQPEDEAEYY**CALWYSNLWV**F

GGGKLTTLV

[0312] The amino acid sequence of hVL4 is shown in SEQ ID NO. 104, of which the encoding nucleic acid is shown in SEQ ID NO. 105, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 102, 103 and 97, respectively.

QAVVTQEPSTLVSPGGTTLTCE**SSDGA**VTT**SNYAN**WVQEKPGQAPRGLI

GGTNKEAPWTPARFSGSLLGGKAALTLSGAQPEDEAEYY**CALWYSNLWV**F

GGGKLTTLV

[0313] The amino acid sequence of hVL5 is shown in SEQ ID NO. 106, of which the encoding nucleic acid is shown in

SEQ ID NO. 107, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 95, 96 and 108, respectively.

QAVVTQEPSTLVSPGGTTLTCSR**SSSTGAVTTSNYAN**WVQEKPGQAPRGLI

GGTNKRAPWTPARFSGSLLGGKAALTITGAQAEDEAEYY**CVLWYSNLWV**F

GGGKLTTLV

[0314] The amino acid sequence of hVL6 is shown in SEQ ID NO. 109, of which the encoding nucleic acid is shown in SEQ ID NO. 110, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 95, 111 and 108, respectively.

QAVVTQEPSTLVSPGGTTLTCSR**SSSTGAVTTSNYAN**WVQEKPGQAPRGLI

YGTNKRAPWTPARFSGSLLGGKAALTLSGAQAEDEAEYY**CVLWYSNLWV**F

GGGKLTTLV

[0315] The sequences of the heavy chain variable region of the humanized CD3 antibody is as follows:

TABLE 4

Heavy chain variable region sequences of CD3 humanized antibodies					
SEQ ID NOs: 112-141					
CD3 humanized antibody	HCVR		HCDR1	HCDR2	HCDR3
	Amino acid sequence	Nucleotide sequence	Amino acid Sequence	Amino acid Sequence	Amino acid Sequence
hVH1	112	113	114	115	116
hVH2	117	118	119	115	116
hVH3	120	121	119	115	122
hVH4	123	124	119	115	125
hVH5	126	127	119	115	128
hVH6	129	130	119	115	131
hVH7	132	133	134	135	116
hVH8	136	137	114	115	116
hVH9	138	139	114	115	116
hVH10	140	141	114	115	116

[0316] The amino acid sequence of hVH1 is shown in SEQ ID NO. 112, of which the encoding nucleic acid is shown in SEQ ID NO. 113, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 114, 115 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFT**FNTYAMN**WVRQAPGKLEWVSR

IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLKTEDTAVYYCVR

HGNFGNSYVSWFAYWGQGLTVTVSS

[0317] The amino acid sequence of hVH2 is shown in SEQ ID NO. 117, of which the encoding nucleic acid is shown in SEQ ID NO. 118, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 119, 115 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFT**FSTYAMN**WVRKAPGKLEWVSR

IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLRATEDTAVYYCVR

HGNFGNSYVSWFAYWGQGLTVTVSS

[0318] The amino acid sequence of hVH3 is shown in SEQ ID NO. 120, of which the encoding nucleic acid is shown in SEQ ID NO. 121, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 119, 115 and 122, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**STYAMN**WVRKAPGKGLEWVSR
IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGESYVSWFAYWGQGLTVTVSS

[0319] The amino acid sequence of hVH4 is shown in SEQ ID NO. 123, of which the encoding nucleic acid is shown in SEQ ID NO. 124, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 119, 115 and 125, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**STYAMN**WVRKAPGKGLEWVSR
IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGQSYVSWFAYWGQGLTVTVSS

[0320] The amino acid sequence of hVH5 is shown in SEQ ID NO. 126, of which the encoding nucleic acid is shown in SEQ ID NO. 127, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 119, 115 and 128, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**STYAMN**WVRKAPGKGLEWVSR
IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGLTVTVSS

[0321] The amino acid sequence of hVH6 is shown in SEQ ID NO. 129, of which the encoding nucleic acid is shown in SEQ ID NO. 130, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 119, 115 and 131, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**STYAMN**WVRKAPGKGLEWVSR
IRSKYNNYATYYADSVKDRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGTSYVSWFAYWGQGLTVTVSS

[0322] The amino acid sequence of hVH7 is shown in SEQ ID NO. 132, of which the encoding nucleic acid is shown in SEQ ID NO. 133, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 134, 135 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**SDYAMN**WVRKAPGKGLEWVSR
IRSKYNNYATYYADSVEDRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGNSYVSWFAYWGQGLTVTVSS

[0323] The amino acid sequence of hVH8 is shown in SEQ ID NO. 136, of which the encoding nucleic acid is shown in SEQ ID NO. 137, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 114, 115 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**NTYAMN**WVRKAPGKGLEWVGR
IRSKYNNYATYYADSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCAR
HGNFGNSYVSWFAYWGQGLTVTVSS

[0324] The amino acid sequence of hVH9 is shown in SEQ ID NO. 138, of which the encoding nucleic acid is shown in SEQ ID NO. 139, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 114, 115 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**NTYAMN**WVRKAPGKGLEWVGR
IRSKYNNYATYYADSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVR
HGNFGNSYVSWFAYWGQGLTVTVSS

[0325] The amino acid sequence of hVH10 is shown in SEQ ID NO. 140, of which the encoding nucleic acid is shown in SEQ ID NO. 141, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 114, 115 and 116, respectively.

EVQLVESGGGLVQPGGSLRLSCAASGFTF**NTYAMN**WVRKAPGKGLEWVAR
IRSKYNNYATYYADSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVR
HGNFGNSYVSWFAYWGQGLTVTVSS

[0326] After the light and heavy chain humanized variants have been respectively subjected to full sequence synthesis, are cloned into the eukaryotic expression vector containing the constant region of the light chain of antibody λ or the constant region CH1-CH3 of the heavy chain of human IgG4, co-transfected into HEK293E cells, and cultured at 37° C. under 120 rpm 5% CO₂ for 5 to 6 days, and then the culture supernatant is collected and purified by Protein A chromatography column.

2. ELISA Affinity Assay

[0327] Human CD3 ϵ protein was coated at 4° C. overnight. After blocking with 2% skim milk, different dilutions of CD3 antibody were added to each well and incubated for 1 h; after the secondary antibody was added to the HPR-labeled goat anti-human IgG Fc and color development of TMB solution, the reaction was stopped with concentrated sulfuric acid and the absorbance was read at 450 nm (the results were shown in FIG. 4).

[0328] FIG. 4 shows a combination of anti-CD3 humanized antibodies (including aCD3-hVH8/VL1, aCD3-hVH9/VL1, aCD3-hVH9/VL2, aCD3-hVH9/VL3, aCD3-hVH1/VL5, aCD3-hVH8/VL5, aCD3-hVH9/VL5) binding to human CD3 ϵ protein, the anti-CD3 humanized antibodies bind to CD3 ϵ recombinant protein with high affinity.

3. Jurkat T Cell Affinity Assay

[0329] Jurkat cells in the logarithmic growth phase were blocked with 3% BSA for 30 min, added to 96-well U-plate at 5×10^4 cells per well, centrifuged to discard the supernatant, and added with 50 μ L of gradient diluted antibody (antibody concentration: from 30 μ g/mL, 3-fold dilution for 5 gradients) per well, and incubated at 4° C. for 1 h. After washing off the primary antibody, a 1:300 dilution of the

secondary antibody, Alexa Fluor647-labeled goat anti-human IgG Fc (Jackson ImmunoResearch, 109-606-170) was added and incubated for 45 min at 4° C. After washing, each well was resuspended in 50 µL PBS for FACS (iQue, Intellicyt) assay (results were shown in FIG. 5).

[0330] FIG. 5 shows a combination of anti-CD3 humanized antibodies bound to Jurkat cells, wherein the CD3 humanized antibody hVH9/VL5 (aCD3-hVH9/VL5) binds to Jurkat cells with moderate affinity.

4. Screening a Combination of BCMA Antibody and CD3 Antibody

[0331] BCMA humanized antibodies h38E2, h23F8, h40C6, h38D9, h29A11, h15H10, and h34G2 (Table 5) were obtained by selecting some BCMA clones with higher affinity and performing CDR transplantation with the human germline genes with the highest homology. Referring to Schaefer W et al. (PNAS, 2011), the combination of humanized BCMA antibody and CD3 antibody aCD3-hVH9/VL5 (the amino acid sequence of the light chain variable region is shown in SEQ ID NO. 106, and the amino acid sequence of the heavy chain variable region is shown in SEQ ID NO. 138) was screened: the light chain variable region of the humanized BCMA antibody was cloned into a eukaryotic expression vector containing the κ light chain constant region, and the heavy chain variable region was cloned into a eukaryotic expression vector containing the mutated human IgG4 constant region; the light chain variable region of humanized CD3 antibody was cloned into a eukaryotic expression vector containing λ light chain constant region as well as heavy chain hinge region and constant regions CH2 and CH3, and the heavy chain variable region of CD3 antibody was cloned into a eukaryotic expression vector containing human CH1 constant region. The four plasmids encoding the light and heavy chain genes were mixed in the ratio of 1:1:1:1 and transfected into HEK293E cells. After expression for 5-6 days, the culture supernatant was collected. After one-step purification by Protein A, SEC-HPLC showed monomer content >55% for relative activity evaluation. BCMA-50 (synthesized according to US201213678247) was constructed as a BCMA50-CD3 bispecific antibody with the CD3 humanized antibody as the method described above, or expressing BCMA50×CD3 (BiTE) with reference to the literature.

[0332] The amino acid sequence of the h38E2 light chain VK is shown in SEQ ID NO. 142, of which the encoding nucleic acid is shown in SEQ ID NO. 143, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 144, 77 and 78, respectively.

DIQMTQSPSSLSASVGRVTITCQASQDVDTAVAWYQKPKGKAPKLLIY

WTSTRHTGVPSRFSGSGSGTDFTFTISSLQPEDIAITYFCQQFSSYPLTF

GQGTKVEIK

Nucleic acid sequence

GACATCCAGATGACACAGAGCCCTAGCAGCCTGTCTGCCAGCGTGGGAG

ACAGAGTGACCATCACCTGTCTAGGCCAGCCAGGATGTGGATACAGCCGT

GGCCTGGTATCAGAAGAAGCCTGGCAAGGCCCTAAGCTGTGATCTAC

TGGACCAGCACCAGACACACAGGCGTGCCCTCTAGATTAGCGGCTCTG

GCTCTGGCACCAGACTTCACCTTCACAATCAGCAGCCTGCAGCCTGAGGA

TATCGCTACCTACTTCTGCCAGCAGTTTACGAGCTACCTCTGACATTC

GGCCAGGGCACCAAGGTGGAAATCAAG

[0333] The amino acid sequence of the h38E2 heavy chain VH is shown in SEQ ID NO. 145, of which the encoding nucleic acid is shown in SEQ ID NO. 146, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 71, 147 and 73, respectively.

QVQLVQSGAEVVKPGSSVKVSCKASGYTFTNYWIGVWREAPGQGLEWMGD

IYPGSGDTNYNGKFKQGRVTLTADISSNTAYMELSSLSRSEDVAVYYCARSY

YRNDAASFAYWGQGLTVTVSS

Nucleic acid sequence

CAGGTTTCAGCTGGTTCAGTCTGGCGCCGAAGTGAAGAAACCTGGCAGCAG

CGTGAAGGTGTCTTGCAAGGCTAGCGGCTACACATTCACCAACTACTGGA

TCCGCTGGGTCCGAGAGGCTCCTGGACAAGGACTTGAGTGGATGGGCGAC

ATCTACCTTCAGCAGCGGCGACACAACTACAACGGCAAGTTCCAGGGCAG

AGTGACCTTGACAGCCGACATCAGCAGCAACACCGCCTACATGGAAGTGA

GCAGCCTGAGAAGCGAGGACACCGCGGTGTACTACTGCGCCAGAAGCTAC

TABLE 5

Humanized BCMA antibody variable region sequences										
BCMA humanized antibody	LCVR		LCDR1	LCDR2	LCDR3	HCVR		HCDR1	HCDR2	HCDR3
	Amino acid sequence	Nucleotide sequence	Amino acid sequence	Amino acid sequence	Amino acid sequence	Amino acid sequence	Nucleotide sequence	Amino acid sequence	Amino acid sequence	Amino acid sequence
h38E2	142	143	144	77	78	145	146	71	147	73
h23F8	148	149	17	40	19	150	151	12	50	14
h40C6	152	153	92	23	19	154	155	12	13	14
h38D9	156	157	83	84	19	158	159	12	13	14
h29A11	160	161	17	40	19	162	163	12	13	14
h15H10	164	165	17	40	19	166	167	12	36	37
h34G2	168	169	17	40	19	170	171	65	66	14

h38E2

-continued

TACAGAAACGACGCCCGCAGCTTCGCTTACTGGGACAGGGAACACTGGT

CACCGTGTCTAGT

h23F8

[0334] The amino acid sequence of the h23F8 light chain VK is shown in SEQ ID NO. 148, of which the encoding nucleic acid is shown in SEQ ID NO. 149, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

DIVLTQSPASLAVSPGQRATITC**RASKSVSTSGYSYMH**WYQKKPGQPPKLLIY**LASNLES**GVPARFSGSGSGTDFTLTINPVEANDTANYYC**QHSRELPW****TFGQGTKVEIK**

Nucleic acid sequence

GACATCGTGTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCTGGCCA

GAGAGCCACCATCACCTGTAGAGCCAGCAAGAGCGTGTCCACCAGCGGCT

ACTCTTACATGCACCTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGCTAGCAACCTCGAAAGCGAGTGCCTGTAGATTTTC

TGGCAGCGGCTCTGGCACCAGCTTCACCCTGACAATCAACCCCGTGAAG

CCAACGACACCGCCAACCTACTACTGCCAGCAGCAGAGAGCTGCCCTGG

ACATTGGCCAGGGCACCAAGGTGGAAATCAAG

[0335] The amino acid sequence of the h23F8 heavy chain VH is shown in SEQ ID NO. 150, of which the encoding nucleic acid is shown in SEQ ID NO. 151, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 50 and 14, respectively.

QVQLVQSGSELKKPGASVkvSCKASGY**FTNF**GMNWVREAPGQGLEWMGW**INTYTGDIYADDFKGR**FAFSLDTSASTAYLQISSLKAEDTAVYFCARGE**IYYGYDVG**FVYWGGQTLTVVSS

Nucleic acid sequence

CAGGTTACAGCTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACACATTACCAACTTCGGCA

TGAAGTGGTCCGAGAGGCTCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGCGACAGATCTACGCCGACGACTTCAAGGGCAG

ATTCCGCTTACAGCTGGACACCTCTGCCAGCAGCTTACCTGCAGATCA

GCAGCTGAAGGCCGAGGATACCGCGGTGTAATTCTGTGCCAGAGGCGAG

ATCTACTACGGCTACGATGTGGGCTTCGTCTACTGGGACAGGGAACACT

GGTCACCGTTAGCTCT

h40C6

[0336] The amino acid sequence of the h40C6 light chain VK is shown in SEQ ID NO. 152, of which the encoding nucleic acid is shown in SEQ ID NO. 153, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 92, 23 and 19, respectively.

DIVLTQSPASLAVSPGQRATITC**RASRSVTTSGYSYIH**WYQKKPGQPPKLLIY**LASDLEAG**GVPARFSGSGSGTDFTLTINPVEANDTANYYC**QHSRELPW****TFGQGTKVEIK**

Nucleic acid sequence

GACATCGTGTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCTGGCCA

GAGAGCCACCATCACCTGTAGAGCCAGCAGAAGCGTGACCACCAGCGGCT

ACTCTTACATCCACTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGCCAGCGATCTGGAAGCTGGCGTGCCAGCTAGATTTTC

TGGCAGCGGCTCTGGCACCAGCTTCACCCTGACAATCAACCCCGTGAAG

CCAACGACACCGCCAACCTACTACTGCCAGCAGCAGAGAGCTGCCCTGG

ACATTGGCCAGGGCACCAAGGTGGAAATCAAG

[0337] The amino acid sequence of the h40C6 heavy chain VH is shown in SEQ ID NO. 154, of which the encoding nucleic acid is shown in SEQ ID NO. 155, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

QVQLVQSGSELKKPGASVkvSCKASGY**FTNF**GMNWVREAPGQGLEWMGW**INTYTGDIYADDFKGR**FAFSLDTSASTAYLQISSLKAEDTAVYFCARGE**IYYGYDVG**FVYWGGQTLTVVSS

Nucleic acid sequence

CAGGTTACAGCTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACATCTTACCAACTTCGGCA

TGAAGTGGTCCGAGAGGCTCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGCGAGCAGATCTACGCCGATGACTTCAAAGGCAG

ATTCGCGTTTACGCTGGACACAGCGCCAGCAGCTTACCTGCAGATCA

GCTCTCTGAAGGCCGAGGATACCGCGGTGTAATTCTGTGCCAGAGGCGAG

ATCTACTACGGCTACGACGTGGGCTTTGTGTACTGGGCGCCAGGGAACACT

GGTCACCGTTAGCTCT

h38D9

[0338] The amino acid sequence of the h38D9 light chain VK is shown in SEQ ID NO. 156, of which the encoding nucleic acid is shown in SEQ ID NO. 157, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 83, 84 and 19, respectively.

DIVLTQSPASLAVSPGQRATITC**RASKSVSTSGYSYMH**WYQKKPGQPPKLLIY**LGSDES**GVPARFSGSGSGTDFTLTINPVEANDTANYYC**QHSRELPW****TFGQGTKVEIK**

Nucleic acid sequence

GACATCGTGTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCTGGCCA

GAGAGCCACCATCACCTGTAGAGCCAGCAAGAGCGTGTCCACCAGCGGCT

ACTCTTACATGCACCTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGGTAGCGACCTCGAAAGCGGAGTGCCTGTAGATTTTC

TGGCAGCGGCTCTGGCACCAGCTTCACCCTGACAATCAACCCCGTGAAG

-continued

CCAACGACACCGCCAACTACTACTGCCAGCACAGCAGAGAGCTGCCCTGG

ACATTTGGCCAGGGCACCAGGTGGAATCAAG

[0339] The amino acid sequence of the h38D9 heavy chain VH is shown in SEQ ID NO. 158, of which the encoding nucleic acid is shown in SEQ ID NO. 159, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

QVQLVQSGSELKKPGASVkvSCKASGYIFTNFGMNWVREAPQGLEWMGWINTYTGQEIYADDFKGRFAFSLDTSASTAYLQISSLKAEDTAVYFCARGEIYYGYDVGFFVYWGQGLTLTVSS

Nucleic acid sequence

CAGGTTGAGCTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACATCTTCACCAACTTCGGCA

TGAAGTGGGTCCGAGAGGCTCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGGCGAGCAGATCTACGCCGATGACTTCAAAGGCAG

ATTGCGCTTCAGCCTGGACACCGCGCCAGCACAGCTTACCTGCAGATCA

GCTCTCTGAAGGCCGAGGATACCGCGTGTACTTCTGTGCCAGAGGCGAG

ATCTACTACGGCTACGAGCTGGGCTTTGTGTACTGGGCGCAGGGAACACT

GGTCACCGTTAGCTCT

h29A11

[0340] The amino acid sequence of the h29A11 light chain VK is shown in SEQ ID NO. 160, of which the encoding nucleic acid is shown in SEQ ID NO. 161, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

DIVLTQSPASLAVSPGQRATITCRASKSVSTSGYSYMHWYQKKPGQPPKLLIYLASNLESGVPARFSGSGSDFTLTINPVEANDTANYYCQHSRELPWTFGQGTKVEIK

Nucleic acid sequence

GACATCGTGCTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCCTGGCCA

GAGAGCCACCATCACCTGTAGAGCCAGCAAGAGCGTGTCCACAGCGGCT

ACTCTTACATGCACTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGCTAGCAACCTCGAAAGCGGAGTGCCTGCTAGATTTTC

TGGCAGCGGCTCTGGCACCAGCTTACCCCTGACAATCAACCCCGTGAAG

CCAACGACACCGCCAACTACTACTGCCAGCACAGCAGAGAGCTGCCCTGG

ACATTTGGCCAGGGCACCAGGTGGAATCAAG

[0341] The amino acid sequence of h29A11 heavy chain VH is shown in SEQ ID NO. 162, of which the encoding nucleic acid is shown in SEQ ID NO. 163, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 13 and 14, respectively.

QVQLVQSGSELKKPGASVkvSCKASGYIFTNFGMNWVREAPQGLEWMGWINTYTGQEIYADDFKGRFAFSLDTSASTAYLQISSLKAEDTAVYFCARGEIYYGYDVGFFVYWGQGLTLTVSS

-continued

Nucleic acid sequence

CAGGTTGAGCTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACATCTTCACCAACTTCGGCA

TGAAGTGGGTCCGAGAGGCTCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGGCGAGCAGATCTACGCCGATGACTTCAAAGGCAG

ATTGCGCTTCAGCCTGGACACCGCGCCAGCACAGCTTACCTGCAGATCA

GCTCTCTGAAGGCCGAGGATACCGCGTGTACTTCTGTGCCAGAGGCGAG

ATCTACTACGGCTACGAGCTGGGCTTTGTGTACTGGGCGCAGGGAACACT

GGTCACCGTTAGCTCT

h15H10

[0342] The amino acid sequence of the h15H10 light chain VK is shown in SEQ ID NO. 164, of which the encoding nucleic acid is shown in SEQ ID NO. 165, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

DIVLTQSPASLAVSPGQRATITCRASKSVSTSGYSYMHWYQKKPGQPPKLLIYLASNLESGVPARFSGSGSDFTLTINPVEANDTANYYCQHSRELPWTFGQGTKVEIK

Nucleic acid sequence

GACATCGTGCTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCCTGGCCA

GAGAGCCACCATCACCTGTAGAGCCAGCAAGAGCGTGTCCACAGCGGCT

ACTCTTACATGCACTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGCTAGCAACCTCGAAAGCGGAGTGCCTGCTAGATTTTC

TGGCAGCGGCTCTGGCACCAGCTTACCCCTGACAATCAACCCCGTGAAG

CCAACGACACCGCCAACTACTACTGCCAGCACAGCAGAGAGCTGCCCTGG

ACATTTGGCCAGGGCACCAGGTGGAATCAAG

[0343] The amino acid sequence of the h15H10 heavy chain VH is shown in SEQ ID NO. 166, of which the encoding nucleic acid is shown in SEQ ID NO. 167, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 12, 36 and 37, respectively.

QVQLVQSGSELKKPGASVkvSCKASGYIFTNFGMNWVREAPQGLEWMGWINTYTGDLIYGDDFKGRFAFSLDTSASTAYLQISSLKAEDTAVYFCARGEIYYGYDVGFAVYWGQGLTLTVSS

Nucleic acid sequence

CAGGTTGAGCTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACACATTACCAACTTCGGCA

TGAAGTGGGTCCGAGAGGCTCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGGCGACCTGATCTACGGCGACGACTTCAAAGGCAG

ATTGCGCTTCAGCCTGGACACCTCTGCCAGCACAGCTTACCTGCAGATCA

GCAGCCTGAAGGCCGAGGATACCGCGTGTACTTCTGTGCCAGAGGCGAG

-continued

ATCTACTACGGCTACGATGTGGGCTTCGCCTACTGGGGACAGGGAACACT

GGTCACCGTTAGCTCT

h34G2

[0344] The amino acid sequence of the h34G2 light chain VK is shown in SEQ ID NO. 168, of which the encoding nucleic acid is shown in SEQ ID NO. 169, and the LCDR1, LCDR2, and LCDR3 are shown in SEQ ID NOs. 17, 40 and 19, respectively.

DIVLTQSPASLAVSPGQRATITCRASKSVSTSGYSYMHWYQKKPGQPPL

LIYLASNLESGVPAFESGSGSGTDFLTINPVEANDTANYYCQHSRELPW

TFGQGTKVEIK

Nucleic acid sequence

GACATCGTGTGACACAGAGCCCTGCTTCTCTGGCTGTGTCTCTGGCCA

GAGAGCCACCATCACTGTAGAGCCAGCAAGAGCGTGTCCACCAGCGGCT

ACTCTTACATGCACTGGTATCAGAAGAAGCCCGGCCAGCCTCCTAAGCTG

CTGATCTACCTGGCTAGCAACCTCGAAAGCGGAGTGCCTGCTAGATTTC

TGGCAGCGGCTCTGGCACCAGCTTCAACCTGACAATCAACCCCGTGAAG

CCAACGACACCGCCAACCTACTACTGCCAGCACAGCAGAGAGCTGCCCTGG

ACATTGGCCAGGGCACCAAGGTGGAATCAAG

[0345] The amino acid sequence of the h34G2 heavy chain VH is shown in SEQ ID NO. 170, of which the encoding nucleic acid is shown in SEQ ID NO. 171, and the HCDR1, HCDR2, and HCDR3 are shown in SEQ ID NOs. 65, 66 and 14, respectively.

QVQLVQSGSELKKPGASVKVSKASGYTFTNYGMNWVREAPQGQLEWMCW

INTYTGRIYADDFKGRFAFSLDTSASTAYLQISSLKAEDTAVYFCARGE

IYYGYDVGFVYWGQGLTVTVSS

Nucleic acid sequence

CAGGTTACAGTGGTGCAGTCTGGCAGCGAGCTGAAGAAACCTGGCGCCTC

TGTGAAGGTGTCTGCAAGGCTAGCGGCTACACATTCAACCACTACGGCA

-continued

TGAACTGGGTCCGAGAGGCTCCTGGACAGGGACTCGAATGGATGGGCTGG

ATCAACACCTACACCGGCGAAAGGATCTACGCCGACGACTTCAAGGGCAG

ATTGCGCTTCAGCCTGGACACCTCTGCCAGCAGCTTACCTGCAGATCA

GCAGCCTGAAGGCCGAGGATACCGCGGTACTTCTGTGCCAGAGGCGAG

ATCTACTACGGCTACGATGTGGGCTTCGTCTACTGGGGACAGGGAACACT

GGTCACCGTTAGCTCT

5. T-Cell-Dependent Cellular Cytotoxicity Assay (TDCC)

[0346] Freshly isolated PBMC was mixed with RPMI-8226 cells in the logarithmic growth phase (effector/target cell=10:1), and 50 μ L of gradient diluted antibody (initial concentration 40 μ g/mL, 10-fold dilution for 7 gradients) was added to each well, and cultured at 37° C. in 5% CO₂ for 24 h. After the end of the culture, centrifuging at 1000 rpm for 3 min, the supernatant was taken to detect the release of LDH, transferred 50 μ L of the supernatant to a black ELISA plate, added with 50 μ L/well of LDH detection substrate, 10 min later, stop the reaction and detect (Biotek Synergy HT), and the killing rate was calculated according to the following formula; after discarding the supernatant, the remaining cells in the wells were washed twice with 4% calf blood, blocked with 100 μ g/mL human IgG for 10 min, and T cell early activation marker CD69 and late marker CD25 were added to detect antibodies (CD25-PE, CD4-APC, CD69-FITC, and CD8-APC) and incubating on ice for 20 min. After the incubation, 200 μ L of 4% calf blood was added to each well for washing three times, the supernatant was discarded, 60 μ L/well propidium iodide (PI) solution (1:200 dilution) was added, incubated on ice for 5 min, and detected by flow cytometer (BD FACS celesta). The formula for calculating the killing rate is as follows:

Killing Rate(%) =

$$\frac{\text{Experimental well} - \text{Target cell spontaneous}}{\text{Target cell maximum release} - \text{Target cell spontaneous}} \times 100\%$$

[0347] The killing activity of the bispecific antibody combination is shown in Table 6.

TABLE 6

Killing activity of bispecific antibodies						
Antibodies\Units	SEC-HPLC	Killing	Upregulation of CD69		Upregulation of CD25	
	Monomer		EC ₅₀		EC ₅₀	
			(%)		(%)	
(nM)	%	EC ₅₀	CD8 ⁺ T	CD4 ⁺ T	CD8 ⁺ T	CD4 ⁺ T
h38E2-CD3	56.2	13.64	2.32(69)	7.40(55)	3.65(38)	3.65(29)
h23F8-CD3	58.4	0.81	0.49(59)	0.44(41)	0.79(34)	0.57(25)
h40C6-CD3	57.3	1.12	0.21(44)	0.63(37)	0.49(28)	0.45(25)
h38D9-CD3	55.1	1.04	0.37(52)	0.52(57)	1.75(35)	0.77(29)
h29A11-CD3	56.3	0.69	3.39(58)	0.37(47)	3.85(36)	0.45(33)
h15H10-CD3	67.6	1.83	0.94(62)	3.39(58)	2.64(41)	3.85(36)
h34G2-CD3	64.0	1.02	0.57(55)	0.68(42)	1.28(33)	1.18(27)
BCMA50-CD3	40.9	3.84	6.51(56)	1.02(34)	9.86(24)	7.94(22)
BCMA50 × CD3(BiTE)	N.D.	0.81	0.16(65)	1.90(52)	0.30(39)	0.64(27)
KLH-CD3	85.8	—	—(11)	—(15)	—(8)	—(10)

6. Construction of BCMA $\times$ CD3 $\kappa\lambda$ Humanized Bispecific Antibody

[0348] Clone 29A11 with higher killing efficiency and clone 6G with homologous sequence were selected for humanized sequence optimization. The heavy chain variable region was derived from murine IGHV9-3, and the light chain variable region was derived from murine IGKV3-12. The light and heavy chain CDR grafts were transplanted into human germline genes IGHV7-4-1*02 and IGVK_7_3, respectively. The humanized BCMA antibody sequences were synthesized and designed. Meanwhile, charge variants were introduced in the BCMA antigen arm and the humanized CD3 antibody arm containing the λ light chain ($V_{\kappa_{BCMA}}$: Gln₄₂Lys; VH_{BCMA} : Gln₃₉Glu; $V\lambda_{CD3}$: Gln₄₀Glu; VH_{CD3} : Gln₃₉Lys) to construct a novel BCMA-CD3 $\kappa\lambda$ humanized bispecific antibody with the native IgG configuration (Table 7). The Fc portion of the bispecific antibody adopts the human IgG4 knob-into-hole structure to achieve

heterodimer pairing (Atwell et al. 1997), and through mutations Ser₂₂₈Pro, Leu₂₃₅Glu, and Pro₃₂₉Ala, the hinge region is kept stable and interactions with Fc γ receptors and C1q are reduced.

[0349] Plasmids encoding the corresponding antibody fragments were mixed in a ratio of κ light chain: λ light chain:heavy chain 1:heavy chain 2=2:2:1:1 with 3 mg/mL PEI, then co-transfected into CHO-S cells and cultured in 500 mL CD CHO AGT medium (Gibco #12490-001) at 37° C., 5% CO₂ 150 rpm. 4% CHO Feed C+ supplement (Gibco #A25031-05) was added on the 2nd, 4th and 6th day of transient transfection, respectively. When the cell viability decreased to about 85%, the fermentation broth was harvested, filtered, and purified by Protein A affinity chromatography, and the SEC-HPLC monomer content was higher than 92%; through Capto S ImpAct ion exchange chromatography, 50-300 mM NaCl/phosphate pH6.0 gradient elution and combined elution peak, the monomer content was further increased to more than 98-99% (Table 8).

TABLE 7

BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibodies		
Bispecific antibody	BCMA antibody	CD3 antibody
BCMA $\times$ CD3 $\kappa\lambda$ 003, introduction of charge variants in the BCMA and CD3 antigen arms: $V_{\kappa_{BCMA}}$: Gln ₄₂ Lys; VH_{BCMA} : Gln ₃₉ Glu; $V\lambda_{CD3}$: Gln ₄₀ Glu; VH_{CD3} : Gln ₃₉ Lys	Light chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 172, and the encoding nucleic acid thereof is shown in SEQ ID NO. 173 DIVLTQSPASLAVSPGQRATIT CRASKSVSTSGYSYMHWYQK KPGQPPKLLIYLASNLESQVP ARFSGSGSGTDFLTINPVEAE DTANYCYCQHSRELPWTFGQG TKVEIKRTVAAPSVFIFPPSDE QLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESV TEQDSKDSSTYSSTLTLSKA DYEKKHVVYACEVTHQGLSSP VTKSFNRGEC Heavy chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 174, and the encoding nucleic acid thereof is shown in SEQ ID NO. 175 QVQLVQSGSELKPGASVKV SCKASGYIFTNFGMNWVREA PGQGLEWMGWINTYTGEQIY ADGFTGRFVFSLDTSASTAYL QISSLKAEDTAVYFCARGEIY YGVDVGFVYWGQGLVTVSS ASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYS LSSVTVTPSSSLGKTYTCNV DHKPSNTKVDKRVESKYGPP CPPCPAPEFEGGPSVFLFPPKP KDTLMISRTPETCVVVDVSDQ EDPEVQFNWYVDGVEVHNA KTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKG LASSIEKTSKAKGQPREPQVC TLPPSQEEMTKNQVSLSCAVK GFYPSDIAVEWESNGQPENNY KTPPVLDSDGSFFLVSKLTV DKSRWQEGNVFSCSVMHREAL HNHYTQKSLSLGK	Light chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 180, and the encoding nucleic acid thereof is shown in SEQ ID NO. 181 QAVVTQEPSTLVSPGGTIVTLTC RSSTGAVTTSTNYANWVQEKPG QAPRGLIGGTNKRAPWTPARF SGSLLGKKAALTITGAQAEDE AEYYCVLWYSLNWVFGGQTK LTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPSPKQ SNNKYAASSYLSLTPEQWKSH RSYSCQVTHEGSTVEKTVAPT ECS Heavy chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 182, and the encoding nucleic acid thereof is shown in SEQ ID NO. 183 EVOLVESGGGLVQPGGSLRLS CAASGFTFTNYAMNWRKAP GKGLEWVGRIRSKYNNYATYY ADSVKDRFTISRDDSKNSLYLQ MNSLKTEDTAVYCVVRHGNF GNSYVSWFAYWGQGLVTVS SASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWNS GALTSGVHTFPAVLQSSGLYS SSVTVTPSSSLGKTYTCNV HKPSNTKVDKRVESKYGPPCP PCPAPEFEGGPSVFLFPPKPKD TLMISRTPETCVVVDVSDQEDP EVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQD WLNKKEYKCKVSNKGLASSIE KTISKAKGQPREPQVYTLPPCQ EEMTKNQVSLWCLVKGFPYPSD IAVEWESNGQPENNYKTPPV LSDGSFFLVSKLTVDKSRWQ EGNVFSCSVMHREALHNHYTQ KSLSLSLGK

TABLE 7-continued

BCMA x CD3 $\kappa\lambda$ bispecific antibodies		
Bispecific antibody	BCMA antibody	CD3 antibody
BCMA x CD3 $\kappa\lambda 004$ introduction of charge variants in the BCMA and CD3 antigen arms: V_{BCMA} : Gln ₄₂ Lys; V_{HBCMA} : Gln ₃₉ Glu; $V_{\lambda CD3}$: Gln ₄₀ Glu; V_{HCD3} : Gln ₃₉ Lys	Light chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 176, and the encoding nucleic acid thereof is shown in SEQ ID NO. 177 DIVLTQSPASLAVSPGQRATIT CRASKSVTTSYGYSYIHWYQK KPGQPPKLLIYLASDLEAGVP ARFSGSGSGTDFTLTINPVEAE DTANYYCQHSRELPWTFGQG TKVEIKRTVAAPSVFIPPPSDE QLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESV TEQDSKDSYSLSTLTLSKA DYEKKHVYACEVTHQGLSSP VTKSFNRGEC Heavy chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 174, and the encoding nucleic acid thereof is shown in SEQ ID NO. 175 QVQLVQSGSEELKKPGASVKV SCKASGYIFTNFGMNWVREA PGQGLEWMGWINTYTGEQIY ADGFTGRFVFLDTSASTAYL QISSLKAEDTAVYFCARGEIY YGYDVGFIYWGQGLVTVSS ASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYS LSSVTVTPSSSLGKTYTCNV DHKPSNTKVDKRVESEKYPGP CPPCPAPFEGGSPVFLPPPKP KDTLMISRTPETVTCVVDVSQ EDPEVQFNWYVDGVEVHNA KTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKGL LASSIEKTIISKAKGPQPREPQVC TLPPSQEEMTKNQVSLTSCAVK GFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLVSKLTV DKSRWQEGNVFSCSV MHEALHNHYTQKSLSLGLGK	Light chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 180, and the encoding nucleic acid thereof is shown in SEQ ID NO. 181 QAVVTQEPSTLVSPGGTVTLTC RSSTGAVTTSNYANWVQEKPG QAPRGLIGGTNKRAPWTPARF SGSLLGGKAALTITGAQAED AEYYCVLWYSLWVFGGGTK LTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPPSKQ SNNKYAASSYLSLTPEQWKSH RSYSCQVTHEGSTVEKTVAPT ECS Heavy chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 182, and the encoding nucleic acid thereof is shown in SEQ ID NO. 183 EVQLVESGGGLVQPGGSLRLS CAASGFTFTNTYAMNWVRKAP GKGLEWVGRIRSKYNNYATYY ADSVKDRFTISRDDSKNSLYLQ MNSLKTEDTAVYYCVRHGNF GNSYVSWFAYWGQGLVTVSS SASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWN GALTSGVHTFPAVLQSSGLYS SSVTVTPSSSLGKTYTCNV HKPSNTKVDKRVESEKYPGP PCPAPFEGGSPVFLPPPKPD TLMISRTPETVTCVVDVSQEDP EVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLASSIE KTIISKAKGPQPREPQVYTLPPCQ EEMTKNQVSLWCLVKGFYPSD IAVEWESNGQPENNYKTTTPV LDSGDSFFLYSKLTVDKSRWQ EGNVFSCSVMEALHNHYTQ KSLSLSLGK
BCMA x CD3 $\kappa\lambda 005$ introduction of charge variants in the BCMA and CD3 antigen arms: V_{BCMA} : Gln ₄₂ Lys; V_{HBCMA} : Gln ₃₉ Glu; $V_{\lambda CD3}$: Gln ₄₀ Glu; V_{HCD3} : Gln ₃₉ Lys	Light chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 172, and the encoding nucleic acid thereof is shown in SEQ ID NO. 173 DIVLTQSPASLAVSPGQRATIT CRASKSVTTSYGYSYIHWYQK KPGQPPKLLIYLASNLESGVP ARFSGSGSGTDFTLTINPVEAE DTANYYCQHSRELPWTFGQG TKVEIKRTVAAPSVFIPPPSDE QLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESV TEQDSKDSYSLSTLTLSKA DYEKKHVYACEVTHQGLSSP VTKSFNRGEC Heavy chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 178, and the encoding nucleic acid thereof is shown in SEQ ID NO. 179 QVQLVQSGSEELKKPGASVKV SCKASGYIFTNFGMNWVREA PGQGLEWMGWINTYTGEQIY ADGFTGRFVFLDTSVSTAYL QISSLKAEDTAVYFCARGEIY YGYDVGFIYWGQGLVTVSS	Light chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 180, and the encoding nucleic acid thereof is shown in SEQ ID NO. 181 QAVVTQEPSTLVSPGGTVTLTC RSSTGAVTTSNYANWVQEKPG QAPRGLIGGTNKRAPWTPARF SGSLLGGKAALTITGAQAED AEYYCVLWYSLWVFGGGTK LTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPPSKQ SNNKYAASSYLSLTPEQWKSH RSYSCQVTHEGSTVEKTVAPT ECS Heavy chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 182, and the encoding nucleic acid thereof is shown in SEQ ID NO. 183 EVQLVESGGGLVQPGGSLRLS CAASGFTFTNTYAMNWVRKAP GKGLEWVGRIRSKYNNYATYY ADSVKDRFTISRDDSKNSLYLQ MNSLKTEDTAVYYCVRHGNF GNSYVSWFAYWGQGLVTVSS

TABLE 7-continued

BCMA × CD3 κλ bispecific antibodies		
Bispecific antibody	BCMA antibody	CD3 antibody
	ASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYS LSSVVTVPSSSLGKTKYTCNV DHKPSNTKVDKRVESKYGPP CPPCPAPEFEGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSQ EDPEVQFNWYVDGVEVHNA KTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKG LASSIEKTIISKAKGQPREPQVC TLPPSQEEMTKNQVSLSCAVK GFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLVSKLTV DKSRWQEGNVFSCVMHEAL HNHYTQKSLSLSLGK	SASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWNS GALTSGVHTFPAVLQSSGLYS SSVVTVPSSSLGKTKYTCNV HKPSNTKVDKRVESKYGPPCP PCPAPEFEGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSQEDP EVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLASSIE KTIISKAKGQPREPQVYTLPPCQ EEMTKNQVSLWCLVKGFYPSD IAVEWESNGQPENNYKTTTPV LDSGDSFFLYSKLTVDKSRWQ EGNVFSCVMHEALHNHYTQ KSLSLSLGK
BCMA × CD3 κλ006	Light chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 176, and the encoding nucleic acid thereof is shown in SEQ ID NO. 177	Light chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 180, and the encoding nucleic acid thereof is shown in SEQ ID NO. 181
introduction of charge variants in the BCMA and CD3 antigen arms:	DIVLTQSPASLAVSPGQRATIT CRASKSVTTSGSYIHQYQK KPGQPPKLLIYLASDLEAGVP ARFSGSGSGTDFTLTINPVEAE DTANYYCQHSRELPWTFGQG TKVEIKRTVAAPSVFIFPPSDE QLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESV TEQDSKDSITYSLSSLTLSKA DYEKKHVVACEVTHQGLSSP VTKSFNRGEC	QAVVTQEPSLTVSPGGTVLTLC RSSTGAVTTSNYANWVQEKPG QAPRGLIGGTNKRAPWTPARF SGSLGGKAALTITGAQAEDE AEYYCVLWYSNLWVFGGGTK LTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVTV AWKADSSPVKAGVETTPSKV SNNKYAASSYLSLTPEQWKSH RSYSCQVTHEGSTVEKTVAPT ECS
V _{BCMA} : Gln ₄₂ Lys; VH _{BCMA} : Gln ₃₉ Glu; V _{CD3} : Gln ₄₀ Glu; VH _{CD3} : Gln ₃₉ Lys	Heavy chain 1: the amino acid sequence thereof is shown in SEQ ID NO. 178, and the encoding nucleic acid thereof is shown in SEQ ID NO. 179 QVQLVQSGSELKKPGASVKV SCKASGYIFTNFGMNWVREA PGQGLEWMGWINTYTGQIY ADGFTGRFVFLDTSVSTAYL QISSLKAEADTAVYFCARGEIY YGYDVG FVYWGQGLTVTVSS ASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYS LSSVVTVPSSSLGKTKYTCNV DHKPSNTKVDKRVESKYGPP CPPCPAPEFEGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSQ EDPEVQFNWYVDGVEVHNA KTKPREEQFNSTYRVVSVLTV LHQDWLNGKEYKCKVSNKG LASSIEKTIISKAKGQPREPQVC TLPPSQEEMTKNQVSLSCAVK GFYPSDIAVEWESNGQPENNY KTTTPVLDSDGSFFLVSKLTV DKSRWQEGNVFSCVMHEAL HNHYTQKSLSLSLGK	Heavy chain 2: the amino acid sequence thereof is shown in SEQ ID NO. 182, and the encoding nucleic acid thereof is shown in SEQ ID NO. 183 EVQLVESGGGLVQPGGSLRLS CAASGFTFTNTYAMNWVRKAP GKGLEWVGRIRSKYNNYATYY ADSVKDRFTISRDDSKNSLYLQ MNSLKTEDTAVYVCVRHGNF GNSYVSWFAYWGQGLTVTVS SASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWNS GALTSGVHTFPAVLQSSGLYS SSVVTVPSSSLGKTKYTCNV HKPSNTKVDKRVESKYGPPCP PCPAPEFEGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSQEDP EVQFNWYVDGVEVHNAKTKP REEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLASSIE KTIISKAKGQPREPQVYTLPPCQ EEMTKNQVSLWCLVKGFYPSD IAVEWESNGQPENNYKTTTPV LDSGDSFFLYSKLTVDKSRWQ EGNVFSCVMHEALHNHYTQ KSLSLSLGK

TABLE 8

Purity of BCMA × CD3 κλ bispecific antibody						
HPLC	HPLC-SEC (after Protein A)			HPLC-SEC (after cation exchange chromatography)		
	Mer	Monomer	Fragment	Mer	Monomer	Fragment
BCMA × CD3 κλ003	2.3%	94.4%	3.3%	1.8%	98.2%	0.0%
BCMA × CD3 κλ004	1.4%	95.2%	3.4%	1.0%	99.0%	0.0%
BCMA × CD3 κλ005	2.0%	91.9%	6.2%	1.6%	98.4%	0.0%
BCMA × CD3 κλ006	0.5%	96.4%	3.1%	0.6%	99.4%	0.0%

Example 4. Binding Activity of BCMA×CD3 κλ Bispecific Antibody

[0350] The affinity of the bispecific antibody BCMA anti-gen arm was determined by measuring binding to BCMA recombinant protein, overexpressing stably transfected cells, or BCMA+ tumor cells, respectively, and the affinity of the bispecific antibody CD3 arm was determined by measuring binding to CD3 recombinant antigen, Jurkat cells, or freshly isolated peripheral blood T cells.

1. Affinity Assay of BCMA×CD3 κλ Bispecific Antibody to Antigen

[0351] 10 μg/mL human or cynomolgus monkey BCMA or CD3εγ recombinant antigen was bound to a CM5 chip (GE healthcare) by amino coupling, controlling the amount of antigen bound to approximately 200 RU. After the baseline plateau, gradient-diluted antibodies (from 10 μg/mL, 2-fold dilutions for 7 gradients) flowed through the chip at a flow rate of 30 μL/min for a binding time of 350 seconds and a dissociation time of 600 seconds. Kinetic constants were fitted using Biacore T200 evaluation soft-ware with a 1:1 binding model. The results of the affinity assay are shown in Table 9.

TABLE 9

Affinity of BCMA × CD3 κλ bispecific antibodies to BCMA and CD3 recombinant antigens				
KD (nM)	Human BCMA	Cynomolgus monkey BCMA	Human CD3εγ	Cynomolgus monkey CD3εγ
BCMA × CD3 κλ003	1.3	0.16	31	43
BCMA × CD3 κλ004	1.0	0.12	32	43
BCMA × CD3 κλ005	1.1	0.09	28	20
BCMA × CD3 κλ006	1.2	0.10	27	33

2. Binding of BCMA×CD3 κλ Bispecific Antibody to CMA Stably Transected Cells

[0352] CHO-human BCMA and CHO-cynomolgus mon-key BCMA stably transfected cells in the logarithmic growth phase prepared in Example 1 were adjusted to 5×10⁵ cells/ml with 4% calf serum (Hyclone, SH30626.06), 100 μl/well cell suspension was added to a 96-well U-plate, centrifuged at 300 g for 5 min, the supernatant was discarded, and 100 μL of gradient diluted antibody (initial concentration 1800 nM, 3-fold dilution for 10 gradients) was added to each well

and incubated at 4° C. for 60 min. 50 μL/well Alexa Fluro647 labeled goat anti-human IgG Fc (1:300 dilution) was added to the secondary antibody, incubated on ice for 20 min, 50 μL/well PI solution (1:300) was added after washing once, incubated for 5 min, and flow cytometer was used to perform detection. FIG. 6 and Table 10 show that the BCMA×CD3 κλ bispecific antibody binds to human, cyno-molgus monkey BCMA stably transfected cells with high affinity.

TABLE 10

Binding of BCMA × CD3 κλ bispecific antibodies to BCMA stably transfected cells		
EC ₅₀ (nM)	Human BCMA-CHO	Cynomolgus monkey BCMA-CHO
BCMA × CD3 κλ003	16	3
BCMA × CD3 κλ004	22	3
BCMA × CD3 κλ005	24	3
BCMA × CD3 κλ006	16	3
KLH × CD3	—	—

3. Binding of BCMA×CD3 xi Bispecific Antibody to BCMA+ Tumor Cells

[0353] NCI-H929 and RPMI-8226 cells in the logarithmic growth phase were added with 200 μg/mL murine IgG (Jackson ImmunoResearch, 115-005-03), blocked with an ice bath for 30 min, adjusted cells to 5×10⁵ cells/mL with 4% calf serum, 100 μL/well was added to 96-well U-plate, centrifuged at 300 g for 5 min, the supernatant was dis-carded, and 100 μL of gradient diluted antibody (initial concentration 1800 nM, 3-fold dilution for 10 gradients) was added to each well and incubated at 4° C. for 60 min. The primary antibody was washed off, 50 μL/well Alexa Fluro647 labeled goat anti-human IgG Fc (1:300 dilution) was added, incubated on ice for 20 min, 50 μL/well PI was added after washing once, incubated for 5 min, and flow cytometer was used to perform detection.

[0354] The results are shown in FIG. 7 and Table 11. the BCMA×CD3 κλ bispecific antibody binds to BCMA+ tumor cells NCI-H929 and RPMI-8226 with high affinity.

TABLE 11

Binding of BCMA × CD3 κλ bispecific antibodies to BCMA+ tumor cells		
EC ₅₀ (nM)	NCI-H929	RPMI-8226
BCMA × CD3 κλ003	34	34
BCMA × CD3 κλ004	36	39

TABLE 11-continued

Binding of BCMA × CD3 κλ bispecific antibodies to BCMA+ tumor cells		
EC ₅₀ (nM)	NCI-H929	RPMI-8226
BCMA × CD3 κλ005	33	38
BCMA × CD3 κλ006	32	19
KLH × CD3	—	—

4. Binding of BCMA×CD3 κλ Bispecific Antibody to Jurkat Cells

[0355] Jurkat cells in the logarithmic growth phase were added with 200 μg/mL murine IgG (Jackson ImmunoResearch, 115-005-03) and incubated on ice for 30 min. Adjusted cells to 5×10⁵ cells/mL with 4% calf serum, 100 μL/well was added to 96-well U-plate, centrifuged at 300 g to remove the supernatant, and 100 μL of gradient diluted antibody (initial concentration 1800 nM, 3-fold dilution for 10 gradients) was added to each well and incubated at 4° C. for 60 min. 50 μL/well Alexa Fluro647 labeled goat anti-human IgG Fc (1:300 dilution) was added to the secondary antibody, incubated on ice for 20 min, 50 μL/well PI was added after washing once, incubated for 5 min, and flow cytometer (BD C6) was used to perform detection. The results are shown in FIG. 8 and Table 12. The BCMA×CD3 κλ bispecific antibody binds to human leukemia T cell line Jurkat cells with moderate affinity.

TABLE 12

Binding of BCMA × CD3 κλ bispecific antibodies to Jurkat cells	
EC ₅₀ (nM)	Jurkat
BCMA × CD3 κλ003	66
BCMA × CD3 κλ004	61
BCMA × CD3 κλ005	51
BCMA × CD3 κλ006	56
KLH × CD3	76

5. Binding of BCMA×CD3 κλ Bispecific Antibody to Peripheral Blood T Cells

[0356] Fresh human or cynomolgus monkey peripheral blood was taken and PBMC was isolated by Ficoll-Paque Plu (GE, 17-1440-03). PBMC was adjusted to 5×10⁵ cells/mL with 4% calf serum (Hyclone, SH30626.06), 100 μL/well was added to a 96-well U-plate, centrifuged to remove the supernatant and 100 μL of gradient diluted antibody (initial concentration 1800 nM, 3-fold dilution, 10 gradients) was added to each well and incubated at 4° C. for 60 min. 50 μL/well Alexa Fluro647 labeled goat anti-human IgG Fc (1:300 dilution) was added to the secondary antibody, ice bath for 20 min, 50 μL/well PI was added after washing once, incubated for 5 min, and flow cytometer (BD C6) was used to perform detection. The control antibody REGN5458 was synthesized and prepared according to US20200024356.

[0357] As shown in FIG. 9 and Table 13, the BCMA×CD3 κλ bispecific antibody recognizes human peripheral blood CD4+ T and CD8+ T cells with an affinity of about 60-97 nM, which is weaker than the binding force of the BCMA

antigen arm to the BCMA receptor, so that the bispecific antibody preferentially enriches into tumor cells.

TABLE 13

Binding of BCMA × CD3 κλ bispecific antibodies to peripheral blood T cells		
EC ₅₀ (nM)	CD4+ T cell	CD8+ T cell
BCMA × CD3 κλ003	87	97
BCMA × CD3 κλ004	77	84
BCMA × CD3 κλ005	66	60
BCMA × CD3 κλ006	91	93
KLH × CD3	57	62

Example 5. BCMA×CD3 κλ Bispecific Antibody-Mediated TDCC

[0358] Freshly isolated PBMC was mixed with target cells, NCI-H929 and RPMI-8226 cells, in the logarithmic growth phase, respectively, with effect/target cell=8:1, 50 μL of gradient diluted antibody (antibody concentration from 66.7 nM, 10-fold dilution for 7 gradients) was added to each well and cultured at 37° C. in 5% CO₂ for 24 h. After the end of the culture, 50 μL of the supernatant was transferred to a new black ELISA plate, 50 μL/well of LDH detection substrate was added, the reaction was stopped after 10 min and LDH release was detected; the remaining cells in the wells were washed twice with 4% calf blood, incubated with 100 μg/mL human IgG for 10 min, and T cell were added to activate and detect antibodies (CD25-PE, CD4-APC, CD69-FITC, and CD8-APC) and incubating on ice for 20 min. The supernatant was washed and discarded, 60 μL/well PI was added, incubated on ice for 5 min, and detected by flow cytometer. FIGS. 10A and 10B show the killing of NCI-H929 cells and activation of T cells by the BCMA×CD3 κλ bispecific antibody, respectively. FIGS. 11A and 11B show the killing of RPMI-8226 cells and activation of T cells by the BCMA×CD3 κλ bispecific antibody, respectively. For tumor cells, NCI-H929 and RPMI-8226 with different expression levels of BCMA, the BCMA×CD3 κλ bispecific antibody can mediate effective killing of T cells, and the killing activity is equivalent to that of the control antibody.

Example 6. Activation of the T Cell Activation Pathway by the BCMA×CD3 κλ Bispecific Antibody

[0359] The luciferase reporter plasmid pGL4.30[luc2P/NFAT-RE/Hygro] (Promega, E8481) was transfected into Jurkat cells and selected by adding 200 μg/ml hygromycin B to obtain Jurkat-NFAT luc stably transfected reporter cells. Jurkat-NFAT-luc reporter cells in the logarithmic growth phase and target cells RPMI-8226 were taken, centrifuged to discard the supernatant, and resuspended to 2×10⁶ cells/ml. 50 μL/well of the target cells were inoculated into a 96-well plate, the supernatant was discarded by centrifugation at 300 g for 5 min, and 50 μL/well of the Jurkat-NFAT-luc reporter cells were inoculated into a 96-well plate, and 50 μL of gradient diluted BCMA×CD3 κλ bispecific antibody or control antibody KLH×CD3 (initial concentration 20 μg/ml, 10-fold dilution for 10 gradients) was added to each well, and cultured at 37° C. for 6 h in 5% CO₂. After the end of the culture, 100 μL of detection reagent was added to each well according to ONE-Glo Luciferase Assay System

instructions, allowed to stand at room temperature for 3 min, and detected on a microplate reader (Biotek Synergy HT). The detection results are shown in FIG. 12. The BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody can activate the NFAT signaling pathway of T cells when RPMI-8226 tumor cells are used as target cells; in the absence of target cells, the NFAT signaling pathway is not activated.

Example 7. Non-Specific Activation of PBMC by BCMA $\times$ CD3 $\kappa\lambda$ Bispecific Antibody

[0360] Freshly isolated PBMC was taken, and 50 μ L of gradient diluted antibody (antibody concentration from 66.7 nM, 10-fold dilution for 7 gradients) was added to each well and cultured at 37° C. in 5% CO₂ for 24 h. After the end of the culture, 50 μ L of the supernatant was transferred to a new black ELISA plate, 50 μ L/well of LDH detection substrate was added, the reaction was stopped after 10 min and LDH release was detected; the remaining cells in the wells were washed twice with 4% calf blood, incubated with 100 μ g/mL human IgG for 10 min, and T cell were added to activate and detect antibodies (CD25-PE, CD4-APC, CD69-FITC, and CD8-APC) and incubating on ice for 20 min. The supernatant was washed and discarded, 60 μ L/well PI was added, incubated on ice for 5 min, and detected by flow cytometer. The results are shown in FIG. 13, in the absence of target cells, the BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody had no activation of peripheral blood T cells, comparable to the negative control KLH $\times$ CD3.

Example 8. Binding of BCMA $\times$ CD3 $\kappa\lambda$ Bispecific Antibodies to Fc Receptors

[0361] 50 Mg/ml His-Tag antibody was amino-coupled to CM5 chip to capture Fc γ RI, Fc γ RIIA_{H131}, and Fc γ RIIIA_{V158} recombinant proteins with His6 tag, respectively, with capture time of 40 s and flow rate of 10 μ L/min, after baseline stable, gradient diluted antibody (initial concentration 37.5 μ g/mL, 2-fold dilution) flowed through the chip with a flow rate of 30 μ L/min, binding time of 120 s, dissociation time of 200 s, and affinity constant was obtained by fitting with Biacore evaluation software. The results are shown in FIG. 14. The BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody did not bind to activated receptors Fc γ RI, FcRIIA_{H131}, and Fc γ RIIIA_{V158}; wild-type IgG4 and control antibodies bound to Fc γ RI, Fc γ RIIA_{H131} or Fc γ RIIIA_{V158}, respectively.

Example 9. Blocking of BCMA Receptor Binding to a Lgand by BCMA $\times$ CD3 I Bispecific Antibody

[0362] 1 μ g/mL human BCMA recombinant antigen was prepared with pH 9.6 carbonic acid buffer, added to a maxisorp96-well plate, and coated overnight at 4° C. After blocking with 5% skim milk/PBS for 1 h at room temperature, 50 μ L antibody per well (4-fold dilution from 30 μ g/mL for a total of 10 gradients) was added and incubated for 30 min at room temperature. 50 μ L of biotinylated human APRIL recombinant protein (Novoprotein, CU89) at a concentration of 240 ng/mL was added and incubation was continued for 30 min. After 3 washes with 0.05% PBST (PBS+0.05% Tween-20) and PBS respectively, 100 μ L of SA-HRP (diluted 1:8000) was added to each well and incubated at room temperature for 45 min. Washing 3 times with 0.05% PBST and PBS respectively, 100 μ L TMB color-developing solution was added into each well, after 10 min, 50 μ L concentrated sulfuric acid was added into each well to stop the reaction, read OD450 nm by a microplate reader, and perform three-parameter fitting curve using Graphpad software. The results are shown in FIG. 15. The BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody effectively blocked the binding of BCMA to the receptor APRIL.

Example 10. Immune Reconstruction of Subcutaneous NCI-H929 Transplanted Tumor Model in Mice

[0363] Six to eight-week-old female B-NGD mice (Bio-Tech Co. Ltd.) were subcutaneously inoculated with 2×10^6 NCI-H929 cells (mixed with Matrigel at 1:1). When the tumors grew to 60 mm³, they were randomly divided into administration group 3.0 mg/kg, administration group 0.6 mg/kg, administration group 0.12 mg/kg and negative control group KLH $\times$ CD3 3 mg/kg. Each mouse received 1×10^7 PBMC cells via tail vein injection. Three days later, the first dose was administered to the mouse once every 5 days for a total of 2 doses. The tumor volume and body weight of the mice were monitored every 2 days. At the end of the experiment, the mice were killed by cervical dislocation, and the tumors were collected and weighed. The results are shown in FIG. 16. The in vivo efficacy of BCMA $\times$ CD3 $\kappa\lambda$ bispecific antibody was dose-related. The tumor inhibition rates of low-dose, medium-dose, and high-dose were 55%, 95%, and 108% (BCMA $\times$ CD3 $\kappa\lambda$.005) and 46%, 94%, and 108% (BCMA $\times$ CD3 $\kappa\lambda$.006), respectively. The effect was significantly better than that of the control antibody. The tumor-bearing mice tolerated the above doses well, without weight loss and other adverse reactions.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 185

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<211> LENGTH: 290

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

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Leu Leu His Ala Cys Ile Pro Cys Gln Leu Arg Cys Ser Ser Asn Thr
20 25 30

-continued

Pro	Pro	Leu	Thr	Cys	Gln	Arg	Tyr	Cys	Asn	Ala	Ser	Val	Thr	Asn	Ser
		35					40					45			
Val	Lys	Gly	Thr	Asn	Ala	Ser	Gly	Ala	Glu	Pro	Arg	Gly	Pro	Thr	Ile
	50					55					60				
Lys	Pro	Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly
65					70				75					80	
Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile
			85					90					95		
Ser	Leu	Ser	Pro	Ile	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	Glu	Asp
			100					105					110		
Asp	Pro	Asp	Val	Gln	Ile	Ser	Trp	Phe	Val	Asn	Asn	Val	Glu	Val	His
	115						120					125			
Thr	Ala	Gln	Thr	Gln	Thr	His	Arg	Glu	Asp	Tyr	Asn	Ser	Thr	Leu	Arg
	130					135					140				
Val	Val	Ser	Ala	Leu	Pro	Ile	Gln	His	Gln	Asp	Trp	Met	Ser	Gly	Lys
145					150					155					160
Glu	Phe	Lys	Cys	Lys	Val	Asn	Asn	Lys	Asp	Leu	Pro	Ala	Pro	Ile	Glu
			165					170						175	
Arg	Thr	Ile	Ser	Lys	Pro	Lys	Gly	Ser	Val	Arg	Ala	Pro	Gln	Val	Tyr
			180					185					190		
Val	Leu	Pro	Pro	Pro	Glu	Glu	Glu	Met	Thr	Lys	Lys	Gln	Val	Thr	Leu
	195						200					205			
Thr	Cys	Met	Val	Thr	Asp	Phe	Met	Pro	Glu	Asp	Ile	Tyr	Val	Glu	Trp
	210					215					220				
Thr	Asn	Asn	Gly	Lys	Thr	Glu	Leu	Asn	Tyr	Lys	Asn	Thr	Glu	Pro	Val
225					230					235					240
Leu	Asp	Ser	Asp	Gly	Ser	Tyr	Phe	Met	Tyr	Ser	Lys	Leu	Arg	Val	Glu
			245						250				255		
Lys	Lys	Asn	Trp	Val	Glu	Arg	Asn	Ser	Tyr	Ser	Cys	Ser	Val	Val	His
			260					265					270		
Glu	Gly	Leu	His	Asn	His	His	Thr	Thr	Lys	Ser	Phe	Ser	Arg	Thr	Pro
	275						280					285			
Gly	Lys														
	290														

<210> SEQ ID NO 2

<211> LENGTH: 290

<212> TYPE: PRT

<213> ORGANISM: Macaca fascicularis

<400> SEQUENCE: 2

Met	Leu	Gln	Met	Ala	Arg	Gln	Cys	Ser	Gln	Asn	Glu	Tyr	Phe	Asp	Ser
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Leu	Leu	His	Asp	Cys	Lys	Pro	Cys	Gln	Leu	Arg	Cys	Ser	Ser	Thr	Pro
		20					25					30			
Pro	Leu	Thr	Cys	Gln	Arg	Tyr	Cys	Asn	Ala	Ser	Met	Thr	Asn	Ser	Val
	35					40					45				
Lys	Gly	Met	Asn	Ala	Ala	Ser	Gly	Ala	Glu	Pro	Arg	Gly	Pro	Thr	Ile
	50					55					60				
Lys	Pro	Cys	Pro	Pro	Cys	Lys	Cys	Pro	Ala	Pro	Asn	Leu	Leu	Gly	Gly
65					70				75					80	
Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Ile	Lys	Asp	Val	Leu	Met	Ile
			85					90					95		

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Ser Leu Ser Pro Ile Val Thr Cys Val Val Val Asp Val Ser Glu Asp
 100 105 110
 Asp Pro Asp Val Gln Ile Ser Trp Phe Val Asn Asn Val Glu Val His
 115 120 125
 Thr Ala Gln Thr Gln Thr His Arg Glu Asp Tyr Asn Ser Thr Leu Arg
 130 135 140
 Val Val Ser Ala Leu Pro Ile Gln His Gln Asp Trp Met Ser Gly Lys
 145 150 155 160
 Glu Phe Lys Cys Lys Val Asn Asn Lys Asp Leu Pro Ala Pro Ile Glu
 165 170 175
 Arg Thr Ile Ser Lys Pro Lys Gly Ser Val Arg Ala Pro Gln Val Tyr
 180 185 190
 Val Leu Pro Pro Pro Glu Glu Glu Met Thr Lys Lys Gln Val Thr Leu
 195 200 205
 Thr Cys Met Val Thr Asp Phe Met Pro Glu Asp Ile Tyr Val Glu Trp
 210 215 220
 Thr Asn Asn Gly Lys Thr Glu Leu Asn Tyr Lys Asn Thr Glu Pro Val
 225 230 235 240
 Leu Asp Ser Asp Gly Ser Tyr Phe Met Tyr Ser Lys Leu Arg Val Glu
 245 250 255
 Lys Lys Asn Trp Val Glu Arg Asn Ser Tyr Ser Cys Ser Val Val His
 260 265 270
 Glu Gly Leu His Asn His His Thr Thr Lys Ser Phe Ser Arg Thr Pro
 275 280 285
 Gly Lys
 290

<210> SEQ ID NO 3
 <211> LENGTH: 304
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 3

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 Leu Leu His Ala Cys Ile Pro Cys Gln Leu Arg Cys Ser Ser Asn Thr
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 Pro Pro Leu Thr Cys Gln Arg Tyr Cys Asn Ala Ser Val Thr Asn Ser
 35 40 45
 Val Lys Gly Thr Asn Ala Ser Gly Gly Gly Gly Ser Asp Lys Thr His
 50 55 60
 Thr Cys Pro Pro Cys Pro Ala Pro Glu Ala Ala Gly Gly Pro Ser Val
 65 70 75 80
 Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 85 90 95
 Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
 100 105 110
 Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
 115 120 125
 Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
 130 135 140
 Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys

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145	150	155	160
Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile Glu Lys Thr Ile	165	170	175
Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro	180	185	190
Pro Cys Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Trp Cys Leu	195	200	205
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn	210	215	220
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser	225	230	235
Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg	245	250	255
Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu	260	265	270
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Gly	275	280	285
Ser Gly Leu Asn Asp Ile Phe Glu Ala Gln Lys Ile Glu Trp His Glu	290	295	300

<210> SEQ ID NO 4
 <211> LENGTH: 303
 <212> TYPE: PRT
 <213> ORGANISM: *Macaca fascicularis*

<400> SEQUENCE: 4

Met Leu Gln Met Ala Arg Gln Cys Ser Gln Asn Glu Tyr Phe Asp Ser	1	5	10	15
Leu Leu His Asp Cys Lys Pro Cys Gln Leu Arg Cys Ser Ser Thr Pro	20	25	30	
Pro Leu Thr Cys Gln Arg Tyr Cys Asn Ala Ser Met Thr Asn Ser Val	35	40	45	
Lys Gly Met Asn Ala Ser Gly Gly Gly Gly Ser Asp Lys Thr His Thr	50	55	60	
Cys Pro Pro Cys Pro Ala Pro Glu Ala Ala Gly Gly Pro Ser Val Phe	65	70	75	80
Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro	85	90	95	
Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val	100	105	110	
Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr	115	120	125	
Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val	130	135	140	
Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys	145	150	155	160
Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile Glu Lys Thr Ile Ser	165	170	175	
Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro	180	185	190	
Cys Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val	195	200	205	

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Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly
 210 215 220
 Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp
 225 230 235 240
 Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp
 245 250 255
 Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His
 260 265 270
 Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Gly Ser
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 Gly Leu Asn Asp Ile Phe Glu Ala Gln Lys Ile Glu Trp His Glu
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<210> SEQ ID NO 5
 <211> LENGTH: 227
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Ala Ala Gly
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 Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
 20 25 30
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
 35 40 45
 Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
 50 55 60
 His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr
 65 70 75 80
 Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
 85 90 95
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile
 100 105 110
 Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
 115 120 125
 Cys Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser
 130 135 140
 Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
 145 150 155 160
 Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
 165 170 175
 Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
 180 185 190
 Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
 195 200 205
 His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
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 Pro Gly Lys
 225

<210> SEQ ID NO 6
 <211> LENGTH: 306
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 6

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 Asp Gly Ser Val Leu Leu Thr Cys Asp Ala Glu Ala Lys Asn Ile Thr
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 Trp Phe Lys Asp Gly Lys Met Ile Gly Phe Leu Thr Glu Asp Lys Lys
 35 40 45
 Lys Trp Asn Leu Gly Ser Asn Ala Lys Asp Pro Arg Gly Met Tyr Gln
 50 55 60
 Cys Lys Gly Ser Gln Asn Lys Ser Lys Pro Leu Gln Val Tyr Tyr Arg
 65 70 75 80
 Met Cys Gln Asn Cys Ile Glu Leu Asn Ala Pro Glu Ala Ala Gly Gly
 85 90 95
 Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
 100 105 110
 Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 115 120 125
 Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 130 135 140
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 145 150 155 160
 Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 165 170 175
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile Glu
 180 185 190
 Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys
 195 200 205
 Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
 210 215 220
 Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 225 230 235 240
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 245 250 255
 Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
 260 265 270
 Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
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 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 290 295 300
 Gly Lys
 305

<210> SEQ ID NO 7

<211> LENGTH: 306

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 7

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 Asp Gly Ser Val Leu Leu Thr Cys Asp Ala Glu Ala Lys Asn Ile Thr
 20 25 30

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Trp Phe Lys Asp Gly Lys Met Ile Gly Phe Leu Thr Glu Asp Lys Lys
 35 40 45
 Lys Trp Asn Leu Gly Ser Asn Ala Lys Asp Pro Arg Gly Met Tyr Gln
 50 55 60
 Cys Lys Gly Ser Gln Asn Lys Ser Lys Pro Leu Gln Val Tyr Tyr Arg
 65 70 75 80
 Met Cys Gln Asn Cys Ile Glu Leu Asn Ala Pro Glu Ala Ala Gly Gly
 85 90 95
 Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
 100 105 110
 Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 115 120 125
 Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 130 135 140
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 145 150 155 160
 Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 165 170 175
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile Glu
 180 185 190
 Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys
 195 200 205
 Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
 210 215 220
 Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 225 230 235 240
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 245 250 255
 Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
 260 265 270
 Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 275 280 285
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 290 295 300
 Gly Lys
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<210> SEQ ID NO 8

<211> LENGTH: 305

<212> TYPE: PRT

<213> ORGANISM: *Macaca fascicularis*

<400> SEQUENCE: 8

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 20 25 30
 Phe Lys Glu Gly Lys Met Ile Asp Ile Leu Thr Ala His Lys Asn Lys
 35 40 45
 Trp Asn Leu Gly Ser Asn Thr Lys Asp Pro Arg Gly Val Tyr Gln Cys
 50 55 60
 Lys Gly Ser Lys Asp Lys Ser Lys Thr Leu Gln Val Tyr Tyr Arg Met

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65	70	75	80
Cys Gln Asn Cys Ile Glu Leu Asn Ala Pro Glu Ala Ala Gly Gly Pro	85	90	95
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser	100	105	110
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp	115	120	125
Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn	130	135	140
Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val	145	150	155
Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu	165	170	175
Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Ser Ile Glu Lys	180	185	190
Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys Thr	195	200	205
Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Ser	210	215	220
Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu	225	230	235
Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu	245	250	255
Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys	260	265	270
Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu	275	280	285
Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly	290	295	300
Lys			
305			

<210> SEQ ID NO 9

<211> LENGTH: 330

<212> TYPE: PRT

<213> ORGANISM: Macaca fascicularis

<400> SEQUENCE: 9

Gln Asp Gly Asn Glu Glu Met Gly Ser Ile Thr Gln Thr Pro Tyr Gln	1	5	10	15
Val Ser Ile Ser Gly Thr Thr Val Ile Leu Thr Cys Ser Gln His Leu	20	25	30	
Gly Ser Glu Ala Gln Trp Gln His Asn Gly Lys Asn Lys Glu Asp Ser	35	40	45	
Gly Asp Arg Leu Phe Leu Pro Glu Phe Ser Glu Met Glu Gln Ser Gly	50	55	60	
Tyr Tyr Val Cys Tyr Pro Arg Gly Ser Asn Pro Glu Asp Ala Ser His	65	70	75	80
His Leu Tyr Leu Lys Ala Arg Val Cys Glu Asn Cys Met Glu Met Asp	85	90	95	
Ala Pro Glu Ala Ala Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys	100	105	110	

-continued

Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
	115						120					125			
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr
	130					135					140				
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
	145				150					155					160
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
			165						170					175	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
		180						185					190		
Ala	Leu	Pro	Ala	Ser	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln
	195						200					205			
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Cys	Arg	Asp	Glu	Leu
	210					215					220				
Thr	Lys	Asn	Gln	Val	Ser	Leu	Trp	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
	225				230					235					240
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn
			245						250					255	
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
			260					265					270		
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
	275						280					285			
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
	290					295					300				
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Gly	Ser	Gly	Leu	Asn	Asp	Ile
	305				310					315					320
Phe	Glu	Ala	Gln	Lys	Ile	Glu	Trp	His	Glu						
			325					330							

<210> SEQ ID NO 10

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 10

Glu	Ile	Gln	Leu	Val	Gln	Ser	Gly	Pro	Glu	Leu	Lys	Lys	Pro	Gly	Glu
1			5					10					15		
Thr	Val	Lys	Ile	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Ile	Phe	Thr	Asn	Phe
	20						25					30			
Gly	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Ala	Leu	Lys	Trp	Met
	35					40					45				
Gly	Trp	Ile	Asn	Thr	Tyr	Thr	Gly	Glu	Gln	Ile	Tyr	Ala	Asp	Asp	Phe
	50				55					60					
Lys	Gly	Arg	Phe	Ala	Phe	Ser	Leu	Glu	Thr	Ser	Ala	Ser	Thr	Ala	Tyr
	65			70					75					80	
Leu	Gln	Ile	Asn	Asn	Leu	Lys	Asn	Glu	Asp	Thr	Ala	Thr	Tyr	Phe	Cys
			85					90					95		
Ala	Arg	Gly	Glu	Ile	Tyr	Tyr	Gly	Tyr	Asp	Val	Gly	Phe	Val	Tyr	Trp
		100						105					110		
Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ala						
	115					120									

<210> SEQ ID NO 11

-continued

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 11

```

gagatccagt tgggtgcagtc tggacctgag ctgaagaagc ctggagagac agtcaagatc      60
tcttgcaagg cttctgggta tatcttcaca aactttggaa tgaactgggt gaggcaggct      120
ccaggaaaagg ctttaaagtg gatgggctgg ataaacacct aactggaga acaaatatat      180
gctgatgact tcaagggacg gttgccttc tctttgaaa cctctgccag cactgcctat      240
ttgcagatca acaacctcaa aaatgaggac acggctacat attctgtgc aagaggggag      300
atctactatg gttacgacgt gggctttgtt tactggggcc aagggactct ggtcactgtc      360
tctgca                                           366

```

<210> SEQ ID NO 12

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 12

```

Thr Asn Phe Gly Met Asn
1              5

```

<210> SEQ ID NO 13

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 13

```

Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Asp Phe Lys
1              5              10              15

```

Gly

<210> SEQ ID NO 14

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 14

```

Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr
1              5              10

```

<210> SEQ ID NO 15

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 15

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Thr Val Ser Leu Gly
1              5              10              15

```

```

Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
20              25              30

```

```

Gly Tyr Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
35              40              45

```

```

Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ser Gly Val Ser Ala
50              55              60

```

```

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His

```

-continued

65		70		75		80									
Pro	Val	Glu	Glu	Glu	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	His	Ser	Arg
				85					90					95	
Glu	Leu	Pro	Trp	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys	
			100						105					110	

<210> SEQ ID NO 16
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 16

gatattgtgc taactcagtc tctgcttcc ttaactgtat ctctggggca gagggccacc	60
atctcatgca gggccagcaa aagtgtcagt acatctggct atagttatat gcactggtac	120
caacagaaac caggacagcc acccaactc ctcactctatc ttgcatccga cctagaatct	180
gggggtctctg ccagggttcag tggcagtggtg tctgggacag acttcacct caacatccat	240
cctgtggagg aggaggatgc tgcaacctat tactgtcaac acagtaggga acttccgtgg	300
acgttcggtg gaggcaccaa gctggaaata aaa	333

<210> SEQ ID NO 17
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 17

Arg	Ala	Ser	Lys	Ser	Val	Ser	Thr	Ser	Gly	Tyr	Ser	Tyr	Met	His
1				5					10				15	

<210> SEQ ID NO 18
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 18

Leu	Ala	Ser	Asp	Leu	Glu	Ser
1				5		

<210> SEQ ID NO 19
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 19

Gln	His	Ser	Arg	Glu	Leu	Pro	Trp	Thr
1				5				

<210> SEQ ID NO 20
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 20

Asp	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Ser	Leu	Ala	Ile	Ser	Leu	Gly
1				5					10					15	

Gln	Arg	Ala	Thr	Ile	Ser	Cys	Arg	Ala	Ser	Lys	Ser	Val	Thr	Thr	Ser
			20					25					30		

Gly	Tyr	Ser	Tyr	Ile	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

-continued

35	40	45	
Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ala Gly Val Pro Ala			
50	55	60	
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His			
65	70	75	80
Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg			
	85	90	95
Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Leu Lys			
	100	105	110

<210> SEQ ID NO 21
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 21

gacattgtgc taacacagtc tectgcttcc ttagctatat ctctggggca gagggccacc	60
atctcatgca gggccagcaa aagtgtcact acatctggct atagttatat aactggtac	120
caacagaaac caggacagcc acccaaactc ctcatctatc ttgcatccga cctagaagct	180
ggggtcctcg ccagggttcag tggcagtggtg tctgggacag acttcacct caacatccat	240
cctgtggagg aagaggatgc tgcaacctat tactgtcagc acagtaggga gcttccgtgg	300
acgttcggtg gaggcaccaa gctggagctg aaa	333

<210> SEQ ID NO 22
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 22

Arg Ala Ser Lys Ser Val Thr Thr Ser Gly Tyr Ser Tyr Ile His	
1	15

<210> SEQ ID NO 23
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 23

Leu Ala Ser Asp Leu Glu Ala	
1	5

<210> SEQ ID NO 24
 <211> LENGTH: 118
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 24

Gln Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala	
1	15
Ser Val Asn Leu Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr	
	30
Val Ile His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile	
35	45
Gly Tyr Thr Ile Pro Tyr Thr Asp Val Thr Lys Tyr Asn Glu Lys Phe	
50	60

-continued

Lys Ala Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ser Thr Ala Phe
65 70 75 80

Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Tyr Asp Phe Asp Gly Tyr Phe Asp Tyr Trp Gly Gln Gly Thr
100 105 110

Thr Leu Thr Val Ser Ser
115

<210> SEQ ID NO 25
 <211> LENGTH: 354
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 25

caggctccagc tgcaacagtc tggacctgag ctggtgaagc ctggggcttc agtgaatttg 60
 tcctgcaagg cttctggata cacattcact aattatgtta tacactgggt aaagcagaag 120
 cctgggcagg gccttgagtg gattggatat actatccctt aactgatgt tactaagtac 180
 aatgagaaat tcaaagccaa ggccacactg acttcagaca aatcctccag cacagccttc 240
 atggagctca gcagcctgac ctctgaggac tctgcggtct attactgtgc aagatatgat 300
 ttcgacggct actttgacta ctggggccaa ggcaccactc tcacagtctc ctca 354

<210> SEQ ID NO 26
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 26

Thr Asn Tyr Val Ile His
 1 5

<210> SEQ ID NO 27
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 27

Tyr Thr Ile Pro Tyr Thr Asp Val Thr Lys Tyr Asn Glu Lys Phe Lys
 1 5 10 15

Ala

<210> SEQ ID NO 28
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 28

Tyr Asp Phe Asp Gly Tyr Phe Asp Tyr
 1 5

<210> SEQ ID NO 29
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 29

Asp Ile Val Met Thr Gln Ala Ala Phe Ser Asn Pro Val Thr Leu Gly

```
<210> SEQ ID NO 34
<211> LENGTH: 122
```

-continued

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 34

```

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Lys Lys Pro Gly Glu
1           5           10           15
Thr Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Phe
           20           25           30
Gly Met Asn Trp Val Arg Gln Thr Pro Gly Lys Ala Leu Lys Trp Met
           35           40           45
Gly Trp Ile Asn Thr Tyr Thr Gly Asp Leu Ile Tyr Gly Asp Asp Phe
           50           55           60
Lys Gly Arg Phe Ala Phe Ser Leu Glu Thr Ser Ala Ser Thr Ala Tyr
65           70           75           80
Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr Tyr Phe Cys
           85           90           95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Ala Tyr Trp
          100          105          110
Gly Gln Gly Thr Leu Val Thr Val Ser Ala
          115          120

```

<210> SEQ ID NO 35

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 35

```

gaggtccagc tgcagcagtc tggacctgag ctgaagaagc ctggagagac agtcaagatc      60
tcctgcaagg cttctgggta taccttcaca aactttggaa tgaactgggt gaggcagact    120
ccaggaaaagg ctttaaagtg gatgggctgg ataaacacct acactggaga cttaatatat    180
ggtgatgact ttaagggacg gtttgccttc tctttggaaa cctctgccag cactgcctat    240
ttacagatca acaacctcaa aaatgaggac acggctacat attctgtgc aagaggggag    300
atctactatg gttacgacgt ggggtttgct tactggggcc aagggaactct ggtcactgtc    360
tctgca                                           366

```

<210> SEQ ID NO 36

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 36

```

Trp Ile Asn Thr Tyr Thr Gly Asp Leu Ile Tyr Gly Asp Asp Phe Lys
1           5           10           15
Gly

```

<210> SEQ ID NO 37

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 37

```

Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Ala Tyr
1           5           10

```

-continued

<210> SEQ ID NO 38
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 38

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
 1 5 10 15
 Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30
 Gly Tyr Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
 35 40 45
 Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
 50 55 60
 Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
 65 70 75 80
 Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg
 85 90 95
 Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 39
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 39

gacattgtgc tgacacagtc tcctgcttcc ttagctgtat ctctggggca gagggccacc 60
 atctcatgca gggccagcaa aagtgtcagt acatctggct atagttatat gcactggtag 120
 caacagaaac caggacagcc acccaaaactc ctcactatc ttgcatccaa cctagaatct 180
 ggggtccctg ccagggttcag tggtagtggg tctgggacag acttcaccct caacatccat 240
 cctgtggagg aggaggatgc tgcaacctat tactgtcagc acagtaggga gcttcctgtg 300
 acgttcggtg gaggcaccaa gctggaaatc aaa 333

<210> SEQ ID NO 40
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 40

Leu Ala Ser Asn Leu Glu Ser
 1 5

<210> SEQ ID NO 41
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 41

Glu Ile Gln Leu Val Gln Ser Gly Pro Glu Leu Lys Lys Pro Gly Glu
 1 5 10 15
 Thr Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
 20 25 30
 Gly Met Asn Trp Val Arg Gln Thr Pro Gly Lys Ala Leu Lys Trp Met
 35 40 45

-continued

Gly	Trp	Ile	Asn	Thr	Tyr	Thr	Gly	Glu	Thr	Ile	Tyr	Ala	Asp	Asp	Phe
50						55					60				

Lys	Gly	Leu	Phe	Ala	Phe	Ser	Leu	Glu	Val	Pro	Ala	Ser	Thr	Ala	Tyr
65					70					75					80

Leu	Gln	Ile	Asn	Asn	Leu	Lys	Asn	Glu	Asp	Thr	Ala	Thr	Tyr	Phe	Cys
			85					90						95	

Ala	Arg	Gly	Glu	Ile	Tyr	Tyr	Gly	Phe	Asp	Val	Gly	Phe	Ala	Tyr	Trp
			100				105						110		

Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ala
	115					120			

<210> SEQ ID NO 42

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 42

gagatccagt	tggtgcagtc	tggacctgag	ctgaagaagc	ctggagagac	agtcaagatc	60
tcttgcaagg	cttctgggta	tattttcaca	aactttggaa	tgaactgggt	gaggcagact	120
ccaggaaagg	ctttaaagtg	gatgggctgg	ataaacacct	acactggaga	aacaatatat	180
gctgatgact	tcaagggact	atttgccttc	tctttggaag	tgectgccag	cactgcctat	240
ttgcagatca	acaacctcaa	aaatgaggac	acggctacat	atttctgtgc	aagaggggag	300
atctactatg	gtttcgacgt	gggatttgct	tactggggcc	aagggaactct	ggtcactgtc	360
tctgca						366

<210> SEQ ID NO 43

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 43

Trp	Ile	Asn	Thr	Tyr	Thr	Gly	Glu	Thr	Ile	Tyr	Ala	Asp	Asp	Phe	Lys
1			5					10						15	

Gly

<210> SEQ ID NO 44

<211> LENGTH: 13

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 44

Gly	Glu	Ile	Tyr	Tyr	Gly	Phe	Asp	Val	Gly	Phe	Ala	Tyr
1			5					10				

<210> SEQ ID NO 45

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 45

Asp	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Ser	Leu	Thr	Val	Ser	Leu	Gly
1			5					10					15		

Gln	Arg	Ala	Thr	Ile	Ser	Cys	Arg	Ala	Ser	Lys	Ser	Val	Ser	Thr	Ser
		20					25					30			

Gly	Tyr	Ser	Tyr	Ile	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

-continued

35	40	45	
Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ser Gly Val Pro Ala			
50	55	60	
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His			
65	70	75	80
Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg			
	85	90	95
Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Leu Lys			
	100	105	110

<210> SEQ ID NO 46
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 46

gacattgtgc taacacagtc tectgtttcc ttaactgtat ctctggggca gagggccacc	60
atctcatgca gggccagcaa aagtgtcagt acatctggct atagttatat aactggtac	120
caacagaaac caggacagcc acccaaactc ctcatctatc ttgcatccga cctagaatct	180
ggggtcctcg ccagggttcag tggcagtggtg tctgggacag acttcacct caacatccat	240
cctgtggagg aggaggatgc tgcaacctat tactgtcaac acagtaggga acttccgtgg	300
acgttcggtg gaggcaccaa gctggagctg aaa	333

<210> SEQ ID NO 47
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 47

Arg Ala Ser Lys Ser Val Ser Thr Ser Gly Tyr Ser Tyr Ile His	
1	15

<210> SEQ ID NO 48
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus
 <400> SEQUENCE: 48

Glu Ile Gln Leu Val Gln Ser Gly Pro Glu Leu Lys Lys Pro Gly Glu	
1	15
Thr Val Lys Ile Ser Cys Thr Ala Ser Gly Tyr Thr Phe Thr Asn Phe	
20	30
Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Ala Leu Arg Trp Met	
35	45
Gly Trp Ile Asn Thr Tyr Thr Gly Asp Gln Ile Tyr Ala Asp Asp Phe	
50	60
Lys Gly Arg Phe Ala Phe Ser Leu Glu Thr Ser Ala Ser Thr Ala Tyr	
65	80
Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr Tyr Phe Cys	
85	95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp	
100	110
Gly Gln Gly Thr Leu Val Thr Val Ser Ala	
115	120

-continued

<210> SEQ ID NO 49
 <211> LENGTH: 366
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 49

```

gagatccagt tgggtcagtc tggacctgag ctgaagaagc ctggagagac agtcaagatc      60
tcctgcacgg cttctgggta taccttcaca aactttgga tgaactgggt gaggcaggct      120
ccaggaaaagg ctttaagggt gatgggctgg ataaacacct aactggaga ccaaatatat      180
gctgatgact tcaagggacg gttgccttc tctttgaaa cctctgccag cactgcctat      240
ttgcagatca acaacctcaa aaatgaggac acggctacat attctgtgc aagaggggag      300
atctactatg gttacgacgt gggctttgtt tattggggcc aagggactct ggctactgtc      360
tctgca                                           366
  
```

<210> SEQ ID NO 50
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 50

```

Trp Ile Asn Thr Tyr Thr Gly Asp Gln Ile Tyr Ala Asp Asp Phe Lys
1             5             10             15
  
```

Gly

<210> SEQ ID NO 51
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 51

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1             5             10             15

Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
20            25            30

Gly Tyr Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
35            40            45

Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
50            55            60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
65            70            75            80

Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg
85            90            95

Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100           105           110
  
```

<210> SEQ ID NO 52
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 52

```

gacattgtgc taacacagtc tctgtcttc ttagctgtat ctctggggca gagggccacc      60
atctcatgca gggccagcaa aagtgtcagt acatctggct atagttatat gcactggtac      120
  
```

-continued

```

caacagaaac caggacagcc acccaaaactc ctcacatc tgcacccaa cctagaatct 180
ggggtcctcg ccagggttcag tggcagtgagg tctgggacag acttcaccct caacatccat 240
cctgtggagg aggaggatgc tgcaacctat tactgtcagc acagtaggga gcttcctgtg 300
acgttcggtg gaggcaccaa gctggaaata aaa 333

```

```

<210> SEQ ID NO 53
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

```

```

<400> SEQUENCE: 53

```

```

Glu Val Gln Leu Gln Gln Ser Gly Thr Val Leu Thr Arg Pro Gly Ala
1      5      10      15
Ser Val Lys Leu Ser Cys Lys Ala Ser Gly Tyr Ser Phe Ser Ser Tyr
20     25     30
Trp Met His Trp Leu Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
35     40     45
Gly Ala Ile Tyr Pro Gly Asn Ser Arg Thr Thr Tyr Asn Gln Lys Phe
50     55     60
Lys Gly Glu Ala Lys Leu Thr Ala Asp Thr Ser Ala Ser Thr Ala Tyr
65     70     75     80
Met Asp Leu Ser Ser Leu Thr Asn Ala Asp Ser Ala Val Tyr Phe Cys
85     90     95
Thr Arg Glu Glu Thr Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser
100    105    110
Ala

```

```

<210> SEQ ID NO 54
<211> LENGTH: 339
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

```

```

<400> SEQUENCE: 54

```

```

gagggttcagc tccagcagtc tgggactgtg ctgacaaggc ctggggcttc cgtgaagttg 60
tcctgcaagg cttctggcta cagcttttcc agctactgga tgcactggct aaaacagagg 120
cctggacagg gtctagaatg gattgtgtct atttacctg gaaatagtcg tactacctac 180
aaccagaagt tcaagggcga ggccaaactg actgcggaca cttccgccag cactgcctac 240
atggacctca gcagcctgac aaatgccgac tctgcggtct atttctgtac aagagaggag 300
acttactggg gccaaaggac tctggtcact gtctctgca 339

```

```

<210> SEQ ID NO 55
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

```

```

<400> SEQUENCE: 55

```

```

Ser Ser Tyr Trp Met His
1      5

```

```

<210> SEQ ID NO 56
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

```

-continued

<400> SEQUENCE: 56

Ala Ile Tyr Pro Gly Asn Ser Arg Thr Thr Tyr Asn Gln Lys Phe Lys
 1 5 10 15

Gly

<210> SEQ ID NO 57

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 57

Thr Arg Glu Glu Thr Tyr
 1 5

<210> SEQ ID NO 58

<211> LENGTH: 112

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 58

Asp Val Val Met Thr Gln Thr Pro Leu Ser Leu Ser Val Thr Ile Gly
 1 5 10 15

Gln Pro Ala Ser Ile Ser Cys Lys Ser Ser Gln Gly Leu Leu Asp Ser
 20 25 30

Tyr Gly Val Thr Tyr Leu Asn Trp Leu Phe Gln Arg Pro Gly Gln Pro
 35 40 45

Pro Lys Arg Leu Ile Tyr Leu Val Ser Glu Leu Asp Ser Gly Val Pro
 50 55 60

Asp Arg Phe Thr Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Arg Ile
 65 70 75 80

Ser Arg Val Glu Ala Glu Asp Leu Gly Ile Tyr Tyr Cys Trp Gln Gly
 85 90 95

Ser His Phe Pro Leu Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
 100 105 110

<210> SEQ ID NO 59

<211> LENGTH: 336

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 59

gatgttgtga tgaccagac tccactctct ttgtcggtta ccattggaca accagcctcc 60
 atctcttgca agtcaagtca gggcctctta gatagttatg gagtgacata tttgaattgg 120
 ttgtttcaga ggccaggcca gcctccaaag cgctaatact atctagtgtc tgaactggac 180
 tctggagtcc ctgacagggt cactggcagt ggatcaggga cagatttcac actgagaatc 240
 agcagagtgg aggctgagga tttgggaatt tattattgct ggcaagggtc acattttcct 300
 ctcacgttcg gtgctggggac caagctggag ctgaaa 336

<210> SEQ ID NO 60

<211> LENGTH: 16

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 60

-continued

Lys Ser Ser Gln Gly Leu Leu Asp Ser Tyr Gly Val Thr Tyr Leu Asn
1 5 10 15

<210> SEQ ID NO 61
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 61

Leu Val Ser Glu Leu Asp Ser
1 5

<210> SEQ ID NO 62
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 62

Trp Gln Gly Ser His Phe Pro Leu Thr
1 5

<210> SEQ ID NO 63
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 63

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Arg Lys Pro Gly Glu
1 5 10 15

Thr Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
20 25 30

Gly Met Asn Trp Val Lys Gln Thr Pro Gly Lys Gly Leu Lys Trp Met
35 40 45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Arg Ile Tyr Ala Asp Asp Phe
50 55 60

Lys Gly Arg Phe Ala Phe Ser Leu Glu Thr Ser Ala Ser Thr Ala Tyr
65 70 75 80

Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr Tyr Phe Cys
85 90 95

Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ala
115 120

<210> SEQ ID NO 64
 <211> LENGTH: 366
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 64

gaggtccagt tgcagcagtc tggacctgag ctgaggaagc ctggagagac agtcaagatc	60
tcctgcaagg cttctgggta taccttcaca aactatggaa tgaactgggt gaagcagact	120
ccaggaaagg gtttaaagtg gatgggctgg ataaacacct aactggaga acgaatatat	180
gctgatgact tcaagggacg gtttgccttc tctttggaaa cctctgccag cactgcctat	240
ttgcagatca acaacctcaa aaatgaggac acggctacat atttctgtgc aagaggggag	300
atctactatg gttacgacgt gggctttgtt tactggggcc aagggaactct ggtcactgtc	360

-continued

tctgca 366

<210> SEQ ID NO 65
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 65

Thr Asn Tyr Gly Met Asn
 1 5

<210> SEQ ID NO 66
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 66

Trp Ile Asn Thr Tyr Thr Gly Glu Arg Ile Tyr Ala Asp Asp Phe Lys
 1 5 10 15

Gly

<210> SEQ ID NO 67
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 67

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Val Val Ser Leu Gly
 1 5 10 15

Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30

Gly Tyr Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
 35 40 45

Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
 50 55 60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
 65 70 75 80

Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg
 85 90 95

Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Arg Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 68
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 68

gacattgtgc tgacacagtc tctgcttcc ttagttgtat ctctggggca gagggccacc 60

atctcatgca gggccagcaa aagtgtcagt acatctggct atagttatat gcactgggtac 120

caacagaaac caggacagcc acccaaacct ctcctctatc ttgcatccaa cctagaatct 180

ggggtccttg ccaggttcag tggcagtggt tctgggacag acttcaccct caacatccat 240

cctgtggagg aggaggatgc tgcaacctat tactgtcagc acagtaggga gcttcctgtg 300

acgttcggtg gaggcaccag actggaaata aaa 333

-continued

<210> SEQ ID NO 69
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 69

```

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Thr
1          5          10          15
Ser Val Lys Met Ser Cys Lys Ala Ala Gly Tyr Thr Phe Thr Asn Tyr
          20          25          30
Trp Ile Gly Trp Val Lys Gln Arg Pro Gly His Gly Leu Glu Trp Ile
          35          40          45
Gly Asp Ile Tyr Pro Gly Ser Gly Asp Thr Asn Tyr Asn Gly Lys Phe
          50          55          60
Lys Gly Lys Ala Thr Leu Ala Ala Asp Ile Ser Ser Asn Ala Ala Tyr
          65          70          75          80
Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys
          85          90          95
Ala Arg Ser Tyr Tyr Arg Asn Asp Ala Ala Ser Phe Ala Tyr Trp Gly
          100          105          110
Gln Gly Thr Leu Val Thr Val Ser Ala
          115          120

```

<210> SEQ ID NO 70
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 70

```

caggctccagc tgcagcagtc tggagctgag ctggttaaggc ctgggacttc agtgaagatg      60
tcctgcaagg ctgctggata cacccttact aactactgga taggttgggt aaagcagagg      120
cctggacatg gccttgagtg gattggagat atttaccctg gaagtgggtga tactaattac      180
aatgggaagt tcaagggcaa ggccacactg gctgcagaca tttcctccaa cgcagcctac      240
atgcagctca gcagcctgac atctgaggac tctgccatct attactgtgc aagaagctac      300
tataggaacg acgcggcctc atttgcttac tggggccaag ggactctggt cactgtctct      360
gca                                                    363

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<210> SEQ ID NO 71
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 71

```

Thr Asn Tyr Trp Ile Gly
1          5

```

<210> SEQ ID NO 72
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 72

```

Asp Ile Tyr Pro Gly Ser Gly Asp Thr Asn Tyr Asn Gly Lys Phe Lys
1          5          10          15

```

-continued

Gly

<210> SEQ ID NO 73
 <211> LENGTH: 12
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 73

Ser Tyr Tyr Arg Asn Asp Ala Ala Ser Phe Ala Tyr
 1 5 10

<210> SEQ ID NO 74
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 74

Asp Ile Val Met Thr Gln Ser His Lys Phe Met Ser Thr Ser Val Gly
 1 5 10 15

Asp Arg Val Asn Ile Thr Cys Lys Ala Ser Gln Asp Val Asp Thr Ala
 20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Ser Gln Ser Pro Lys Leu Leu Ile
 35 40 45

Tyr Trp Thr Ser Thr Arg His Thr Gly Val Pro Asp Arg Phe Thr Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Val Gln Ser
 65 70 75 80

Glu Asp Leu Ala Asp Tyr Phe Cys Gln Gln Phe Ser Ser Tyr Pro Leu
 85 90 95

Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys
 100 105

<210> SEQ ID NO 75
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 75

gacattgtga tgaccagtc tcacaaattc atgtccacat cagtaggaga cagggccaac 60
 atcacctgca aggccagtc ggaatgtgat actgctgtag cctgggtatca acagaaacca 120
 agtcaatctc ctaaaactact gatttactgg acatccaccc ggcacactgg agtcctgat 180
 cgcttcacag gcagtgatc tgggacagat ttcactctca ccattagcaa tgtgcagtct 240
 gaagacttgg cagattatct ctgtcaacaa tttagtagct atcctctcac gttegggtgct 300
 gggaccaagc tggagctgaa a 321

<210> SEQ ID NO 76
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 76

Lys Ala Ser Gln Asp Val Asp Thr Ala Val Ala
 1 5 10

<210> SEQ ID NO 77
 <211> LENGTH: 7

-continued

<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 77

Trp Thr Ser Thr Arg His Thr
1 5

<210> SEQ ID NO 78
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 78

Gln Gln Phe Ser Ser Tyr Pro Leu Thr
1 5

<210> SEQ ID NO 79
<211> LENGTH: 122
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 79

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Lys Lys Pro Gly Glu
1 5 10 15

Thr Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
20 25 30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Ala Leu Lys Trp Met
35 40 45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Asp Phe
50 55 60

Lys Gly Arg Phe Ala Phe Ser Leu Glu Thr Ser Ala Ser Thr Ala Tyr
65 70 75 80

Leu Gln Ile Asn Asn Leu Lys Asn Glu Asp Thr Ala Thr Tyr Phe Cys
85 90 95

Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ala
115 120

<210> SEQ ID NO 80
<211> LENGTH: 366
<212> TYPE: DNA
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 80

gaggtcacgt tgcagcagtc tggacctgag ctgaagaagc ctggagagac agtcaagatc 60

tcctgcaagg cttctgggta tatcttcacg aactttggaa tgaactgggt gaggcaggct 120

ccaggaaagg ctttaaagtg gatgggctgg ataaacacct aactggaga acaaatatat 180

gctgatgact tcaagggacg gtttgccctc tctttggaaa cctctgccag cactgcctat 240

ttgcagatca acaacctcaa aaatgaggac acggctacat atttctgtgc aagaggggag 300

atctactatg gttacgacgt gggctttgtt tactggggcc aagggactct ggteactgtc 360

tctgca 366

<210> SEQ ID NO 81
<211> LENGTH: 111

-continued

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 81

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Thr Val Ser Leu Gly
1 5 10 15
Gln Arg Ala Thr Ile Ser Cys Gly Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30
Gly Tyr Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
 35 40 45
Lys Leu Leu Ile Tyr Leu Gly Ser Asp Leu Glu Ser Gly Val Pro Ala
 50 55 60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
65 70 75 80
Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg
 85 90 95
Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
 100 105 110

<210> SEQ ID NO 82

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 82

gacattgtgc tgacacagtc tctgcttcc ttaactgtat ctctggggca gagggccacc 60
atctcatgcg gggccagcaa aagtgtcagt acatctggct atagttatat gcactggtac 120
caacagaaac caggacagcc acccaaactc ctcatctatc ttggatccga cctagaatct 180
gggggtccctg ccagggttcag tggcagtggt tctgggacag acttcacct caacatccat 240
cctgtggagg aggaggatgc tgcaacctat tactgtcaac acagtaggga acttccgtgg 300
acgttcggtg gaggcaccaa gctggaaatc aaa 333

<210> SEQ ID NO 83

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 83

Gly Ala Ser Lys Ser Val Ser Thr Ser Gly Tyr Ser Tyr Met His
1 5 10 15

<210> SEQ ID NO 84

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 84

Leu Gly Ser Asp Leu Glu Ser
1 5

<210> SEQ ID NO 85

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 85

-continued

Asn	Ile	Val	Leu	Thr	Gln	Ser	Pro	Ala	Ser	Leu	Ala	Val	Ser	Leu	Gly
1				5					10					15	
Gln	Arg	Ala	Thr	Ile	Ser	Cys	Arg	Ala	Ser	Glu	Ser	Val	Asp	Ser	Ser
			20					25					30		
Gly	Asn	Ser	Phe	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Pro	Pro
		35					40					45			
Lys	Leu	Leu	Ile	Tyr	Leu	Glu	Ser	Asn	Leu	Glu	Ser	Gly	Val	Pro	Ala
	50					55					60				
Arg	Phe	Ser	Gly	Ser	Gly	Ser	Arg	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Glu
65					70					75					80
Pro	Val	Glu	Ala	Asp	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Ser	Asn
				85					90					95	
Glu	Asp	Pro	Trp	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys	
			100					105					110		

<210> SEQ ID NO 86

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 86

aacattgtgc tgacccaatc tccagcttct ttggtgtgt ctctagggca gagggccacc	60
atatcctgca gagccagtga aagtgttgat agttctggca atagttttat gcactggtac	120
cagcagaaac caggacagcc acccaaaactc ctcattctatc ttgaatccaa cctagaatct	180
ggggccctcg ccagggttcag tggcagtggt tctaggacag acttcaccct caccattgaa	240
cctgtggagg ctgatgatgc tgcaacctat tactgtcagc aaagtaatga ggatccgtgg	300
acgttcggtg gaggcaccaa gctggaaatc aaa	333

<210> SEQ ID NO 87

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 87

Arg	Ala	Ser	Glu	Ser	Val	Asp	Ser	Ser	Gly	Asn	Ser	Phe	Met	His
1			5						10				15	

<210> SEQ ID NO 88

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 88

Leu	Glu	Ser	Asn	Leu	Glu	Ser
1			5			

<210> SEQ ID NO 89

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 89

Gln	Gln	Ser	Asn	Glu	Asp	Pro	Trp	Thr
1				5				

<210> SEQ ID NO 90

-continued

<211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 90

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Ile Ser Leu Gly
1             5             10             15

Gln Arg Ala Thr Ile Ser Cys Arg Ala Ser Arg Ser Val Thr Thr Ser
                20             25             30

Gly Tyr Ser Tyr Ile His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
            35             40             45

Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ala Gly Val Pro Ala
            50             55             60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile His
65             70             75             80

Pro Val Glu Glu Glu Asp Ala Ala Thr Tyr Tyr Cys Gln His Ser Arg
            85             90             95

Glu Leu Pro Trp Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
            100            105            110

```

<210> SEQ ID NO 91
 <211> LENGTH: 333
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 91

```

gacattgtgc tgacacagtc tctgcttcc ttagctatat ctctggggca gagggccacc      60
atctcatgca gggccagcag aagtgtcact acatctggct atagttatat acactggtac      120
caacagaaac caggacagcc acccaaactc ctcattctatc ttgcatccga cctagaagct      180
gggggccctg ccagggtcag tggcagtggt tctgggacag acttcaccct caacatccat      240
cctgtggagg aagaggatgc tgcaacctat tactgtcagc acagtaggga gcttcctgtg      300
acgttcggtg gaggcaccaa gctggaaata aaa                                333

```

<210> SEQ ID NO 92
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 92

```

Arg Ala Ser Arg Ser Val Thr Thr Ser Gly Tyr Ser Tyr Ile His
1             5             10             15

```

<210> SEQ ID NO 93
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 93

```

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1             5             10             15

Thr Val Thr Leu Thr Cys Arg Ser Ser Thr Gly Ala Val Thr Thr Ser
            20             25             30

Asn Tyr Ala Asn Trp Val Gln Gln Lys Pro Gly Gln Ala Pro Arg Gly
            35             40             45

Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Trp Thr Pro Ala Arg Phe

```

-continued

50	55	60	
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Ala			
65	70	75	80
Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn			
	85	90	95
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu			
	100	105	

<210> SEQ ID NO 94
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 94

caggetgtgg tcacacaaga gcctagcctg acagtgtctc ctggcggcac agtgaccctg	60
acctgtagat cttctacagg cgccgtgacc accagcaact acgctaattg ggtgcagcag	120
aagcccggcc aggetcctag aggactgac gcggaacaa acaagagagc cccttgga	180
cccgccagat tctctggac tctgctggc ggaaaggccg ctctgacact ttctggtgct	240
cagcctgagg acgaggccga gtactattgt gccctgtggt acagcaacct gtgggtgttc	300
ggcggaggca ccaaactgac agttctg	327

<210> SEQ ID NO 95
 <211> LENGTH: 13
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 95

Ser Ser Thr Gly Ala Val Thr Thr Ser Asn Tyr Ala Asn
1 5 10

<210> SEQ ID NO 96
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 96

Gly Gly Thr Asn Lys Arg Ala Pro
1 5

<210> SEQ ID NO 97
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 97

Ala Leu Trp Tyr Ser Asn Leu Trp Val
1 5

<210> SEQ ID NO 98
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 98

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15
Thr Val Thr Leu Thr Cys Arg Ser Ser Thr Gly Ala Val Thr Thr Ser

-continued

20	25	30	
Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Pro Arg Gly			
35	40	45	
Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Trp Thr Pro Ala Arg Phe			
50	55	60	
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Ala			
65	70	75	80
Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn			
85	90	95	
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu			
100	105		

<210> SEQ ID NO 99
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 99

caggetgtgg tcacacaaga gcttagcctg acagtgtctc ctggcggcac agtgaccctg	60
acctgtagat cttctacagg cgccgtgacc accagcaact acgctaattg ggtgcaggag	120
aagcccggcc aggtctctag aggactgac gccggaacaa acaagagagc cccttgga	180
cccgccagat tctctggatc tctgtcggc ggaaaggccg ctctgacact ttctggtgct	240
cagcctgagg acgaggccga gtactattgt gccctgtggt acagcaacct gtgggtgttc	300
ggcggaggca ccaaactgac agttctg	327

<210> SEQ ID NO 100
 <211> LENGTH: 109
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 100

Glu Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly	
1	15
Thr Val Thr Leu Thr Cys Glu Ser Ser Asp Gly Ala Val Thr Thr Ser	
20	30
Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Pro Arg Gly	
35	45
Leu Ile Gly Gly Thr Asn Lys Glu Ala Pro Trp Thr Pro Ala Arg Phe	
50	60
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Ala	
65	80
Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn	
85	95
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu	
100	105

<210> SEQ ID NO 101
 <211> LENGTH: 327
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 101

gaggetgtgg tcacacaaga gcttagcctg acagtgtctc ctggcggcac agtgaccctg	60
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acctgtgagt cttctgacgg cgccgtgacc accagcaact acgctaattg ggtgcaggag 120
aagccccggcc aggtctctag aggactgacg gccggaacaa acaaggaggc cccttggaca 180
cccgccagat tctctggatc tctgctcggc ggaaaggccg ctctgacact ttctggtgct 240
cagcctgagg acgaggccga gtactattgt gccctgtggt acagcaacct gtgggtgttc 300
ggcggaggca ccaaactgac agttctg 327

```

```

<210> SEQ ID NO 102
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 102

```

```

Ser Ser Asp Gly Ala Val Thr Thr Ser Asn Tyr Ala Asn
1          5              10

```

```

<210> SEQ ID NO 103
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 103

```

```

Gly Gly Thr Asn Lys Glu Ala Pro
1          5

```

```

<210> SEQ ID NO 104
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 104

```

```

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1          5              10              15
Thr Val Thr Leu Thr Cys Glu Ser Ser Asp Gly Ala Val Thr Thr Ser
                20              25              30
Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Pro Arg Gly
                35              40              45
Leu Ile Gly Gly Thr Asn Lys Glu Ala Pro Trp Thr Pro Ala Arg Phe
                50              55              60
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Leu Ser Gly Ala
        65              70              75              80
Gln Pro Glu Asp Glu Ala Glu Tyr Tyr Cys Ala Leu Trp Tyr Ser Asn
                85              90              95
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
        100              105

```

```

<210> SEQ ID NO 105
<211> LENGTH: 327
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 105

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```

caggctgtgg tcacacaaga gcctagcctg acagtgtctc ctggcggcac agtgaccctg 60
acctgtgagt cttctgacgg cgccgtgacc accagcaact acgctaattg ggtgcaggag 120
aagccccggcc aggtctctag aggactgacg gccggaacaa acaaggaggc cccttggaca 180
cccgccagat tctctggatc tctgctcggc ggaaaggccg ctctgacact ttctggtgct 240

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-continued

cagcctgagg acgaggccga gtactattgt gccctgtggt acagcaacct gtgggtgttc 300

ggcggaggca ccaaactgac agttctg 327

<210> SEQ ID NO 106

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 106

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15Thr Val Thr Leu Thr Cys Arg Ser Ser Thr Gly Ala Val Thr Thr Ser
20 25 30Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Pro Arg Gly
35 40 45Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Trp Thr Pro Ala Arg Phe
50 55 60Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Ile Thr Gly Ala
65 70 75 80Gln Ala Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Trp Tyr Ser Asn
85 90 95Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
100 105

<210> SEQ ID NO 107

<211> LENGTH: 327

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 107

caggctgtgg tcacacaaga gcctagcctg acagtgtctc ctggcggcac agtgaccctg 60

acctgtagat cttctacagg cgccgtgacc accagcaact acgctaattg ggtgcaggag 120

aagcccggcc aggtctctag aggactgac gccggaacaa acaagagagc cccttgga 180

cccgccagat tctctggatc tctgctcggc ggaaaggccg ctctgacaat cactggtgct 240

caggctgagg acgaggccga gtactattgt gtgctgtggt acagcaacct gtgggtgttc 300

ggcggaggca ccaaactgac agttctg 327

<210> SEQ ID NO 108

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 108

Val Leu Trp Tyr Ser Asn Leu Trp Val
1 5

<210> SEQ ID NO 109

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 109

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15

-continued

Thr	Val	Thr	Leu	Thr	Cys	Arg	Ser	Ser	Thr	Gly	Ala	Val	Thr	Thr	Ser
			20					25					30		
Asn	Tyr	Ala	Asn	Trp	Phe	Gln	Glu	Lys	Pro	Gly	Gln	Ala	Pro	Arg	Gly
	35						40				45				
Leu	Ile	Tyr	Gly	Thr	Asn	Lys	Arg	Ala	Pro	Trp	Thr	Pro	Ala	Arg	Phe
	50				55						60				
Ser	Gly	Ser	Leu	Leu	Gly	Gly	Lys	Ala	Ala	Leu	Thr	Leu	Ser	Gly	Ala
65					70					75				80	
Gln	Ala	Glu	Asp	Glu	Ala	Glu	Tyr	Tyr	Cys	Val	Leu	Trp	Tyr	Ser	Asn
			85					90						95	
Leu	Trp	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Thr	Val	Leu			
		100						105							

<210> SEQ ID NO 110

<211> LENGTH: 327

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 110

caggctgtgg tcacacaaga gcctagcctg acagtgtctc ctggcggcac agtgaccctg	60
acctgtgatg attctacagg cgccgtgacc accagcaact acgctaattg gttccaggag	120
aagcccggcc aggctcctag aggactgata tacggaacaa acaagagagc cccttggaca	180
cccgccagat tctctggatc tctgctcggc ggaaaggccg ctctgacact ttctggtgct	240
caggctgagg acgaggccga gtactattgt gtctctgggt acagcaacct gtgggtgttc	300
ggcggaggca ccaaactgac agttctg	327

<210> SEQ ID NO 111

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 111

Tyr	Gly	Thr	Asn	Lys	Arg	Ala	Pro
1			5				

<210> SEQ ID NO 112

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 112

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly
1			5					10						15	
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Asn	Thr	Tyr
		20						25					30		
Ala	Met	Asn	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
	35					40						45			
Ser	Arg	Ile	Arg	Ser	Lys	Tyr	Asn	Asn	Tyr	Ala	Thr	Tyr	Tyr	Ala	Asp
	50				55					60					
Ser	Val	Lys	Asp	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Ser	Lys	Asn	Thr	
65				70					75				80		
Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Lys	Thr	Glu	Asp	Thr	Ala	Val	Tyr
			85					90					95		
Tyr	Cys	Val	Arg	His	Gly	Asn	Phe	Gly	Asn	Ser	Tyr	Val	Ser	Trp	Phe

-continued

100	105	110	
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser			
115	120	125	
 <210> SEQ ID NO 113			
<211> LENGTH: 375			
<212> TYPE: DNA			
<213> ORGANISM: Homo sapiens			
 <400> SEQUENCE: 113			
gaggtgcagc tggttgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg			60
tcttgtgccg ccagcggcgtt caccttcaac acctacgcta tgaactgggt ccgacaggcc			120
cctggcaaag gactggaatg ggtgtccaga atcagggtcca agtacaacaa ctacgccacc			180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc			240
ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga			300
cacggcaact tcggcaacag ctatgtgtct tggtttgctt actggggcca gggcacactg			360
gtcacagtta gctct			375

<210> SEQ ID NO 114
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 114

Asn Thr Tyr Ala Met Asn
1 5

<210> SEQ ID NO 115
 <211> LENGTH: 18
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 115

Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ser Val
1 5 10 15

Lys Asp

<210> SEQ ID NO 116
 <211> LENGTH: 14
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 116

His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe Ala Tyr
1 5 10

<210> SEQ ID NO 117
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <400> SEQUENCE: 117

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
20 25 30

-continued

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 118
 <211> LENGTH: 375
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 118

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg 60
 tcttgtgccg ccagcgggett caccttctcc acctacgcta tgaactgggt ccgaaaggcc 120
 cctggcaaag gactggaatg ggtgtccaga atcaggtcca agtacaacaa ctacgccacc 180
 tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc 240
 ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga 300
 cacggcaact tcggcaacag ctatgtgtct tggtttgctt actggggcca gggcacactg 360
 gtcacagtta gctct 375

<210> SEQ ID NO 119
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 119

Ser Thr Tyr Ala Met Asn
 1 5

<210> SEQ ID NO 120
 <211> LENGTH: 125
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 120

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
 20 25 30

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95

-continued

Tyr Cys Val Arg His Gly Asn Phe Gly Glu Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 121

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 121

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg 60
 tcttgtgccg ccagcgggtt caccttctcc acctacgcta tgaactgggt ccgaaaggcc 120
 cctggcaaag gactggaatg ggtgtccaga atcaggtcca agtacaacaa ctacgccacc 180
 tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc 240
 ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga 300
 cacggcaact tcggcgagag ctatgtgtct tggtttcct actggggcca gggcacactg 360
 gtcacagtta gctct 375

<210> SEQ ID NO 122

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 122

His Gly Asn Phe Gly Glu Ser Tyr Val Ser Trp Phe Ala Tyr
 1 5 10

<210> SEQ ID NO 123

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 123

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
 20 25 30
 Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60
 Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80
 Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95
 Tyr Cys Val Arg His Gly Asn Phe Gly Gln Ser Tyr Val Ser Trp Phe
 100 105 110
 Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 124

<211> LENGTH: 375

<212> TYPE: DNA

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 124

```

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg      60
tcttgtgccg ccagcgggtt caccttctcc acctacgcta tgaactgggt ccgaaaggcc      120
cctggcaaaag gactggaatg ggtgtccaga atcaggtcca agtacaacaa ctacgccacc      180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc      240
ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga      300
cacggcaact tcggccagag ctatgtgtct tggtttgctt actggggcca gggcacactg      360
gtcacagtta gctct                                           375

```

<210> SEQ ID NO 125

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 125

```

His Gly Asn Phe Gly Gln Ser Tyr Val Ser Trp Phe Ala Tyr
1           5           10

```

<210> SEQ ID NO 126

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 126

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
20           25           30
Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
35           40           45
Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50           55           60
Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65           70           75           80
Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
85           90           95
Tyr Cys Val Arg His Gly Asn Phe Gly Asp Ser Tyr Val Ser Trp Phe
100          105          110
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115          120          125

```

<210> SEQ ID NO 127

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 127

```

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg      60
tcttgtgccg ccagcgggtt caccttctcc acctacgcta tgaactgggt ccgaaaggcc      120
cctggcaaaag gactggaatg ggtgtccaga atcaggtcca agtacaacaa ctacgccacc      180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc      240

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-continued

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ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga    300
cacggcaact tcggcgacag ctatgtgtct tggtttgctt actggggcca gggcacactg    360
gtcacagtta gctct                                                    375

```

```

<210> SEQ ID NO 128
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 128

```

```

His Gly Asn Phe Gly Asp Ser Tyr Val Ser Trp Phe Ala Tyr
1           5           10

```

```

<210> SEQ ID NO 129
<211> LENGTH: 125
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 129

```

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
20          25          30
Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50          55          60
Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65          70          75          80
Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
85          90          95
Tyr Cys Val Arg His Gly Asn Phe Gly Thr Ser Tyr Val Ser Trp Phe
100         105         110
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120         125

```

```

<210> SEQ ID NO 130
<211> LENGTH: 375
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 130

```

```

gagggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg    60
tcttgtgccg ccagcggctt caccttctcc acctacgcta tgaactgggt ccgaaaggcc    120
cctggcaaag gactggaatg ggtgtccaga atcagggtcca agtacaacaa ctacgccacc    180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacacc    240
ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga    300
cacggcaact tcggcaccag ctatgtgtct tggtttgctt actggggcca gggcacactg    360
gtcacagtta gctct                                                    375

```

```

<210> SEQ ID NO 131
<211> LENGTH: 14
<212> TYPE: PRT

```

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 131

His Gly Asn Phe Gly Thr Ser Tyr Val Ser Trp Phe Ala Tyr
 1 5 10

<210> SEQ ID NO 132

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 132

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
 20 25 30

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ser Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Glu Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 133

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 133

gagggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg 60
 tcttgtgccg ccagcggcgtt caccttctcc gactacgcta tgaactgggt ccgaaaggcc 120
 cctggcaaag gactggaatg ggtgtccaga atcaggtcca agtacaacaa ctacgccacc 180
 tactacgccg acagcgtgga ggacagattc accatcagca gggacgacag caagaacacc 240
 ctgtacctgc agatgaacag cctgagagcc gaggacaccg ccgtgtacta ctgtgtcaga 300
 cacggcaact tcggcaacag ctatgtgtct tggtttgccct actggggcca gggcacactg 360
 gtcacagtta gctct 375

<210> SEQ ID NO 134

<211> LENGTH: 6

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 134

Ser Asp Tyr Ala Met Asn
 1 5

<210> SEQ ID NO 135

<211> LENGTH: 18

<212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 135

Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp Ser Val
 1 5 10 15

Glu Asp

<210> SEQ ID NO 136

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 136

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
 20 25 30

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Gly Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Ser
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Ala Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 137

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 137

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg 60

tcttgtgccg ccagcggcgt caccctcaac acctacgcta tgaactgggt ccgaaaggcc 120

cctggcaaag gactggaatg ggtgggaaga atcaggtcca agtacaacaa ctacgccacc 180

tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacagc 240

ctgtacctgc agatgaacag cctgaaaacc gaggacaccg ccgtgtacta ctgtgccaga 300

cacggcaact tcggcaacag ctatgtgtct tggtttgctt actggggcca gggcacactg 360

gtcacagtta gctct 375

<210> SEQ ID NO 138

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 138

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
 20 25 30

-continued

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Gly Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Ser
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 139

<211> LENGTH: 375

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 139

```

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg      60
tcttgtgccg ccagcgggtt caccttcaac acctacgcta tgaactgggt ccgaaaggcc      120
cctggcaaag gactggaatg ggtgggaaga atcaggtcca agtacaacaa ctacgccacc      180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacagc      240
ctgtacctgc agatgaacag cctgaaaacc gaggacaccg ccgtgtacta ctgtgtcaga      300
cacggcaact tcggcaacag ctatgtgtct tggtttgctt actggggcca gggcacactg      360
gtcacagtta gctct

```

<210> SEQ ID NO 140

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 140

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
 20 25 30

Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Ser
 65 70 75 80

Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
 85 90 95

Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 141

<211> LENGTH: 375

-continued

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 141

```

gaggtgcagc tgggtgaatc tggcggagga ctggttcagc ctggcggatc tctgagactg      60
tcttgtgccg ccagcggett caccttcaac acctacgcta tgaactgggt ccgaaaggcc      120
cctggcaaag gactggaatg ggtggccaga atcaggtcca agtacaacaa ctacgccacc      180
tactacgccg acagcgtgaa ggacagattc accatcagca gggacgacag caagaacagc      240
ctgtacctgc agatgaacag cctgaaaacc gaggacaccg ccgtgtacta ctgtgtcaga      300
cacggcaact tcggcaacag ctatgtgtct tggtttgctt actggggcca gggcacactg      360
gtcacagtta gctct                                     375

```

<210> SEQ ID NO 142

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 142

```

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1             5             10             15
Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Val Asp Thr Ala
20            25            30
Val Ala Trp Tyr Gln Lys Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35            40            45
Tyr Trp Thr Ser Thr Arg His Thr Gly Val Pro Ser Arg Phe Ser Gly
50            55            60
Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile Ser Ser Leu Gln Pro
65            70            75            80
Glu Asp Ile Ala Thr Tyr Phe Cys Gln Gln Phe Ser Ser Tyr Pro Leu
85            90            95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100           105

```

<210> SEQ ID NO 143

<211> LENGTH: 321

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 143

```

gacatccaga tgacacagag ccctagcagc ctgtctgccca gcgtgggaga cagagtgacc      60
atcacctgtc aggccagcca ggatgtggat acagccgtgg cctgggtatca gaagaagcct      120
ggcaaggccc ctaagctgct gatctactgg accagcacca gacacacagg cgtgccctct      180
agattcagcg gctctggctc tggcacccgac ttcaccttca caatcagcag cctgcagcct      240
gaggatatcg ctacctactt ctgccagcag ttcagcagct accctctgac attcggccag      300
ggcaccaagg tggaaatcaa g                                     321

```

<210> SEQ ID NO 144

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 144

-continued

Gln Ala Ser Gln Asp Val Asp Thr Ala Val Ala
1 5 10

<210> SEQ ID NO 145
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 145

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ser
1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
20 25 30
 Trp Ile Gly Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35 40 45
 Gly Asp Ile Tyr Pro Gly Ser Gly Asp Thr Asn Tyr Asn Gly Lys Phe
50 55 60
 Gln Gly Arg Val Thr Leu Thr Ala Asp Ile Ser Ser Asn Thr Ala Tyr
65 70 75 80
 Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95
 Ala Arg Ser Tyr Tyr Arg Asn Asp Ala Ala Ser Phe Ala Tyr Trp Gly
100 105 110
 Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 146
 <211> LENGTH: 363
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 146

cagggttcagc tgggttcagtc tggcgccgaa gtgaagaaac ctggcagcag cgtgaaggtg 60
 tcctgcaagg ctageggcta cacattcacc aactactgga tcggetgggt ccgagaggct 120
 cctggacaag gacttgagtg gatgggagac atctaccctg gcagcggcga cacaactac 180
 aacggcaagt tccagggcag agtgaccctg acagccgaca tcagcagcaa caccgcctac 240
 atggaactga gcagcctgag aagcgaggac accgcccgtg actactgcgc cagaagctac 300
 tacagaaaacg acgcccag cttcgcttac tggggacagg gaacactggt caccgtgtct 360
 agt 363

<210> SEQ ID NO 147
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 147

Asp Ile Tyr Pro Gly Ser Gly Asp Thr Asn Tyr Asn Gly Lys Phe Gln
1 5 10 15

Gly

<210> SEQ ID NO 148
 <211> LENGTH: 111
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 148

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
1           5           10           15
Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
20          25          30
Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
35          40          45
Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
50          55          60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
65          70          75          80
Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85          90          95
Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100         105         110

```

<210> SEQ ID NO 149

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 149

```

gacatcgtgc tgacacagag cctgtcttct ctggctgtgt ctctggcca gagagccacc      60
atcacctgta gagccagcaa gagcgtgtcc accagcggct actcttacat gcactggtat      120
cagaagaagc ccggccagcc tcctaagctg ctgatctacc tggctagcaa cctcgaaagc      180
ggagtgcctg ctagattttc tggcagcggc tctggcacog acttcacct gacaatcaac      240
cccgtggaag ccaacgacac cgccaactac tactgccagc acagcagaga gctgccctgg      300
acatttggcc agggcaccaa ggtggaaatc aag                                     333

```

<210> SEQ ID NO 150

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 150

```

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
1           5           10           15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Phe
20          25          30
Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35          40          45
Gly Trp Ile Asn Thr Tyr Thr Gly Asp Gln Ile Tyr Ala Asp Asp Phe
50          55          60
Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
65          70          75          80
Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
85          90          95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100         105         110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

-continued

<210> SEQ ID NO 151

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 151

```

caggttcagc tggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaaggtg      60
tcctgcaagg ctagcggcta cacattcacc aacttcggca tgaactgggt ccgagaggct      120
cctggacagg gactcgaatg gatgggcttg atcaacacct acaccggcga ccagatctac      180
gccgacgact tcaaggcgag attcgccttc agcctggaca cctctgccag cacagcttac      240
ctgcagatca gcagcctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag      300
atctactacg gctacgatgt gggcttcgtc tactggggac agggaacact ggtcaccgtt      360
agctct

```

<210> SEQ ID NO 152

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 152

```

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
1         5         10        15
Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Arg Ser Val Thr Thr Ser
20        25        30
Gly Tyr Ser Tyr Ile His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
35        40        45
Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ala Gly Val Pro Ala
50        55        60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
65        70        75        80
Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85        90        95
Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100       105       110

```

<210> SEQ ID NO 153

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 153

```

gacatcgtgc tgacacagag ccctgcttct ctggctgtgt ctctggcca gagagccacc      60
atcacctgta gagccagcag aagcgtgacc accagcggct actcttacat ccactggtat      120
cagaagaagc ccggccagcc tcctaagctg ctgatctacc tggccagcga tctggaagct      180
ggcgtgccag ctagatttct tggcagcggc tctggcaccg acttcacct gacaatcaac      240
cccgtggaag ccaacgacac cgccaactac tactgccagc acagcagaga gctgccttgg      300
acatttggcc agggcaccaa ggtggaaatc aag

```

<210> SEQ ID NO 154

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

-continued

<400> SEQUENCE: 154

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
 20 25 30
 Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45
 Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Asp Phe
 50 55 60
 Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80
 Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95
 Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 155

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 155

cagggttcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaaggtg 60
 tcctgcaagg ctageggcta catcttcacc aacttcggca tgaactgggt ccgagaggct 120
 cctggacagg gactcgaatg gatgggcttg atcaacacct acaccggcga gcagatctac 180
 gccgatgact tcaaaggcag attcgcgttc agcctggaca ccagcgccag cacagcttac 240
 ctgcagatca gctctctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag 300
 atctactacg gctacgacgt gggctttgtg tactggggcc agggaacact ggtcaccgtt 360
 agctct 366

<210> SEQ ID NO 156

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 156

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
 1 5 10 15
 Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30
 Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
 35 40 45
 Lys Leu Leu Ile Tyr Leu Gly Ser Asp Leu Glu Ser Gly Val Pro Ala
 50 55 60
 Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
 65 70 75 80
 Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
 85 90 95
 Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105 110

-continued

<210> SEQ ID NO 157

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 157

```

gacatcgtgc tgacacagag cctgcttct ctggtgtgt ctctggcca gagagccacc    60
atcacctgta gagccagcaa gagcgtgtcc accagcggct actcttacat gcactgggat    120
cagaagaagc cgggccagcc tctaagctg ctgatctacc tgggtagcga cctcgaaagc    180
ggagtgcctg ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac    240
ccgtggaag ccaacgacac cgccaactac tactgccagc acagcagaga gctgccttgg    300
acatttgccc agggcaccaa ggtggaaatc aag                                333

```

<210> SEQ ID NO 158

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 158

```

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
1      5      10      15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
20     25     30
Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35     40     45
Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Asp Phe
50     55     60
Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
65     70     75     80
Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
85     90     95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100    105    110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115    120

```

<210> SEQ ID NO 159

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 159

```

caggttcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgcctc tgtgaaggtg    60
tcctgcaagg ctacgggcta catcttcacc aacttcggca tgaactgggt ccgagaggct    120
cctggacagg gactogaatg gatgggctgg atcaaacact acaccggcga gcagatctac    180
gccgatgact tcaaaggcag attcgcgttc agcctggaca ccagcgccag cacagcttac    240
ctgcagatca gctctctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag    300
atctactacg gctacgacgt gggctttgtg tactggggcc agggaaactt ggtaaccgtt    360
agctct

```

-continued

<210> SEQ ID NO 160

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 160

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
 1 5 10 15

Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30

Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
 35 40 45

Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
 50 55 60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
 65 70 75 80

Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
 85 90 95

Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 161

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 161

gacatcgtgc tgacacagag ccctgcttct ctggctgtgt ctcttgcca gagagccacc 60

atcacctgta gagccagcaa gagcgtgtcc accagcggt actcttacat gcactgggtat 120

cagaagaagc ccggccagcc tctaagctg ctgatctacc tggctagcaa cctcgaaagc 180

ggagtgcctg ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac 240

cccgagggaag ccaacgcac cgccaactac tactgccagc acagcagaga gctgccttgg 300

acatttggcc agggcaccaa ggtggaaatc aag 333

<210> SEQ ID NO 162

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 162

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
 20 25 30

Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Asp Phe
 50 55 60

Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95

Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
 100 105 110

-continued

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 163

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 163

```
caggttcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaagggtg    60
tcctgcaagg ctagcggcta catcttcacc aacttcggca tgaactgggt ccgagagggt    120
cctggacagg gactcgaatg gatgggctgg atcaaacacct acaccggcga gcagatctac    180
gccgatgact tcaaaggcag attcgcgttc agcctggaca ccagcgccag cacagcttac    240
ctgcagatca gctctctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag    300
atctactacg gctacgacgt gggctttgtg tactggggcc agggaaactt ggtcaccgtt    360
agctct
```

<210> SEQ ID NO 164

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 164

```
Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
1      5      10     15
Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
20     25     30
Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
35     40     45
Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
50     55     60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
65     70     75     80
Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85     90     95
Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100    105    110
```

<210> SEQ ID NO 165

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 165

```
gacatcgctg tgacacagag cctgcttct ctggctgtgt ctctggcca gagagccacc    60
atcacctgta gagccagcaa gagcgtgtcc accagcggct actcttacat gcactggtat    120
cagaagaagc cggccagacc tcctaagctg ctgatctacc tggctagcaa cctcgaaagc    180
ggagtgcctg ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac    240
cccggtgaag ccaacgacac cgccaactac tactgccagc acagcagaga gctgccctgg    300
acatttggcc agggcaccaa ggtggaaatc aag
```

-continued

<210> SEQ ID NO 166

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 166

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
 1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Phe
 20 25 30

Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
 35 40 45

Gly Trp Ile Asn Thr Tyr Thr Gly Asp Leu Ile Tyr Gly Asp Asp Phe
 50 55 60

Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
 85 90 95

Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Ala Tyr Trp
 100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 167

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 167

cagggtcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaagggtg 60
 tcctgcaagg ctagcggcta cacattcacc aacttcggca tgaactgggt ccgagaggct 120
 cctggacagg gactogaatg gatgggctgg atcaacacct acaccggcga cctgatctac 180
 ggcgacgact tcaagggcag attcgccctc agcctggaca cctctgccag cacagcttac 240
 ctgcagatca gcagcctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag 300
 atctactacg gctacgatgt gggcttcgcc tactggggac agggaaact ggtcaccgtt 360
 agctct 366

<210> SEQ ID NO 168

<211> LENGTH: 111

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 168

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
 1 5 10 15

Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
 20 25 30

Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
 35 40 45

Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
 50 55 60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
 65 70 75 80

-continued

Pro Val Glu Ala Asn Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85 90 95

Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105 110

<210> SEQ ID NO 169

<211> LENGTH: 333

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 169

```

gacatcgctgc tgacacagag ccctgcttct ctggctgtgt ctctggcca gagagccacc      60
atcacctgta gagccagcaa gagcgtgtcc accagcggtc actcttacat gcactggtat      120
cagaagaagc cgggccagcc tcctaagctg ctgatctacc tggctagcaa cctcgaaagc      180
ggagtgcctg ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac      240
cccgtggaag ccaacgacac cgccaactac tactgccagc acagcagaga gctgccctgg      300
acatttggcc agggcaccaa ggtggaaatc aag                                     333

```

<210> SEQ ID NO 170

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 170

```

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
1      5      10     15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
20     25     30
Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35     40     45
Gly Trp Ile Asn Thr Tyr Thr Gly Glu Arg Ile Tyr Ala Asp Asp Phe
50     55     60
Lys Gly Arg Phe Ala Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
65     70     75     80
Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
85     90     95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100    105    110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115    120

```

<210> SEQ ID NO 171

<211> LENGTH: 366

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 171

```

caggttcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaagggtg      60
tcctgcaagg ctagcggcta cacattcacc aactacggca tgaactgggt ccgagaggct      120
cctggacagg gactcgaatg gatgggctgg atcaaacacct acaccggcga aaggatctac      180
gccgacgact tcaagggcag attcgccctc agcctggaca cctctgccag cacagcttac      240
ctgcagatca gcagcctgaa ggccgaggat accgccgtgt acttctgtgc cagaggcgag      300

```

-continued

```
atctactacg gctacgatgt gggttcgtc tactggggac agggaacact ggtcaccgtt 360
agctctt 366
```

```
<210> SEQ ID NO 172
<211> LENGTH: 218
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
```

```
<400> SEQUENCE: 172
```

```
Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
1           5           10          15
Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Ser Thr Ser
20          25          30
Gly Tyr Ser Tyr Met His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
35          40          45
Lys Leu Leu Ile Tyr Leu Ala Ser Asn Leu Glu Ser Gly Val Pro Ala
50          55          60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
65          70          75          80
Pro Val Glu Ala Glu Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85          90          95
Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
100         105         110
Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
115         120         125
Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
130         135         140
Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
145         150         155         160
Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
165         170         175
Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys
180         185         190
His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro
195         200         205
Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
210         215
```

```
<210> SEQ ID NO 173
<211> LENGTH: 654
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens
```

```
<400> SEQUENCE: 173
```

```
gacatcgtgc tgacacagag ccttgccttct ctggctgtgt ctctggcca gagagccacc 60
atcacctgta gagccagcaa gagcgtgtcc accagcggct actcttacat gcaactggtat 120
cagaagaagc ccggccagcc tcctaagctg ctgatctacc tggctagcaa cctcgaaagc 180
ggagtgcctg ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac 240
cccgtggaag ccgaagacac cgccaactac tactgccagc acagcagaga gctgccctgg 300
acatttgccc agggcaccaa ggtggaatc aagcgaactg tggctgcacc atctgtcttc 360
atcttcccg cactgatga gcagttgaaa tctggaactg cctctgttgt gtgcctgctg 420
```

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```

aataacttct atcccagaga ggccaaagta cagtggaagg tggataacgc cctccaatcg      480
ggtaactccc aggagagtgt cacagagcag gacagcaagg acagcaccta cagcctcagc      540
agcaccttga cgctgagcaa agcagactac gagaacaca aagtctacgc ctgcgaagtc      600
acccatcagg gcctgagctc gcccgtcaca aagagcttca acaggggaga gtgt          654

```

<210> SEQ ID NO 174

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 174

```

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
20          25          30
Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35          40          45
Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Gly Phe
50          55          60
Thr Gly Arg Phe Val Phe Ser Leu Asp Thr Ser Ala Ser Thr Ala Tyr
65          70          75          80
Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
85          90          95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100         105         110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
115         120         125
Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr
130         135         140
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
145         150         155         160
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
165         170         175
Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
180         185         190
Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp
195         200         205
His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr
210         215         220
Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Glu Gly Gly Pro
225         230         235         240
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
245         250         255
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp
260         265         270
Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
275         280         285
Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val
290         295         300
Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
305         310         315         320

```

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Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Ala Ser Ser Ile Glu Lys
 325 330 335
 Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys Thr
 340 345 350
 Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser
 355 360 365
 Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
 370 375 380
 Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu
 385 390 395 400
 Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys
 405 410 415
 Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu
 420 425 430
 Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly
 435 440 445
 Lys

<210> SEQ ID NO 175

<211> LENGTH: 1347

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 175

```

caggttcagc tgggtcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaaggtg      60
tcctgcaagg ctagcggtcta catcttcacc aacttcggca tgaactgggt ccgagaggct      120
cctggacagg gactcgaatg gatgggctgg atcaacacct acaccggcga gcagatctac      180
gocgatggct tcacaggcag attcgtgttc agcctggaca ccagcgccag cacagcttac      240
ctgcagatca gctctctgaa ggccgaggat accgcccgtg acttotgtgc cagaggcgag      300
atctactacg gctacgacgt gggctttgtg tactggggcc agggaacact ggtcaccgtt      360
agctctgcta gcaccaaggg ccccgcgctg tccccctgg ccccttgacg cagaagcacc      420
agcgagagca cagccgcctt gggctgcctg gtgaaggact acttccccga gcccgtagcc      480
gtgtcctgga acagcggcgc tctgaccagc ggcgtgcata ccttccccgc cgtgctccag      540
agcagcggac tgtactccct gagcagcgtg gtgaccgtgc cttccagcag cctgggcacc      600
aagacctaca cctgcaacgt ggaccacaag cccagcaaca ccaagggtga caagagagtg      660
gagagcaagt acggccctcc ctgccccctt tgccctgccc ccgagttcga gggcggacct      720
agcgtgttcc tgttcccccc caagcccaag gacaccctga tgatcagcag aacccccgag      780
gtgacctgcy tgggtggtgga cgtgtcccag gaggaccccc aggtccagtt taattggtac      840
gtggacggcg tggaagtgca taacgccaag accaagccca gagaggagca gttcaacagc      900
acctacagag tggtgtccgt gctgaccgtg ctgcaccagg actggctgaa cggaaggaa      960
tacaagtgca aggtctccaa caagggcctg gccagcagca tcgagaagac catcagcaag     1020
gccaaaggcc agccacggga gccccaggtc tgcaccctgc cacctagcca agaggagatg     1080
accaagaacc aggtgtccct gagctgtgcc gtgaaaggct tctatcccag cgatatcgcc     1140
gtggagtggg agagcaacgg ccagcccagag aacaactaca agaccacccc cctgtgtgctg     1200
gacagcgacg gcagcttctt cctggtttcc aagctgaccg tggacaagtc cagatggcag     1260

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gaggggcaacg tcttcagctg ctcctgtgatg cactgaggccc tgcacaacca ctacaccag 1320
aagtcacctga gcctgagcct gggcaag 1347

<210> SEQ ID NO 176

<211> LENGTH: 218

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 176

Asp Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Pro Gly
1 5 10 15
Gln Arg Ala Thr Ile Thr Cys Arg Ala Ser Lys Ser Val Thr Thr Ser
20 25 30
Gly Tyr Ser Tyr Ile His Trp Tyr Gln Lys Lys Pro Gly Gln Pro Pro
35 40 45
Lys Leu Leu Ile Tyr Leu Ala Ser Asp Leu Glu Ala Gly Val Pro Ala
50 55 60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn
65 70 75 80
Pro Val Glu Ala Glu Asp Thr Ala Asn Tyr Tyr Cys Gln His Ser Arg
85 90 95
Glu Leu Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
100 105 110
Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
115 120 125
Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
130 135 140
Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
145 150 155 160
Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
165 170 175
Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys
180 185 190
His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro
195 200 205
Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
210 215

<210> SEQ ID NO 177

<211> LENGTH: 654

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 177

gacatcgctg tgacacagag cctgtcttct ctggctgtgt ctctggcca gagagccacc 60
atcacctgta gagccagcaa gagcgtgacc accagcggt actcttacat cactgggtat 120
cagaagaagc ccggccagcc tcctaagctg ctgatctacc tggccagcga tctggaagct 180
ggcgtgccag ctagattttc tggcagcggc tctggcaccg acttcaccct gacaatcaac 240
cccgtggaag ccgaagacac cgccaactac tactgccagc acagcagaga gctgccctgg 300
acatttgccc agggcaccaa ggtggaaatc aagcgaactg tggctgcacc atctgtcttc 360
atcttcccg catctgatga gcagttgaaa tctggaactg cctctgttgt gtgcctgctg 420

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aataacttct atcccagaga ggccaaagta cagtgggaagg tggataacgc cctccaatcg      480
ggtaactccc aggagagtgt cacagagcag gacagcaagg acagcaccta cagcctcagc      540
agcacccctga cgctgagcaa agcagactac gagaacaca aagtctacgc ctgcgaagtc      600
acccatcagg gcctgagctc gcccgtcaca aagagcttca acaggggaga gtgt          654

```

<210> SEQ ID NO 178

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 178

```

Gln Val Gln Leu Val Gln Ser Gly Ser Glu Leu Lys Lys Pro Gly Ala
1          5          10          15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Ile Phe Thr Asn Phe
20        25        30
Gly Met Asn Trp Val Arg Glu Ala Pro Gly Gln Gly Leu Glu Trp Met
35        40        45
Gly Trp Ile Asn Thr Tyr Thr Gly Glu Gln Ile Tyr Ala Asp Gly Phe
50        55        60
Thr Gly Arg Phe Val Phe Ser Leu Asp Thr Ser Val Ser Thr Ala Tyr
65        70        75        80
Leu Gln Ile Ser Ser Leu Lys Ala Glu Asp Thr Ala Val Tyr Phe Cys
85        90        95
Ala Arg Gly Glu Ile Tyr Tyr Gly Tyr Asp Val Gly Phe Val Tyr Trp
100       105       110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
115       120       125
Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr
130       135       140
Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
145       150       155       160
Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
165       170       175
Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
180       185       190
Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp
195       200       205
His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr
210       215       220
Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Glu Gly Gly Pro
225       230       235       240
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
245       250       255
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp
260       265       270
Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
275       280       285
Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val
290       295       300
Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
305       310       315       320

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Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Ala Ser Ser Ile Glu Lys
325 330 335

Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys Thr
340 345 350

Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser
355 360 365

Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
370 375 380

Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu
385 390 395 400

Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys
405 410 415

Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu
420 425 430

Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly
435 440 445

Lys

<210> SEQ ID NO 179

<211> LENGTH: 1347

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 179

cagggttcagc tgggtgcagtc tggcagcgag ctgaagaaac ctggcgccctc tgtgaaggtg 60
tcctgcaagg ctagcggcta catcttcacc aacttcggca tgaactgggt ccgagaggct 120
cctggacagg gactcgaatg gatgggctgg atcaacacct acaccggcga gcagatctac 180
gocgatggct tcacaggcag attcgtgttc agcctggaca ccagcgtcag cacagcttac 240
ctgcagatca gctctctgaa ggccgaggat accgccgtgt acttotgtgc cagaggcgag 300
atctactacg gctacgacgt gggctttgtg tactggggcc agggaaact ggtaaccgtt 360
agctctgcta gcaccaaggg cccagcgtg tccccctgg ccccttgca cagaagcacc 420
agcgagagca cagccgccct gggtgcctg gtgaaggact acttccccga gcccgtagcc 480
gtgtcctgga acagcggcgc tctgaccagc ggcgtgcata ccttccccgc cgtgctccag 540
agcagcggac tgtactccct gagcagcgtg gtgaccgtgc cttccagcag cctgggcacc 600
aagacctaca cctgcaacgt ggaccacaag cccagcaaca ccaagggtga caagagagtg 660
gagagcaagt acggccctcc ctgccccctc tgccctgccc ccgagttcga gggcggacct 720
agcgtgttcc tgttcccccc caagcccaag gacaccctga tgatcagcag aacccccgag 780
gtgacctgac tgggtggtgga cgtgtcccag gaggaccccc aggtccagtt taattggtac 840
gtggacggcg tggaagtgca taacgccaag accaagcccc gagaggagca gttcaacagc 900
acctacagag tgggtgtccgt gctgaccgtg ctgcaccagg actgggtgaa cggaaggaa 960
tacaagtgca aggtctccaa caagggcctg gccagcagca tcgagaagac catcagcaag 1020
gccaaaggcc agccacggga gccccaggtc tgcaccctgc cacctagcca agaggagatg 1080
accaagaacc aggtgtccct gagctgtgcc gtgaaaggct tctatcccag cgatatcgcc 1140
gtggagtggg agagcaacgg ccagcccgag aacaactaca agaccacccc cctgtgtgctg 1200
gacagcgacg gcagcttctt cctggtttcc aagctgaccg tggacaagtc cagatggcag 1260

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gaggggcaacg tcttcagctg ctcogtcatg cactgagggcc tgcacaacca ctacaccag 1320
aagtcctctga gcctgagcct gggcaag 1347

<210> SEQ ID NO 180
<211> LENGTH: 215
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 180

Gln Ala Val Val Thr Gln Glu Pro Ser Leu Thr Val Ser Pro Gly Gly
1 5 10 15
Thr Val Thr Leu Thr Cys Arg Ser Ser Thr Gly Ala Val Thr Thr Ser
20 25 30
Asn Tyr Ala Asn Trp Val Gln Glu Lys Pro Gly Gln Ala Pro Arg Gly
35 40 45
Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Trp Thr Pro Ala Arg Phe
50 55 60
Ser Gly Ser Leu Leu Gly Gly Lys Ala Ala Leu Thr Ile Thr Gly Ala
65 70 75 80
Gln Ala Glu Asp Glu Ala Glu Tyr Tyr Cys Val Leu Trp Tyr Ser Asn
85 90 95
Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln Pro
100 105 110
Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu
115 120 125
Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro
130 135 140
Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys Ala
145 150 155 160
Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala
165 170 175
Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg
180 185 190
Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr
195 200 205
Val Ala Pro Thr Glu Cys Ser
210 215

<210> SEQ ID NO 181
<211> LENGTH: 645
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 181

caggctgtgg tcacacaaga gcctagcctg acagtgtctc ctggcggcac agtgaccctg 60
acctgtagat cttctacagg cgccgtgacc accagcaact acgctaattg ggtgcaggag 120
aagcccgccc aggtctctag aggactgac gccggaacaa acaagagagc cccttgagaca 180
cccgccagat tctctggatc tctgtcggc ggaaaggccg ctctgacaat cactggtgct 240
caggctgagg acgaggccga gtactattgt gtgctgtggt acagcaacct gtgggtgttc 300
ggcggaggca ccaactgac agttctgggt cagcccaagg cggcgcctc ggtaactctg 360
ttccgcct cctctgagga gcttcaagcc aacaaggcca cactggtgtg tctcataagt 420

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gacttctatc cgggagccgt gacagtggcc tggaaggcag atagcagccc cgtaaggcg      480
ggagtggaga ccaccacacc ctccaacaa agcaacaaca agtacgcggc cagcagctac      540
ctgagcctga cgctgagca gtggaagtcc cacagaagct acagctgccca ggtaacgcat      600
gaagggagca ccgtggagaa gacagtggcc cctacagaat gttca                      645

```

<210> SEQ ID NO 182

<211> LENGTH: 452

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 182

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
20          25          30
Ala Met Asn Trp Val Arg Lys Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Gly Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
50          55          60
Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Ser
65          70          75          80
Leu Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
85          90          95
Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
100         105         110
Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr
115         120         125
Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser
130         135         140
Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
145         150         155         160
Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
165         170         175
Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
180         185         190
Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys
195         200         205
Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu
210         215         220
Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Glu
225         230         235         240
Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
245         250         255
Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
260         265         270
Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu
275         280         285
Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr
290         295         300
Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
305         310         315         320

```

```
<210> SEQ ID NO 183
<211> LENGTH: 1356
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 183
```

gaggtgcagc	tggttgaatc	tggcggagga	ctggttcagc	ctggcggatc	tctgagactg	60
tcttgtgccg	ccagcggcct	caccttcaac	acctacgcta	tgaactgggt	ccgaaaggcc	120
cctggcaaag	gactggaatg	ggtgggaaga	atcagggtcca	agtacaacaa	ctacgccacc	180
tactacgccg	acagcgtgaa	ggacagattc	accatcagca	gggacgacag	caagaacagc	240
ctgtacctgc	agatgaacag	cctgaaaacc	gaggacaccg	ccgtgtacta	ctgtgtcaga	300
cacggcaact	tcggaacatg	ctatgtgtct	tggtttgcct	actggggcca	gggcacactg	360
gtcacagtta	gctctgctag	caccaagggc	cccagcgtgt	tccccctggc	cccttgcagc	420
agaagcacca	gcgagagcac	agccgccctg	ggctgcctgg	tgaaggacta	cttccccgag	480
cccgtgaccg	tgtcctggaa	cagcggcgct	ctgaccacg	gcgtgcatac	cttccccgcc	540
gtgctccaga	gcagcggact	gtactccctg	agcagcgtgg	tgaccgtgcc	ttccagcagc	600
ctgggcacca	agacctacac	ctgcaacgtg	gaccacaagc	ccagcaacac	caaggtggac	660
aagagagtgg	agagcaagta	cggccctccc	tgccccctt	gccctgcccc	cgagttcgag	720
ggcggaccta	gcgtgttctt	gttccccccc	aagcccaagg	acacctgat	gatcagcaga	780
acccccgagg	tgacctgcgt	ggtggtggac	gtgtcccagg	aggaccccca	ggtccagttt	840
aattggtagc	tggacggcgt	ggaagtgcac	aacgccaa	ccaagcccag	agaggagcag	900
ttcaacagca	cctacagagt	ggtgtccctg	ctgaccgtgc	tgcaccagga	ctggctgaac	960
ggcaggaat	acaagtgcaa	ggtctccaac	aagggccttg	ccagcagcat	cgaagaagacc	1020
atcagcaagg	ccaagggccca	gccacgggag	ccccagggtc	acacctgcc	accttgtcaa	1080
gaggagatga	ccaagaacca	ggtgtccctg	tgggtgtctg	tgaaggcctt	ctatcccagc	1140
gatatcgccg	tggagtggga	gagcaacggc	cagcccagga	acaactacaa	gaccaccccc	1200
cctgtctctg	acacgcacgc	caqcttcttc	ctgtactcca	agctgaccct	ggacaagtcc	1260

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 agatggcagg agggcaacgt cttcagctgc tccgtgatgc acgaggccct gcacaaccac 1320

tacacccaga agtccttgag cctgagcctg ggcaag 1356

<210> SEQ ID NO 184

<211> LENGTH: 109

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 184

 Gln Ala Val Val Thr Gln Glu Ser Ala Leu Thr Thr Ser Pro Gly Glu
 1 5 10 15

 Thr Val Thr Leu Thr Cys Arg Ser Ser Thr Gly Ala Val Thr Thr Ser
 20 25 30

 Asn Tyr Ala Asn Trp Val Gln Gln Lys Pro Asp His Leu Phe Thr Gly
 35 40 45

 Leu Ile Gly Gly Thr Asn Lys Arg Ala Pro Gly Val Pro Ala Arg Phe
 50 55 60

 Ser Gly Ser Leu Ile Gly Asp Lys Ala Ala Leu Thr Ile Thr Gly Ala
 65 70 75 80

 Gln Thr Glu Asp Glu Ala Ile Tyr Phe Cys Ala Leu Trp Tyr Ser Asn
 85 90 95

 Leu Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu
 100 105

<210> SEQ ID NO 185

<211> LENGTH: 125

<212> TYPE: PRT

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 185

 Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Lys Gly
 1 5 10 15

 Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asn Thr Tyr
 20 25 30

 Ala Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45

 Ala Arg Ile Arg Ser Lys Tyr Asn Asn Tyr Ala Thr Tyr Tyr Ala Asp
 50 55 60

 Ser Val Lys Asp Arg Phe Thr Ile Ser Arg Asp Asp Ser Gln Ser Ile
 65 70 75 80

 Leu Tyr Leu Gln Met Asn Asn Leu Lys Thr Glu Asp Thr Ala Met Tyr
 85 90 95

 Tyr Cys Val Arg His Gly Asn Phe Gly Asn Ser Tyr Val Ser Trp Phe
 100 105 110

 Ala Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120 125

1. A bispecific antibody or antigen-binding portion thereof comprising:

- (a) a first antigen-binding portion or an antigen-binding fragment thereof bound to a BCMA antigen on the surface of a target cell, the first antigen-binding portion comprises a first heavy chain and a first light chain, the first antigen-binding portion comprises a first binding domain that binds to the first antigen, wherein the first

binding domain comprises a heavy chain CDR selected from the amino acid sequences of SEQ ID NOS: 12, 13, 14, 26, 27, 28, 36, 37, 43, 44, 50, 55, 56, 57, 65, 66, 71, 72, 73, 147 and 186, or any variant thereof, and/or a light chain CDR selected from the amino acid sequences of SEQ ID Nos: 17, 18, 19, 22, 23, 31, 32, 33, 40, 47, 60, 61, 62, 76, 77, 78, 83, 84, 87, 88, 89, 92 and 144 or any variant; and

(b) a second antigen-binding portion or antigen-binding fragment thereof bound to a CD3 antigen on the surface of a T cell, the second antigen-binding portion comprises a second heavy chain and a second light chain, the second antigen-binding portion comprises a second binding domain that binds to a second antigen.

2. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein the first binding domain comprises a heavy chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 12, 26, 55, 65, 71 or any variant thereof, a heavy chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 13, 27, 36, 43, 50, 56, 66, 72, 147, 186 or any variant thereof, a heavy chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 14, 28, 37, 44, 57, 73 or any variant thereof; and/or the first binding domain comprises a light chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 17, 22, 31, 47, 60, 76, 83, 87, 92, 144 or any variant thereof, a light chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 18, 23, 32, 40, 61, 77, 84, 88 or any variant thereof, a light chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 19, 33, 62, 78, 89 or any variant thereof.

3. The bispecific antibody or antigen-binding portion thereof of claim 2, wherein,

the heavy chain CDR1, CDR2, and CDR3 sequences of the first binding domain are selected from: first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 186 and 14, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 26, 27 and 28, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 36 and 37, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 43 and 44, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 50 and 14, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 55, 56 and 57, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 65, 66 and 14, first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 72 and 73; first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 147 and 73; and the light chain CDR1, CDR2, and CDR3 sequences of the first binding domain are selected from: first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 18 and 19, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 22, 23 and 19, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 31, 32 and 33, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 47, 18 and 19, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 60, 61 and 62, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 76, 77 and

78, first light chain CDR1, CDR2 and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 83, 84 and 19, first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 87, 88 and 89, first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 92, 23 and 19, first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 144, 77 and 78;

preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14, and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 18 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 22, 23 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 26, 27, and 28 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 31, 32, and 33; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 36 and 37 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 43 and 44 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 47, 18 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 50 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 55, 56 and 57 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 60, 61 and 62; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 65, 66 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 17, 40 and 19; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 71, 72, and 73 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 76, 77, and 78; preferably, the first binding domain comprises first heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 12, 13 and 14 and first light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 83, 84 and 19;

of SEQ ID NO: 166 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 164 or any variant thereof;

preferably, the first binding domain comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 170 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 168 or any variant thereof;

preferably, the first binding domain comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 187 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 188 or any variant thereof.

5. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein, the second binding domain comprises a heavy chain CDR selected from the amino acid sequences of SEQ ID NOs: 114, 115, 116, 119, 122, 125, 128, 131, 134, 135 or any variant thereof; and/or a light chain CDR selected from the amino acid sequences of SEQ ID NOs: 95, 96, 97, 102, 103, 108, 111 or any variant thereof;

preferably, the second binding domain comprises a second heavy chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 114, 119, 134 or any variant thereof, a second heavy chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 115, 135 or any variant thereof, a second heavy chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 116, 122, 125, 128, 131 or any variant thereof, and/or a second light chain CDR1 selected from the amino acid sequences of SEQ ID NOs: 95, 102 or any variant thereof, a second light chain CDR2 selected from the amino acid sequences of SEQ ID NOs: 96, 103, 111 or any variant thereof, a second light chain CDR3 selected from the amino acid sequences of SEQ ID NOs: 91, 108 or any variant thereof;

preferably, the heavy chain CDR1, CDR2, and CDR3 sequences of the second binding domain are selected from: second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115 and 116, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 116, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 122, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 125, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 128, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 119, 115 and 131, second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 134, 135 and 116; and the light chain CDR1, CDR2, and CDR3 sequences of the second binding domain are selected from: second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96 and 97, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 102, 103 and 97, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96

and 108, second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 111 and 108;

more preferably, the second binding domain comprises second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115, 116 and second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96, 97;

more preferably, the second binding domain comprises second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115, 116 and second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 102, 103, 97;

more preferably, the second binding domain comprises second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115, 116 and second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 96, and 108;

more preferably, the second binding domain comprises second heavy chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 114, 115, 116 and second light chain CDR1, CDR2, and CDR3 sequences of the amino acid sequences of SEQ ID NOs: 95, 111, and 108;

preferably, the second binding domain comprises a second heavy chain variable region selected from the amino acid sequences of SEQ ID NOs: 112, 117, 120, 123, 126, 129, 132, 136, 138, 140 or any variant thereof, and/or comprises a second light chain variable region selected from the amino acid sequences of SEQ ID NOs: 93, 98, 100, 104, 106, 109 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 136 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 93 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 93 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 98 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 100 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 112 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid

sequence of SEQ ID NO: 136 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

more preferably, the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof.

6. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein, the first light chain of the first antigen-binding portion is a κ type light chain, the second light chain of the second antigen-binding portion is a λ type light chain,

preferably, the second light chain variable region of the second antigen-binding portion has a Gln₄₀Glu mutation (V λ_{CD3} :Gln₄₀Glu); the second heavy chain variable region of the second antigen-binding portion has a Gln₃₉Lys mutation (VH $_{CD3}$:Gln₃₉Lys);

the first light chain variable region of the first antigen-binding portion has a Gln₄₂Lys mutation (VK $_{BCMA}$:Gln₄₂Lys); more preferably, the first heavy chain variable region of the first antigen-binding portion has a Gln₃₉Glu mutation (VH $_{BCMA}$:Gln₃₉Glu);

preferably, the Fc portion of the first antigen-binding portion and the second antigen-binding portion of the bispecific antibody adopts a knob-into-hole structure; preferably, a human IgG4 knob-into-hole structure is adopted.

7. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein the first heavy chain comprises a heavy chain selected from SEQ ID NOs: 174, 178 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NOs: 172, 176 or any variant thereof;

preferably, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 174 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 172 or any variant thereof;

preferably, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 174 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 176 or any variant thereof;

preferably, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 172 or any variant thereof;

preferably, the first heavy chain comprises a heavy chain selected from SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain selected from SEQ ID NO: 176 or any variant thereof.

8. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 145 or any

variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 142 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 150 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 148 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 154 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 152 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 158 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 156 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 162 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 160 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

preferably, the first binding domain of the bispecific antibody comprises a first heavy chain variable region of the amino acid sequence of SEQ ID NO: 166 or any variant thereof, and a first light chain variable region of the amino acid sequence of SEQ ID NO: 164 or any variant thereof; the second binding domain comprises a second heavy chain variable region of the amino acid sequence of SEQ ID NO: 138 or any variant thereof, and a second light chain variable region of the amino acid sequence of SEQ ID NO: 106 or any variant thereof;

more preferably, the first heavy chain of the bispecific antibody comprises a heavy chain of SEQ ID NO: 174 or any variant thereof, the first light chain comprises a light chain of SEQ ID NO: 172 or any variant thereof; the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof;

more preferably, the first heavy chain of the bispecific antibody comprises a heavy chain of SEQ ID NO: 174 or any variant thereof, the first light chain comprises a light chain of SEQ ID NO: 176 or any variant thereof; the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof;

more preferably, the first heavy chain of the bispecific antibody comprises a heavy chain of SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain of SEQ ID NO: 172 or any variant thereof; the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof;

more preferably, the first heavy chain of the bispecific antibody comprises a heavy chain of SEQ ID NO: 178 or any variant thereof, the first light chain comprises a light chain of SEQ ID NO: 176 or any variant thereof; the second heavy chain comprises a heavy chain of SEQ ID NO: 182 or any variant thereof, the second light chain comprises a light chain of SEQ ID NO: 180 or any variant thereof.

9. A BCMA-binding antibody or antigen-binding portion thereof, the antibody comprises the first antigen-binding portion of claim 1.

10. A nucleic acid encoding the bispecific antibody or antigen-binding portion thereof of claim 1 or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion;

preferably, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion is selected from the nucleotide sequences of SEQ ID NOs: 11, 25, 35, 42, 49, 54, 64, 70, 80, 146, 151, 155, 159, 163, 167, 171, and 189, or any variant thereof; and/or the nucleic acid encoding the first light chain variable region of the first antigen-binding portion is selected from the nucleotide sequences of SEQ ID NOs: 16, 21, 30, 39, 46, 52, 59, 68, 75, 82, 86, 91, 143, 149, 153, 157, 161, 165, 169, and 190, or any variant thereof;

more preferably, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 11; and

[illegible]

of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107;

more preferably, the nucleic acid encoding the first heavy chain variable region of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 171; and the nucleic acid encoding the first light chain variable region is the nucleotide sequence of SEQ ID NO: 169; the nucleic acid encoding the second heavy chain variable region of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 139; and the nucleic acid encoding the second light chain variable region is the nucleotide sequence of SEQ ID NO: 107;

preferably, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 175, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 173, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181;

preferably, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 175, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 177, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181;

preferably, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 179, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 173, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181;

preferably, the nucleic acid encoding the first heavy chain of the first antigen-binding portion of the bispecific antibody is the nucleotide sequence of SEQ ID NO: 179, and the nucleic acid encoding the first light chain of the first antigen-binding portion is the nucleotide sequence of SEQ ID NO: 177, the nucleic acid encoding the second heavy chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 183, and the nucleic acid encoding the second light chain of the second antigen-binding portion is the nucleotide sequence of SEQ ID NO: 181.

11. A vector comprising the nucleic acid of claim 10.
12. A cell comprising the nucleic acid of claim 10 or a vector comprising the nucleic acid.
13. A composition comprising the bispecific antibody or antigen-binding portion thereof of claim 1, a BCMA-binding agent, and a cell comprising the nucleic acid of claim 10.

ing antibody or antigen-binding portion thereof comprising the first antigen-binding portion, a nucleic acid encoding the bispecific antibody or antigen-binding portion thereof or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, a vector comprising the nucleic acid and/or a cell comprising the nucleic acid or the vector.

14. An antibody-drug conjugate comprising the bispecific antibody or antigen-binding portion thereof of claim 1 or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion of claim 1 covalently attached to a therapeutic moiety;

preferably, the therapeutic moiety is selected from a cytotoxic moiety, a chemotherapeutic agent, a cytokine, an immunosuppressant, an immunostimulant, a lytic peptide, or a radioisotope;

preferably, the cytotoxic moiety is selected from paclitaxel; cytochalasin B; gramicidin D; ethidium bromide; emetine; mitomycin; etoposide; teniposide; vincristine; vinblastine; colchicine; doxorubicin; daunorubicin; dihydroxy anthracenedione; tubulin inhibitors such as maytansine or analogs or derivatives thereof, antimetabolic agents such as monomethyl auristatin E or F or analogues or derivatives thereof, dolastatin 10 or 15 or an analogue thereof, irinotecan or an analog thereof, mitoxantrone; mithramycin; actinomycin D; 1-dehydrotestosterone; glucocorticoids; procaine; tetracaine; lidocaine; propranolol; puromycin; calicheamicin or an analogue or derivative thereof, antimetabolites such as methotrexate, 6 mercaptopurine, 6 thioguanine, cytarabine, fludarabine, 5 fluorouracil, decadiazine, hydroxyurea, asparaginase, gemcitabine or cladribine; alkylating agents such as mechlorethamine, thiopurine, chlorambucil, melphalan, carmustine (BSNU), lomustine (CCNU), cyclophosphamide, busulfan, dibromomannitol, streptozotocin, dacarbazine (DTIC), procarbazine, mitomycin C; platinum derivatives such as cisplatin or carboplatin; duocarmycin A, duocarmycin SA, rachelmycin (CC-1065) or analogs or derivatives thereof, antibiotics such as actinomycin, bleomycin, daunorubicin, doxorubicin, idarubicin, mithramycin, mitomycin, mitoxantrone, plicomycin, anthramycin (AMC); pyrrolo [2,1-c][1,4]-benzodiazepine (PDB); diphtheria toxin and related molecules such as diphtheria A chain and active fragments thereof and hybrid molecules, ricin toxin such as ricin A or deglycosylated ricin A chain toxin, cholera toxin, Shiga-like toxin such as SLT I, SLT II, SLT IIV, LT toxin, C3 toxin, Shiga toxin, pertussis toxin, tetanus toxin, soybean Bowman-Birk protease inhibitor, *pseudomonas* exotoxin, azlocillin, saporin, modeccin, gelsolin, abrin A chain, modeccin A chain, α -sarcin, *Aleurites fordii* proteins, dianthin proteins, *Phytolacca americana* proteins such as PAPI, PAPI, and PAP-S, *Momordica charantia* inhibitor, curcumin, crotonin, *Saponaire officinalis* inhibitor, gelonin, mitomycin, restrictocin, phenomycin, and enomycin toxins; ribonuclease (RNase); DNase I, staphylococcal endotoxin A; pokeweed antiviral protein; diphtheria toxin and *pseudomonas* endotoxin;

preferably, the cytokine is selected from IL-2, IL-4, IL-6, IL-7, IL-10, IL-12, IL-13, IL-15, IL-18, IL-23, IL-24, IL-27, IL-28a, IL-28b, IL-29, KGF, IFN α , IFN β , IFN γ , GM-CSF, CD40L, Flt3 ligand, stem cell factor, ances-tim, and TNF α ;

preferably, the radioisotope is selected from ^3H , ^{14}C , ^{15}N , ^{35}S , ^{67}Cu , ^{90}Y , ^{99}Tc , ^{125}I , ^{131}I , ^{186}Re , ^{188}Re , ^{211}At , ^{212}Bi , ^{212}Pb , ^{213}Bi , ^{225}Ac , and ^{227}Th .

15. A kit comprising the bispecific antibody or antigen-binding portion thereof of claim 1 or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, or a nucleic acid encoding the bispecific antibody or antigen-binding portion thereof or the BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, or a vector comprising the nucleic acid, or a cell comprising the nucleic acid or the vector.

16. A method of diagnosing, treating or preventing a disease associated with BCMA, wherein the method includes a step of: administering to a subject a therapeutically effective amount of the bispecific antibody or antigen-binding portion thereof of claim 1, a BCMA-binding antibody or antigen-binding portion thereof of comprising the first antigen-binding portion, or a nucleic acid molecule encoding the bispecific antibody or antigen-binding portion thereof or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, or a vector comprising the nucleic acid, or a cell comprising the nucleic acid, or a composition, or an antibody-drug conjugate,

wherein the composition comprises the bispecific antibody or antigen-binding portion thereof, a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, a nucleic acid encoding the bispecific antibody or antigen-binding portion thereof or encoding a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion, a vector comprising the nucleic acid and/or a cell comprising the nucleic acid or the vector,

wherein the antibody-drug conjugate comprises the bispecific antibody or antigen-binding portion thereof or a BCMA-binding antibody or antigen-binding portion thereof comprising the first antigen-binding portion covalently attached to a therapeutic moiety;

preferably, the disease associated with BCMA comprises B cell disease;

preferably, the disease is cancer; more preferably, the cancer is a B-cell related cancer selected from the group consisting of multiple myeloma, malignant plasmacytoma, hodgkin lymphoma, Nodular lymphocyte predominant Hodgkin lymphoma, kahler's disease and myeloid leukemia, plasma cell leukemia, plasmacytoma, B-cell prolymphocytic leukemia, hairy cell leukemia, B-cell non-Hodgkin lymphoma (NHL), acute myelogenous leukemia (AML), chronic lymphocytic leukemia (CLL), acute lymphocytic leukemia (ALL), chronic myelogenous leukemia (CML), follicular lymphoma, burkitt lymphoma, marginal zone lymphoma, mantle cell lymphoma, large cell lymphoma, precursor B-lymphocyte lymphoma, myeloid leukemia, waldenstrom's macroglobulinaemia, diffuse large B cell lymphoma, follicular lymphoma, marginal zone lymphoma, mucosa-associated lymphoid tissue lymphoma, small cell lymphocytic lymphoma, mantle cell lymphoma, burkitt lymphoma, primary mediastinal (thymic) large B cell lymphoma, lymphoplasmacytic lymphoma, waldenstrom's macroglobulinaemia, lymph node marginal zone B cell lymphoma, splenic marginal

zone lymphoma, intravascular large B-cell lymphoma, primary effusion lymphoma, lymphomatoid granulomatosis, large B-cell lymphoma rich in T cells/histiocytes, primary central nervous system lymphoma, primary cutaneous diffuse large B-cell lymphoma (leg type), EBV-positive diffuse large B-cell lymphoma of the elderly, diffuse large B-cell lymphoma, intravascular large B-cell lymphoma associated with inflammation, ALK-positive large B-cell lymphoma, plasmablastic lymphoma, large B-cell lymphoma arising in HHV8-associated multicenter Castleman's disease, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and Burkitt's lymphoma, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and classic Hodgkin lymphoma, and other B-cell-related lymphomas;

more preferably, the B cell disease is a B cell disorder; preferably, the plasma cell disorder is selected from multiple myeloma, plasmacytoma, plasma cell leukemia, macroglobulinaemia, amyloidosis, waldenstrom's macroglobulinaemia, solitary plasmacytoma of the bone, extramedullary plasmacytoma, osteosclerotic myeloma, heavy chain disease, monoclonal gammopathy of indetermined significance, and multiple myeloma of stasis type;

the disease is an autoimmune disorder, such as systemic lupus erythematosus or rheumatoid arthritis.

17. A kit comprising the composition of claim 13.

18. A kit comprising the antibody-drug conjugate of claim 14.

19. The bispecific antibody or antigen-binding portion thereof of claim 1, wherein the first binding domain comprises a heavy chain variable region (VH) comprising VH CDR1, VH CDR2, and VH CDR3 that are shown in SEQ ID NOs: 174 or 178, and a light chain variable region (VL) comprising VL CDR1, VL CDR2, and VL CDR3 that are shown in SEQ ID NOs: 172 or 176;

preferably, the heavy chain comprises a heavy chain selected from SEQ ID NO: 174, the light chain comprises a light chain selected from SEQ ID NO: 172;

preferably, the heavy chain comprises a heavy chain selected from SEQ ID NO: 174, the light chain comprises a light chain selected from SEQ ID NO: 176;

preferably, the heavy chain comprises a heavy chain selected from SEQ ID NO: 178, the light chain comprises a light chain selected from SEQ ID NO: 172;

preferably, the heavy chain comprises a heavy chain selected from SEQ ID NO: 178, the light chain comprises a light chain selected from SEQ ID NO: 176.

20. A nucleic acid encoding the BCMA-binding antibody or antigen-binding fragment thereof of claim 19.

21. A vector comprising the nucleic acid of claim 20.

22. A cell comprising the nucleic acid of claim 20.

23. A vector comprising the vector of claim 21.

24. A composition comprising the antibody or antigen-binding portion thereof of claim 19, the nucleic acid encoding the BCMA-binding antibody or antigen-binding fragment thereof, the vector comprising the nucleic acid and/or the cell comprising the nucleic acid or the vector.

25. An antibody-drug conjugate comprising the bispecific antibody or antigen-binding portion thereof of claim 19 covalently attached to a therapeutic moiety.

26. A kit comprising the antibody or antigen-binding portion thereof of claim 19, the nucleic acid encoding the BCMA-binding antibody or antigen-binding fragment thereof, the vector comprising the nucleic acid, the cell comprising the nucleic acid or the vector, the composition comprising the antibody or antigen-binding portion thereof, the nucleic acid, the vector, the cell and/or the antibody-drug conjugate comprising the bispecific antibody or antigen-binding portion thereof of claim 19 covalently attached to a therapeutic moiety;

preferably, the therapeutic moiety is selected from a cytotoxic moiety, a chemotherapeutic agent, a cytokine, an immunosuppressant, an immunostimulant, a lytic peptide, or a radioisotope;

preferably, the cytotoxic moiety is selected from paclitaxel; cytochalasin B; gramicidin D; ethidium bromide; emetine; mitomycin; etoposide; teniposide; vincristine; vinblastine; colchicine; doxorubicin; daunorubicin; dihydroxy anthracenedione; tubulin inhibitors such as maytansine or analogs or derivatives thereof, antimetabolic agents such as monomethyl auristatin E or F or analogues or derivatives thereof, dolastatin 10 or 15 or an analogue thereof, irinotecan or an analog thereof, mitoxantrone; mithramycin; actinomycin D; 1-dehydrotestosterone; glucocorticoids; procaine; tetracaine; lidocaine; propranolol; puromycin; calicheamicin or an analogue or derivative thereof, antimetabolites such as methotrexate, 6 mercaptopurine, 6 thioguanine, cytarabine, fludarabine, 5 fluorouracil, decadiazine, hydroxyurea, asparaginase, gemcitabine or cladribine; alkylating agents such as mechlorethamine, thiopurine, chlorambucil, melphalan, carmustine (BSNU), lomustine (CCNU), cyclophosphamide, busulfan, dibromomannitol, streptozotocin, dacarbazine (DTIC), procarbazine, mitomycin C; platinum derivatives such as cisplatin or carboplatin; duocarmycin A, duocarmycin SA, rachelmycin (CC-1065) or analogs or derivatives thereof, antibiotics such as actinomycin, bleomycin, daunorubicin, doxorubicin, idarubicin, mithramycin, mitomycin, mitoxantrone, plicomycin, anthramycin (AMC); pyrrolo [2,1-c][1,4]-benzodiazepine (PDB); diphtheria toxin and related molecules such as diphtheria A chain and active fragments thereof and hybrid molecules, ricin toxin such as ricin A or deglycosylated ricin A chain toxin, cholera toxin, Shiga-like toxin such as SLT I, SLT II, SLT IIV, LT toxin, C3 toxin, Shiga toxin, pertussis toxin, tetanus toxin, soybean Bowman-Birk protease inhibitor, *pseudomonas* exotoxin, azlocillin, saporin, modeccin, gelsolin, abrin A chain, modeccin A chain, α -sarcin, *Aleurites fordii* proteins, dianthin proteins, *Phytolacca americana* proteins such as PAPI, PAPII, and PAP-S, *Momordica charantia* inhibitor, curcumin, croton, *Sapaonaria officinalis* inhibitor, gelonin, mitomycin, restrictocin, phenomycin, and enomycin toxins; ribonuclease (RNase); DNase I, staphylococcal endotoxin A; pokeweed antiviral protein; diphtheria toxin and *pseudomonas* endotoxin;

preferably, the cytokine is selected from IL-2, IL-4, IL-6, IL-7, IL-10, IL-12, IL-13, IL-15, IL-18, IL-23, IL-24, IL-27, IL-28a, IL-28b, IL-29, KGF, IFN α , IFN β , IFN γ , GM-CSF, CD40L, Flt3 ligand, stem cell factor, ances-tim, and TNF α ;

preferably, the radioisotope is selected from ^3H , ^{14}C , ^{15}N , ^{35}S , ^{67}Cu , ^{90}Y , ^{99}Tc , ^{125}I , ^{131}I , ^{186}Re , ^{188}Re , ^{211}At , ^{212}Bi , ^{212}Pb , ^{213}Bi , ^{225}Ac , and ^{227}Th .

27. Use of the specific antibody or antigen-binding portion thereof of any of claim 19, the nucleic acid encoding the BCMA-binding antibody or antigen-binding fragment thereof, the vector comprising the nucleic acid, the cell comprising the nucleic acid or the vector, the composition comprising the antibody or antigen-binding portion thereof, the nucleic acid, the vector, the cell and/or the antibody-drug conjugate comprising the bispecific antibody or antigen-binding portion thereof of claim 19 covalently attached to a therapeutic moiety in diagnosing, treatment or prevention of a disease associated with BCMA;

preferably, the disease associated with BCMA comprises B cell disease;

preferably, the disease is cancer; more preferably, the cancer is a B-cell related cancer selected from the group consisting of multiple myeloma, malignant plasmacytoma, hodgkin lymphoma, Nodular lymphocyte predominant Hodgkin lymphoma, kahler's disease and myeloid leukemia, plasma cell leukemia, plasmacytoma, B-cell prolymphocytic leukemia, hairy cell leukemia, B-cell non-Hodgkin lymphoma (NHL), acute myelogenous leukemia (AML), chronic lymphocytic leukemia (CLL), acute lymphocytic leukemia (ALL), chronic myelogenous leukemia (CML), follicular lymphoma, burkitt lymphoma, marginal zone lymphoma, mantle cell lymphoma, large cell lymphoma, precursor B-lymphocyte lymphoma, myeloid leukemia, waldenstrom's macroglobulinaemia, diffuse large B cell lymphoma, follicular lymphoma, marginal zone lymphoma, mucosa-associated lymphoid tissue lymphoma, small cell lymphocytic lymphoma, mantle cell lymphoma,

burkitt lymphoma, primary mediastinal (thymic) large B cell lymphoma, lymphoplasmacytic lymphoma, waldenstrom's macroglobulinaemia, lymph node marginal zone B cell lymphoma, splenic marginal zone lymphoma, intravascular large B-cell lymphoma, primary effusion lymphoma, lymphomatoid granulomatosis, large B-cell lymphoma rich in T cells/histiocytes, primary central nervous system lymphoma, primary cutaneous diffuse large B-cell lymphoma (leg type), EBV-positive diffuse large B-cell lymphoma of the elderly, diffuse large B-cell lymphoma, intravascular large B-cell lymphoma associated with inflammation, ALK-positive large B-cell lymphoma, plasmablastic lymphoma, large B-cell lymphoma arising in HHV8-associated multicenter Castleman's disease, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and Burkitt's lymphoma, unclassified B-cell lymphoma with intermediate features between diffuse large B-cell lymphoma and classic Hodgkin lymphoma, and other B-cell-related lymphomas;

more preferably, the B cell disease is a B cell disorder; preferably, the plasma cell disorder is selected from multiple myeloma, plasmacytoma, plasma cell leukemia, macroglobulinaemia, amyloidosis, waldenstrom's macroglobulinaemia, solitary plasmacytoma of the bone, extramedullary plasmacytoma, osteosclerotic myeloma, heavy chain disease, monoclonal gammopathy of indetermined significance, and multiple myeloma of stasis type; or

the disease is an autoimmune disorder, such as systemic lupus erythematosus or rheumatoid arthritis.

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